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Dynamic movement primitives in robotics: A tutorial survey

Matteo Saveriano ¹, Fares J Abu-Dakka ², Aljaž Kramberger ³ and Luka Peternel ⁴

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Abstract

Biological systems, including human beings, have the innate ability to perform complex tasks in a versatile and agile manner. Researchers in sensorimotor control have aimed to comprehend and formally define this innate characteristic. The idea, supported by several experimental findings, that biological systems are able to combine and adapt basic units of motion into complex tasks finally leads to the formulation of the motor primitives' theory. In this respect, Dynamic Movement Primitives (DMPs) represent an elegant mathematical formulation of the motor primitives as stable dynamical systems and are well suited to generate motor commands for artificial systems like robots. In the last decades, DMPs have inspired researchers in different robotic fields including imitation and reinforcement learning, optimal control, physical interaction, and human–robot co-working, resulting in a considerable amount of published papers. The goal of this tutorial survey is two-fold. On one side, we present the existing DMP formulations in rigorous mathematical terms and discuss the advantages and limitations of each approach as well as practical implementation details. In the tutorial vein, we also search for existing implementations of presented approaches and release several others. On the other side, we provide a systematic and comprehensive review of existing literature and categorize state-of-the-art work on DMP. The paper concludes with a discussion on the limitations of DMPs and an outline of possible research directions.

Keywords

Motor control of artificial systems, movement primitives' theory, dynamic movement primitives, learning from demonstration

1. Introduction

How do biological systems, like humans and animals, execute complex movements in a versatile and creative manner?

In the past decades, researchers of neurobiology and motor control have made a significant effort trying to answer this research question, and their experimental findings lead to the formulation of the motor or motion primitives theory. The motion primitives' theory explains the execution of complex motion with the ability of biological systems of sequencing and adapting units of actions, the so-called motion primitives (Flash and Hochner, 2005; Mussa-Ivaldi, 1999).

Dynamic Movement Primitives (DMPs) have their roots in the motor control of biological systems and can be seen as a rigorous mathematical formulation of the motion primitives as stable nonlinear dynamical systems (Schaal, 2006a, 2006b). In this respect, DMPs represent one of the first attempts to answer the research question:

How artificial systems, like (humanoid) robots, can execute complex movements in a versatile and creative manner?

Beyond their biological motivation, DMPs have a simple and elegant formulation, guarantee convergence to a given target, are sufficiently flexible to create complex behaviors, are capable of reacting to external perturbations in real-time, and can be learned from data using efficient algorithms. These properties explain the “success” of DMPs in robotic applications, where they have been established as a prominent tool for learning and generation of motor commands. Since their formulation in the pioneering work from Ijspeert et al., 2002c; Schaal, 2006), DMPs have been successfully exploited in a variety of applications, becoming de facto the first approach that novices in the Imitation Learning (IL) field use on their robots.

1.1. Existing surveys and tutorials

The popularity of DMPs resulted in a large amount of work that use, modify, or extend the original formulation of Ijspeert and colleagues. The pioneering work (Ijspeert et al., 2001) computes the desired position and velocity as a mixture of multiple dynamical systems with different time scales. One year later, (Ijspeert et al., 2002c; Schaal, 2006) propose a different formulation that became popular and that we name, in this paper, the classical DMPs. As shown in Table 1, some tutorials and surveys already tried to categorize and review existing work on DMPs.

Schaal et al. (2007) presented the classical DMPs as an attempt to unify nonlinear dynamical systems and optimal control theory, that is, the two prominent frameworks used to derive computational models of neurobiological motor theories (Flash and Hochner, 2005; Mussa-Ivaldi, 1999). Ijspeert et al. (2013) presented a homogeneous formulation of rhythmic and discrete DMPs together with some extensions including coupling terms, generalization to different goals, and online adaptation for collision avoidance. They also described possible applications in IL and motion recognition methods. In the same year, Pastor et al. (2013) presented an extension to classical DMPs with a special focus on online adaptation of the DMP attractor landscape by integrating the perceptual information into the action generation process. Later on, Deniša et al. (2016b) reviewed the so-called Compliant Movement Primitive (CMP), which was first introduced by Petrič et al. (2014a). CMPs combine classical DMP to generate the desired kinematic path and torque primitives—a weighted summation of Gaussian basis functions—to generate task-specific dynamics. As shown in the Deniša et al. review, CMPs are capable of accurately tracking the kinematic path in a compliant manner, which makes them well suited for tasks that require interaction of the robot with the environment.

However, the abovementioned papers, reviews, and tutorials primarily focused on the methods and advancements within their respective research group and/or focused on a specific problem or field of application. On the other hand, the DMP-related literature is extensive and broad, with contributions from many research groups that made advancements in several important fields of application. Therefore, the proposed survey and tutorial on DMPs aim to scan a wider range and present a tutorial with unified and structured formulations for various DMP methods and advancements up to date. This should make it clearer for the users to see the differences and connections between various methods and can contribute to easier application. In addition, we provide a more comprehensive and categorized survey of all major DMP application areas in robotics. This can help to inspire the readers to apply the DMPs in various areas.

In the tutorial part, we present mathematical formulations, implementation details, and potential issues of existing DMP formulations starting from the classical DMPs presented in (Ijspeert et al., 2002c; Schaal, 2006) up to recent extensions of DMPs to Riemannian geometry and Symmetric Positive Definite (SPD) matrices (Abu-Dakka and Kyrki, 2020). In the survey part, we meticulously review existing literature on DMPs in a comprehensive and methodological manner by focusing on the quality and significance of their continuations without putting a bias on any particular research group. Details on the systematic review procedure are given as follows.

1.2. Systematic review process

We performed an automatic search for documents containing the string

Dynamic Movement Primitive

in Scopus on 25 November 2020 that returned 1223 papers. We started the search from 2001 when preliminary work on DMPs was published. We further refined the search on 20 June 2023 to include last-minute papers.

We manually inspect all the papers and removed the ones that do not explicitly use DMPs and that only compare against DMP in their literature review. The first and foremost selection criteria were the technical quality of work and the significance of the contribution with respect to the DMP state of the art prior to the publication of any particular paper. In other words, we asked the question “did the paper make a significant step change in the field?”. Therefore, we discarded papers that presented similar (or same) ideas multiple times or that made insignificant improvements to the state of the art. If multiple papers presented the same/similar idea, we included the one with the most comprehensive technical quality, and if the quality was similar, the next deciding factors were publication in more prestigious journals/venues or the most cited ones. This manual selection led to 321 papers on DMPs (out of a total of 373 references) analyzed in this work.

1.3. A taxonomy of DMP-related research

The systematic review of DMP literature led to the taxonomy shown in Figure 1, which also describes the structure of this paper. DMPs are placed at the root of the tree and branch into two nodes, namely, the tutorial and the survey. In the tutorial part, we present different DMP formulations and extensions in rigorous mathematical terms.

The tutorial part spans Sections 2 and 3. Section 2 embraces DMP formulations for discrete and periodic motions, orientation trajectories, and SPD matrices. Section 3 discusses extensions of the DMP formalism to account for skills generalization, joining of multiple primitives, online adaptation based on force feedback, or reference velocity. The section ends with a short description of DMP-related formulations.

The survey part spans Sections 4 and 5. Section 4 presents DMP integration in larger executive frameworks for manipulation and variable impedance tasks, reinforcement, deep, and

Table 1. Comparison between existing papers, reviews, and tutorials about DMPs and our tutorial survey.

Paper/survey/tutorial	Topics	Description
Schaal et al. (2007)	· Classical DMPs · Online adaptation · Optimization	A tutorial that provides a unifying view of the two main approaches used to develop computational motor control theories, namely, differential equations and optimal control. In this work, discrete and rhythmic DMPs (Ijspeert et al., 2002c; Schaal, 2006) are presented as a computational model of the motor primitives' theory (Mussa-Ivaldi 1999) that unifies nonlinear differential equations and optimal control. The tutorial has a section dedicated to DMP parameter optimization beyond ILs. Schaal et al. show how to optimize DMP parameters to minimize various costs describing, for instance, the total jerk of the trajectory or the endpoint variance.
Ijspeert et al. (2013)	· Classical DMPs · Online adaptation · Coupling terms · Generalization	A paper on classical DMPs that in addition to its scientific contribution, it presents both discrete and rhythmic formulations, mostly developed in (Ijspeert et al., 2002c, 2002; Schaal, 2006), and their application in IL and movement recognition. The paper also presents extensions of the classical DMP formulation to prevent high accelerations at the beginning of the motion, to avoid collisions with unforeseen obstacles (Pastor et al. 2009), and to generalize both in space (e.g., reach a different goal) and time (e.g., produce longer/shorter trajectories).
Pastor et al. (2013)	· Classical DMPs · Online adaptation · Coupling terms · Impedance learning	A paper on classical DMPs that in addition to its scientific contribution, it presents both discrete and rhythmic formulations, mostly developed in (Ijspeert et al., 2002c, 2002; Schaal, 2006). The paper also presents extensions of the classical DMP formulation to avoid collisions with unforeseen obstacles (Pastor et al. 2009) and to learn impedance control policies via Reinforcement Learning (RL) (Buchli et al. 2011b). The key difference between Ijspeert et al. (2013) and Pastor et al. (2013) is the section dedicated to the sensory association and online, context-aware adaptation of DMP trajectories using the associative skill memory framework developed in Pastor et al. (2011) and Pastor et al. (2011a).
Deniša et al. (2016b)	· Classical DMPs · Compliant Movement Primitives (CMPs)	A tutorial on CMPs, a framework developed to generate compliant robot behaviors that accurately track a reference trajectory. CMPs exploit classical DMPs to generate the desired kinematic landscape and encode task-dependent dynamics as a combination of Gaussian basis functions (torque primitives). The tutorial shows how to learn torque primitives from training data, how to generalize CMPs to new situations, and how to combine existing CMPs to synthesize new robot motions.
Survey and tutorial	Topics	Description
This paper	DMP tutorial · Classical · Orientation · SPD · Joining · Generalization · Online adaptation · DMP survey · (Co-)Manipulation · Variable impedance · Physical interaction · Rehabilitation · Teleoperation · Motion recognition · Reinforcement, deep, and lifelong learning	This tutorial survey conducts a wide scan of the existing DMP literature with the aim of categorizing and presenting the published work in the field. The main objective of this comprehensive literature review is to give the reader an exhausting overview on DMP-related research, on its major achievements, as well as on open issues and possible research directions. Our tutorial survey also provides a structured and unified formulation for different methods developed starting from the classical DMPs proposed by (Ijspeert et al., 2002c; Schaal, 2006). We believe that such formulation contributes to an easier understanding of different methods and extension that can be found in the literature, clarifying connections and differences among the existing approaches. The tutorial survey also provides an analysis on pros and cons of various methods and a discussion with guidelines for different application scenarios.

life-long learning. Section 5 presents DMPs in different robotic applications including physical interaction, co-manipulation, rehabilitation, teleoperation, motion recognition, humanoids and field robotics, and autonomous driving.

The paper ends with a discussion (Section 6) of presented approaches with the aim of providing, where possible, guidelines to select the most suitable DMP approach for specific needs. We have also collected available DMP implementations (see Table 4) and contributed to the community with further open-source implementations available at <https://gitlab.com/dmp-codes-collection>. Section 6 terminates with a discussion on open issues and possible research directions.

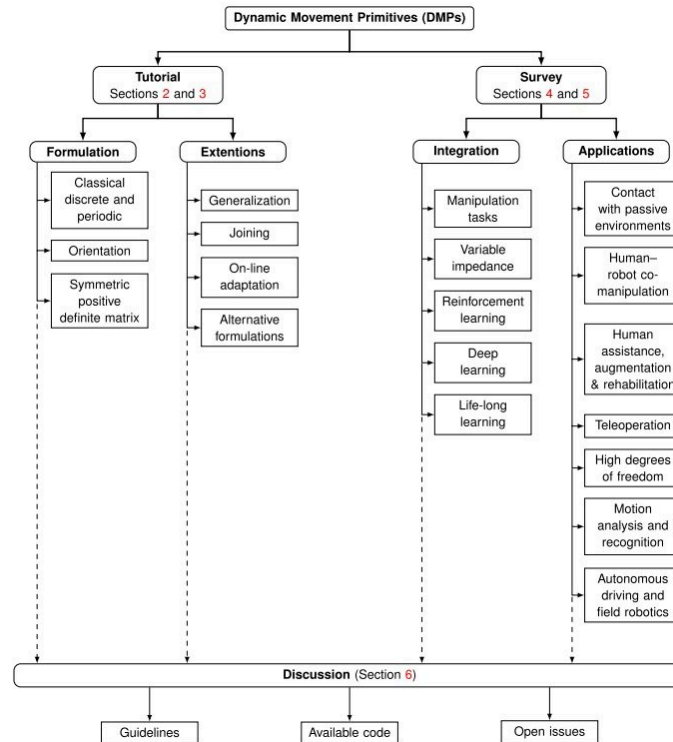


Figure 1. Structure of this tutorial survey on DMPs.

1.4. Contribution overview

Our paper has several key contributions that are summarized as follows.

Concerning the tutorial part:

- We present the classical DMP formulation and existing variations of this formulation in a unified manner with rigorous mathematical terms, providing implementation details and discussing the advantages and limitations of different approaches (Section 2).
- We describe advanced approaches where DMPs are integrated into sophisticated control and/or larger executive frameworks (Section 3).
- We release to the community several implementations of described approaches. Detailed information on these code repositories is provided in Table 4 and Section 6. Moreover, we search for existing open-source implementations of the presented formulations and list them in our repository (Section 6.2).

Concerning the survey part:

- We perform a systematic literature search to provide a comprehensive and unbiased review of the topic (Sections 4 and 5).
- We categorize existing work on DMPs into different streams and highlight prominent approaches in each category (Figure 1 and Sections 4 and 5).

- We present guidelines to select the most suitable approach for different applications, discuss limitations inherent to the DMP formalism, and highlight open issues and possible research directions (Section 6).

2. Formulation of DMP types

In this section, we will provide a complete description of the standard formulation of DMPs. Specifically, point attractors formulation—to encode discrete point-to-point motions—in Section 2.1 and cycle attractors' formulation—to encode rhythmic-patterns motions—in Section 2.2. For a better

understanding, we have summarized the key notations and the used abbreviations in Table 2.

2.1. Discrete DMP

The discrete DMP is used to encode a point-to-point motion into a stable dynamical system. In the following subsections, we will go through the formulation and main features of discrete DMPs starting with the classical one operating in $\mathbb{R}^1$ space (Section 2.1.1), then passing by Cartesian space— $\mathcal{S}^3$ and $\mathcal{SO}(3)$ —in Section 2.1.2, and ending by DMP formulation for SPD space $\mathcal{S}_{++}^m$ in Section 2.1.3.

2.1.1. Classical DMP. The classical discrete DMPs, first introduced by Ijspeert et al. (2002c), encapsulate training data into a linear, second-order dynamics (a mass–spring–damper system) with an additive, nonlinear forcing term learned from a single demonstration. A DMP for a single DoF trajectory y of a discrete movement (point-to-point) is defined by the following set of nonlinear differential equations (Ijspeert et al., 2002c, 2013)

$$\dot{z} = a_z(\beta_z(g-y)-z) + f(x),$$

$$\dot{y} = z,$$

$$\dot{x} = -a_x x$$

where x is the phase variable and z is an auxiliary variable. Parameters α_z and β_z define the behavior of the second-order system described by (1) and (2). With the choice $\tau > 0$, $\alpha_z = 4\beta_z$, and $\alpha_x > 0$, the convergence of the underlying dynamic system to a unique attractor point at $(y = g)$, $(z = 0)$ is ensured (Ijspeert et al., 2013). Alternatively, the gains α_z and β_z can be learned from training data while preserving the convergence of the system (Tan et al., 2016). In the DMP literature, equations (1) and (2), as well as their periodic counterpart (34)–(35), are called the transformation system, while (3) (or (36)) is the canonical system. $f(x)$ is defined as a linear combination of N nonlinear Radial Basis Functions (RBFs), which enables the robot to follow any smooth trajectory from the initial position (y_0) to the final configuration (g) .

$$f(x) = \frac{\sum_{i=1}^N w_i \Psi_i(x)}{\sum_{i=1}^N \Psi_i(x)},$$

$$\Psi_i(x) = \exp(-h_i(x - c_i)^2)$$

where c_i are the centers of Gaussian basis functions distributed along the phase of the movement and h_i their widths. For a given N and setting τ equal to the total duration of the desired movement, we can define $c_i = \exp(-\alpha_x x_i - 1/N - 1)$, $h_i = 1/(c_{i+1} - c_i)^2$, and $h_N = h_{N-1}$ where $i = 1, \dots, N$. For each DoF, the weights w_i should be adjusted from the measured data so that the desired behavior is achieved. The selection of the number of weights should be based on the desired resolution of the trajectory. For controlling a robotic system with more than one DoF, we represent the movement of every DoF with its own equation system (1)–(2), but with the common phase (3) to synchronize them.

2.1.1.1. Learning the forcing term. For a discrete motion, given a demonstrated trajectory $\{y_d(t_J)\}$, $\{t_J = 1, \dots, \mathcal{T}\}$, and its time derivatives $\{\dot{y}_d(t_J)\}$ and $\{\ddot{y}_d(t_J)\}$, it is possible to invert (1) and approximate the desired shape $\{f_d\}$ as follows

$$\{f_d(t_J)\} = \tau^2 \{\ddot{y}_d(t_J)\} - a_z \{ \beta_z (g - y_d(t_J)) - \tau \dot{y}_d(t_J) \}$$

By stacking each $\{f_d(t_J)\}$ and $\{w_i\}$ into the column vectors $\{\mathbf{F} = [f_d(t_1), \dots, f_d(t_{\mathcal{T}})]^{\text{top}}\}$ and $\{\mathbf{w} = [w_1, \dots, w_N]^{\text{top}}\}$, we obtain the following linear system

$$\{\mathbf{\Phi} \mathbf{\Phi}^{\text{top}} \mathbf{w}\} = \mathbf{V}$$

where

$$\mathbf{\Phi} = \begin{bmatrix} \frac{\Psi_1(x_1)}{\sum_{i=1}^N \Psi_i(x_1)} x_1 & \dots & \frac{\Psi_N(x_1)}{\sum_{i=1}^N \Psi_i(x_1)} x_1 \\ \vdots & \ddots & \vdots \\ \frac{\Psi_1(x_{\mathcal{T}})}{\sum_{i=1}^N \Psi_i(x_{\mathcal{T}})} x_{\mathcal{T}} & \dots & \frac{\Psi_N(x_{\mathcal{T}})}{\sum_{i=1}^N \Psi_i(x_{\mathcal{T}})} x_{\mathcal{T}} \end{bmatrix}$$

Locally Weighted Regression (LWR) (Atkeson et al., 1997; Schaal and Atkeson, 1998; Ude et al., 2010) is a popular approach used to update the weights $\{w_i\}$. LWR with the recursive least squares method uses the error between the desired trajectory shape and the currently learned shape and a forgetting factor $\{\lambda\}$ to incrementally update the weights as follows

$$\{\mathbf{P}_J\} = \frac{1}{\lambda} \left(\{\mathbf{P}_{J-1}\} - \frac{\{\mathbf{P}_{J-1}\} \mathbf{\varphi}_J^{\text{top}} \{\mathbf{P}_{J-1}\}}{\lambda + \{\mathbf{P}_{J-1}\} \mathbf{\varphi}_J^{\text{top}} \{\mathbf{P}_{J-1}\}} \right)$$

$$\{\mathbf{w}_J\} = \{\mathbf{w}_{J-1}\} + (f_d(t_J) - \{\mathbf{P}_{J-1}\} \mathbf{\varphi}_J^{\text{top}} \{\mathbf{w}_{J-1}\}) \{\mathbf{P}_J\} \mathbf{\varphi}_J$$

In the previous equations, $\{\mathbf{w}_J = \mathbf{w}(t, j)\}$ and $\{\mathbf{\varphi}_J\}$ is the column vector obtained by transposing the J -th row of $\{\mathbf{\Phi}\}$. The initial value of the parameters is $\{\mathbf{P}_0 = \mathbf{P}_0\}$, $\{\mathbf{w}_0 = \mathbf{w}_0\}$. It is worth mentioning that the update rules in (9)–(10) represent an incremental version of LWR. A batch version of LWR is presented in Ijspeert et al. (2013), and the difference between the two approaches is highlighted in Schaal and Atkeson (1998) on page 9. A discrete DMP learned on synthetic data is shown in Figure 2.

LWR has been the standard method to learn the weights of DMPs and therefore $\{f(x)\}$. As an alternative to LWR, Krug and Dimitrovz (2013) have shown that learning a forcing term defined as in (4) can be formulated as a quadratic optimization problem and efficiently solved.

In general, the problem of learning and retrieving $\{f(x)\}$ can be in principle solved with any regression technique (Stulp et al., 2013). For instance, Wang et al. (2016) modified $\{f(x)\}$ in (4) by considering a bias term $\{b_i\}$, that is, $\{w_i x + b_i\}$, and used truncated kernels ($\{\Psi_i\}$ vanishes if $\{x - c_i\}$ is smaller than a

Table 2. Description of key notations and abbreviations. Indices, super/subscripts, constants, and variables have the same meaning over the whole text.

N	$\triangleleft$	# of nonlinear basis functions	i	$\triangleleft$ index: $(i = 1, 2, \dots, N)$
J	$\triangleleft$	# of joints or Degree of Freedoms (DoFs)	j	$\triangleleft$ index: $(j = 1, 2, \dots, J)$
L	$\triangleleft$	# of demonstrations or DMPs	l	$\triangleleft$ index: $(l = 1, 2, \dots, L)$
V	$\triangleleft$	# of via-points or via-goals	v	$\triangleleft$ index: $(v = 1, 2, \dots, V)$
$\mathcal{T}$	$\triangleleft$	# of datapoints	j	$\triangleleft$ index: $(j = 1, 2, \dots, \mathcal{T})$
m	$\triangleleft$	Dimensions of $\mathcal{S}_{++}^m$	n	$\triangleleft$ Dimensions of $\mathbb{R}^n$
$\mathbf{d}$	$\triangleleft$	Subscript for desired value	$\mathbf{q}$ or $\mathbf{q}$	$\triangleleft$ Quaternion-related variable
$\mathbf{R}$ or $\mathbf{R}$	$\triangleleft$	Rotation matrix-related variable	$\mathbf{R}_{++}$, $\mathbf{R}_{++}$ or $\mathbf{R}_{++}$	$\triangleleft$ SPD-related variable
$\mathbf{g}$	$\triangleleft$	Subscript for goal value	a_z , β_z , a_x , a_y , a_{yx} , a_{qg}	$\triangleleft$ Positive gains
τ	$\triangleleft$	Time modulation parameter	c_i , h_i	$\triangleleft$ Centers and widths of Gaussians
T	$\triangleleft$	Time duration	t	$\triangleleft$ Continuous time
λ	$\triangleleft$	Forgetting factor	r	$\triangleleft$ Amplitude modulation parameter
x	$\triangleleft$	Phase variable	y , $\dot{y}$	$\triangleleft$ Trajectory data and its 1st derivative
s	$\triangleleft$	Sigmoidal decay phase	z , $\dot{z}$	$\triangleleft$ Scaled velocity and acceleration
p	$\triangleleft$	Piece-wise linear phase	$\mathbf{g}_q$, $\mathbf{g}_{++}$	$\triangleleft$ Attractor point (goal) in different spaces
ω	$\triangleleft$	Angular velocity	$\hat{\mathbf{g}}$, $\hat{\mathbf{g}}_q$ and $\tilde{\mathbf{g}}$, $\tilde{\mathbf{g}}_q$	$\triangleleft$ Moving target and delayed goal function in different spaces
$\mathbf{Q}$, $\dot{\mathbf{Q}}$	$\triangleleft$	Joint position, its 1st time-derivative	$\mathbf{g}_v$	$\triangleleft$ Intermediate attractor (via-goal)
$\mathbf{q}$, $\dot{\mathbf{q}}$	$\triangleleft$	Unit quaternion, its 1st time-derivative	$\mathbf{R}$, $\dot{\mathbf{R}}$	$\triangleleft$ Rotation matrix, its 1st time-derivative
$\mathbf{f}$, $\mathbf{f}_q$, $\mathbf{f}_R$, $\mathbf{f}_q$, $\mathcal{F}_{++}$	$\triangleleft$	Forcing term for different spaces	$\mathbf{w}_i$	$\triangleleft$ Adjustable weights
Ψ_i	$\triangleleft$	Basis functions	θ and ϑ	$\triangleleft$ An angle and learnable parameters
$\mathcal{S}_{++}^m$	$\triangleleft$	$(m \times m)$ SPD manifold	$\text{Sym } \mathcal{M}$	$\triangleleft$ $(m \times m)$ symmetric matrix space

M	$\triangleleft$	A Riemannian manifold	X	$\triangleleft$ An arbitrary SPD matrix
$T_{\lambda} M$	$\triangleleft$	A tangent space of M at an arbitrary point λ	M	$\triangleleft$ The mean of $\{X_{t=1}^{\Sigma}\}$
$\log_{\lambda}(\mathbf{q})$	$\triangleleft$	$M \xrightarrow{\log} T_{\lambda} M$, maps an arbitrary point $\mathbf{q} \in \mathcal{M}$ into $\mathbf{q} \in \mathcal{T}_{\lambda} \mathcal{M}$	$\exp_{\lambda}(\mathbf{q})$	$\triangleleft T_{\lambda} M \xrightarrow{\exp} M$, maps $\mathbf{q} \in \mathcal{T}_{\lambda} \mathcal{M}$ into $\mathbf{q} \in \mathcal{M}$
$\text{vec}(\cdot)$	$\triangleleft$	A function transforms $\text{Sym}(\mathbf{m})$ into $\mathbb{R}^n$ using Mandel's notation.	$\text{mat}(\cdot)$	$\triangleleft$ A function transforms $\mathbb{R}^n$ into $\text{Sym}(\mathbf{m})$ using Mandel's notation.
$\mathbf{k}, \mathbf{K}, \mathbf{K}(\mathbf{p}), \mathbf{K}(\mathbf{o})$	$\triangleleft$	Different forms of stiffness gains	$\mathbf{D}, \mathbf{D}^V, \mathbf{D}^W$	$\triangleleft$ Different forms of damping gains
$\mathbf{p}$ and $\mathbf{I}$	$\triangleleft$	Mass and inertia matrices	$\mathbf{F}, \mathbf{f}^e, \mathbf{r}^e$	$\triangleleft$ Forces and external forces and torques
DMP	Dynamic Movement Primitive	IL	Imitation Learning	
CMP	Compliant Movement Primitive	UAV	Unmanned Aerial Vehicle	
RL	Reinforcement Learning	SPD	Symmetric Positive Definite	
DoF	Degree of Freedom	RBF	Radial Basis Function	
LWR	Locally Weighted Regression	GMM	Gaussian Mixture Model	
GMR	Gaussian Mixture Regression	GP	Gaussian Process	
NN	Neural Network	VMP	Via-points Movement Primitive	
ProMP	Probabilistic Movement Primitives	LfD	Learning from Demonstration	
GPR	Gaussian Process Regression	MoMP	Mixture of Motor Primitives	
EMG	Electromyography	ILC	Iterative Learning Control	
VIC	Variable Impedance Control	VILC	Variable Impedance Learning Control	

PI $\mathbb{V}^{\{2\}}$	Policy Improvement With Path Integrals	CM $\mathbb{V}^{\{\dot{M}\}}$ S	Covariance Matrix Adaptation-Evolution Strategies	
CC-DMP	Coordinate Change-DMPs	RBF-NN	Radial Basis Function-Neural Network	
AL-DMP	Arc-Length-DMP	HRL	Hierarchical RL	
AEDMP	AutoEncoded DMP	CNN	Convolutional Neural Network	
GPDMP	Global Parametric Dynamic Movement Primitive	PoWER	Policy Learning by Weighting Exploration with the Returns	

threshold). This formulation, called DMP+, produces more accurate trajectories than the original DMP. Moreover, a learned trajectory can be modified by updating only a subset of the weights. However, DMP+ has double the parameters of the original DMP, and the truncated kernel introduces a discontinuity at the truncation point. To remedy this issue, Ginesi et al. (2021b) exploited the use of Molifier-like and Wendland basis functions with promising results. Rouse and Daltorio (2021) substituted the underlying kernels of stable

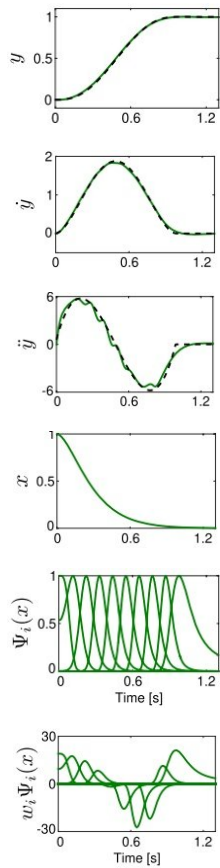


Figure 2. A classical DMP is used to generate a discrete motion connecting $\mathbb{V}(x_0 = 0)$ and $\mathbb{V}(g = 1)$ (green line in the top left panel). The training data (black dashed lines) are obtained from a minimum jerk trajectory connecting $\mathbb{V}(x_0)$ and $\mathbb{V}(g)$ in $\mathbb{V}(T = 1)$ s and used to learn the weights $\mathbb{V}(w_i)$ of 10 Gaussian basis functions equally distributed in time. The results of the parameter learning procedure are shown in

the bottom right panel. The exponentially decaying phase variable is used as shown in the middle right panel. Results are obtained with the open-source implementation available at <https://gitlab.com/dmp-codes-collection/dmp-classical>.

attractor points or limit cycles in DMPs with the underlying kernels of saddle points, known as Structurally Homogeneous Centers (SHCs). They demonstrated that the application of SHCs in the DMP framework was straightforward and yielded comparable results to the original approach in terms of ease of use and similarity of outcomes. Other work focused on using multiple demonstrations to increase the generalization power of the learned primitive. To learn a suitable forcing term from multiple demonstrations, some authors used Gaussian Mixture Model (GMM) (Yin and Chen, 2014; Pervez et al., 2017a) and Gaussian Mixture Regression (GMR) (Cohn et al., 1996), while others adopted Gaussian Process (GP) (Fanger et al., 2016; Rasmussen and Williams, 2006; Umlauf et al., 2017), or exploited a deep Neural Network (NN) (Pervez et al., 2017b; Pahić et al., 2020) developed originally in LeCun et al. (2015).

2.1.1.2. Phase stopping and goal switching. The phase variable x in (3) provides the ability to manipulate time during the execution of DMP equations. Moreover, DMP provides the ability to slow down or even stop the execution through the phase-stopping mechanism (Ijspeert et al., 2002c)

$$\dot{\tau}x = -\frac{a_x x}{1 + a_{yx} \tilde{y} - y}$$

where $\tilde{y}$ is the measured state and y is the one generated from the DMP. Alternatively, Anand et al. (2021) have shown that it is possible to modulate the time scaling factor τ as follows

$$\dot{\tau} = -k_{\tau}(\tau - u_{\tau})$$

where control input u_{τ} is designed such that the decay rate of the canonical system in (3) is slowed down if the desired execution time is increased or sped up if the desired execution time is reduced.

DMPs also provide an elegant way to adapt the trajectory generation in real-time through goal-switching mechanisms (Ijspeert et al., 2013)

$$\dot{\tau}g = a_g(g_0 - g)$$

where g_0 is the original goal and g is the new one to reach.

DMPs in their standard formulation are not suitable for direct encoding of skills with specific geometry constraints, such as orientation profiles (represented in either unit quaternions or rotation matrices), stiffness/damping, and manipulability profiles (encapsulated in full SPD matrices). For instance, direct integration of unit quaternions does not ensure the unity of the quaternion norm. Any representation of orientation that does not contain singularities is non-minimal, which means that additional constraints need to be taken into account during integration.

2.1.1.3. Alternative phase variables. Equation (3) describes an exponential decaying phase variable that has been widely used in the DMP literature. The main drawback of the exponential decaying phase is that it rapidly drops to very small values toward the end of the motion. This “forces” the learning algorithm to exploit relatively high weights w_i to accurately reproduce the last part of the demonstration (Samant et al., 2016). As an example, in Figure 3 the exponential decaying phase (brown dot-dashed line) is very small already after 0.6s, while the expected time duration of the motion is $T = 1s$.

To overcome this limitation, Kulvicius et al. (2011) propose the sigmoidal decay phase s (green solid line in Figure 3), obtained by integrating

$$\dot{s} = \frac{a_s e^{\frac{s}{\Delta t}} (\tau - T)}{1 + e^{\frac{s}{\Delta t}} (\tau - T)^2}$$

where α defines the steepness of s centered at time T and Δt is the sampling time. As shown in Figure 3, $s = 1$ for $t < T - \Delta t$, where the time Δt depends on the steepness α , and then it decays to $s = 0$.

The sigmoidal decay in Figure 3 has a tail effect since it vanishes after $(T + \delta_s)$, where (δ_s) depends on the tunable parameter (α_s) . The piece-wise linear phase (p) (blue-dashed line in Figure 3), proposed by Samant et al. (2016), linearly decays from 1 to 0 in exactly T_s and then remains constant. (p) is obtained by integrating

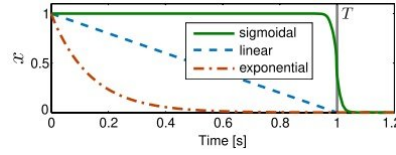


Figure 3. Possible phase variables used in different discrete DM formulations. All the different possibilities ensure that $(x, s, p \rightarrow 0)$ for $(t \rightarrow +\infty)$ (for $t > T$ in practice).

$$\dot{p} = \begin{cases} \frac{1}{T} & \text{if } 0 \leq t \leq T \\ 0 & \text{otherwise} \end{cases}$$

where $(p(0) = 1)$ and T is the time duration of the motion.

2.1.2. Orientation DMP. The classical DMP formulation described in Section 2.1.1 applies to single DoF motions. Multidimensional motions are generated independently and synchronized with a common phase. In other words, equations (1) and (2) are repeated for each DoF while the phase variable in (3) is shared. This works when the evolution of different DoFs is independent, like for joint space or Cartesian position trajectories. Unlike Cartesian position, the elements of orientation representations like unit quaternion or rotation matrix are constrained. In this section, we present approaches that extend the classical DMP formulation to represent Cartesian orientations.

2.1.2.1. Quaternion DMP. Unit quaternion $(\mathbf{q} = \mathbf{v} + \mathbf{u} \in \mathcal{S}^3)$ provides a representation of the orientation of the robot's end-effector (Chiaverini and Siciliano, 1999). $(\mathcal{S}^3)$ is a unit sphere in $(\mathbb{R}^4)$, $(\mathbf{v} \in \mathbb{R})$, and $(\mathbf{u} \in \mathbb{R}^3)$. Abu-Dakka et al. (2015a) rewrote DMP equations (1) and (2) for direct unit quaternion encoding as follows

$$\begin{aligned} \dot{\mathbf{q}} &= \mathbf{a}_z \mathbf{q} \mathbf{a}_z^* \\ \mathbf{g} &= \mathbf{q} \mathbf{g}_q \mathbf{q}^* \end{aligned}$$

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \mathbf{v} \mathbf{q}^*$$

where $(\mathbf{g}_q \in \mathcal{S}^3)$ denotes the goal orientation, the quaternion conjugation is defined as $(\overline{\mathbf{q}} = \overline{\mathbf{v}} + \overline{\mathbf{u}} = \mathbf{v} - \mathbf{u})$, and $(*)$ denotes the quaternion product

$$\begin{aligned} \mathbf{q}_1 * \mathbf{q}_2 &= (\mathbf{v}_1 + \mathbf{u}_1) * (\mathbf{v}_2 + \mathbf{u}_2) = (\mathbf{v}_1 \mathbf{v}_2 - \mathbf{u}_1^{\top} \mathbf{u}_2) + \\ &+ (\mathbf{v}_1 \mathbf{u}_2 + \mathbf{v}_2 \mathbf{u}_1 + \mathbf{u}_1^{\top} \mathbf{u}_2 \mathbf{v}_1) \mathbf{v}_2 \end{aligned}$$

$(\mathbf{v} \in \mathbb{R})$ is the scaled angular velocity (ω) and treated as unit quaternion with zero scalar $(v = 0)$ in (17). The function $(\text{Log}^q : \mathcal{S}^3 \mapsto \mathbb{R}^3)$ is given as follows

$$\begin{aligned} \text{Log}^q(\mathbf{q}) &= \begin{cases} \arccos(\mathbf{v}) \frac{\mathbf{u}}{\|\mathbf{u}\|} \\ \left[\frac{\mathbf{u}}{\|\mathbf{u}\|} \right] \times \mathbf{u} \end{cases} \end{aligned}$$

where $(\|\cdot\|)$ denotes (ℓ_2) norm. In practice, to avoid numerical instabilities, $(\|\mathbf{u}\|)$ is compared with a small (ϵ) before normalizing $(\mathbf{u})$.

An early attempt to encode unit quaternion profiles using DMP was presented by Pastor et al. (2011). Unlike Abu-Dakka et al.'s formulation, Pastor et al.'s formulation does not take into account the geometry of $\mathcal{SO}(3)$ as they just used the vector part of the quaternion product $(\mathbf{g}_q^* \overline{\mathbf{q}})$ in (16) instead of $(2 \log^q(\mathbf{g}_q^* \overline{\mathbf{q}}))$ which defines the angular velocity ω that rotates quaternion $\mathbf{q}$ into $\mathbf{g}_q$ within a unit sampling time.

Equation (17) can be integrated as follows

$$\mathbf{q}(t+\delta t) = \exp\left\{\mathbf{q} \left(\frac{\delta t}{2} \frac{\boldsymbol{\eta}(t)}{\tau}\right)^* \mathbf{q}(t)\right\}$$

$$[\delta t > 0]$$

$$\{\mathbf{E} \times \mathbf{p}\}^{\mathbf{q}}(\cdot): \mathbb{R}^3 \mapsto \mathcal{S}^3$$

$$\begin{aligned} \{\mathbf{Exp}\}^{\mathbf{q}}(\omega) = & \left\{ \begin{aligned} & \cos(\|\omega\|) + \sin(\|\omega\|) \frac{\omega}{\|\omega\|} \\ & , \|\omega\| \neq 0 \\ & , 1 + \left[0.00\right]^{\text{top}} \text{ \& otherwise.} \end{aligned} \right. \end{aligned}$$

where $\|\omega\|$ is compared with a small ϵ before normalizing ω to avoid numerical instabilities.

Both mappings become one-to-one, continuously differentiable, and inverse to each other if the input domain of the mapping $(\log^q(\cdot))$ is restricted to $\mathcal{S}^3$ except for $(-1 + [0 \ 0 \ 0]^T)$, while the input domain of the mapping $(\exp^q(\omega))$ should fulfill the constraint $\|\omega\| < \pi$ (Abu-Dakka et al., 2015a). An exemplar unit quaternion DMP is shown in Figure 4.

Phase-stopping (11) can be rewritten as follows

$$\tau \dot{\mathbf{x}} = \frac{\mathbf{a}_x}{1 + \mathbf{a}_x \mathbf{q}^* \mathbf{d}(\tilde{\mathbf{q}}, \mathbf{q})}$$

where

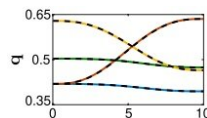
$$\begin{aligned} \mathbf{d}(\tilde{\mathbf{q}}, \mathbf{q}) = & \begin{cases} 2\pi, & 2\pi \leq \mathbf{Log}^{\mathbf{q}}(\tilde{\mathbf{q}}_1^* \overline{\mathbf{q}}_2) \\ \text{otherwise} \end{cases} \\ \mathbf{d}(\tilde{\mathbf{q}}, \mathbf{q}) = & \begin{cases} 2\pi, & 2\pi \leq \mathbf{Log}^{\mathbf{q}}(\tilde{\mathbf{q}}_1^* \overline{\mathbf{q}}_2) \\ \text{otherwise} \end{cases} \end{aligned}$$

Ude et al. (2014) extended DMP quaternion-based formulation by rewriting (13) to include goal switching mechanism

$$\tau \dot{\mathbf{g}}_q = \mathbf{a}_g \mathbf{L}_g^{\mathbf{q}} \left(\mathbf{g}_{q, \text{new}} - \overline{\mathbf{g}}_q \right)^* \mathbf{g}_q$$

so that $\mathbf{g}_q$ is continuously changing onto $\mathbf{g}_{q, \text{new}}$ in real-time. Equation (22) should be integrated using (20) along with (16) and (17).

As shown by Saveriano et al. (2019) using Lyapunov arguments, both the quaternion DMP formulations in Pastor et al. (2011) and in Abu-Dakka et al. (2015a) and Ude et al. (2014) asymptotically converge to the target quaternion $\mathbf{g}_q$ with zero velocity.



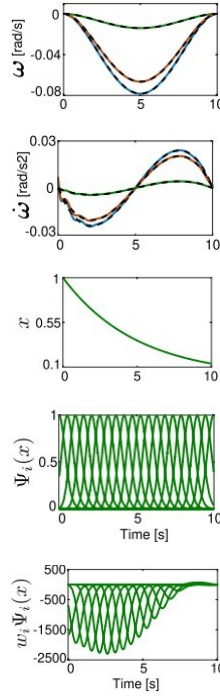


Figure 4. A unit quaternion DMP is used to generate a discrete motion connecting $\mathbf{q}_1$ and $\mathbf{q}_g$. The training data (black-dashed lines) are obtained from a minimum jerk trajectory connecting $\mathbf{q}_1$ and $\mathbf{q}_g$ in $(T = 10s)$ and used to learn the weights $\{w_i\}$ of 20 Gaussian basis functions equally distributed in time. The results of the parameter learning procedure are shown in the bottom right panel.

The exponentially decaying phase variable is used as shown in the middle right panel. Results are obtained with the open-source implementation available at <https://gitlab.com/dmp-codes-collection>.

2.1.2.2. Rotation matrix DMP. In their work on orientation DMPs, Ude et al. (2014) extended DMP formulation in order to encode orientation trajectories represented in the form of rotation matrices $\mathbf{R}(t) \in \mathcal{SO}(3)$. Therefore, they rewrote (1) and (2) in the following form

$$\begin{bmatrix} \tau \dot{\mathbf{pmb}}\{\eta\} \\ \mathbf{a}_z \mathbf{big}(\beta_z) \mathbf{L o g}^{\mathbf{R}} \mathbf{big}(\mathbf{R}_g) \mathbf{mathbf{R}}^{\mathbf{top}} \mathbf{big}(\mathbf{pmb}\{\eta\} \mathbf{big}) + \mathbf{mathbf{f}}_{\mathbf{R}}(x) \end{bmatrix}$$

$$\begin{bmatrix} \tau \dot{\mathbf{R}} \\ \mathbf{[\eta]}_{\mathbf{times}} \mathbf{R} \end{bmatrix}$$

where $\mathbf{R}_g$ represents the goal orientation. $[\eta]_{\mathbf{times}}$ is a skew symmetric matrix, such as $[\eta]_{\mathbf{times}}^{\mathbf{top}} = -[\eta]_{\mathbf{times}}$. The relation between the angular velocity and the 1st-time-derivative of the rotation matrix is given by

$$\begin{bmatrix} \omega_z \\ \omega_y \\ \omega_x \end{bmatrix} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ \omega_y & \omega_x & 0 \end{bmatrix} \dot{\mathbf{R}} \mathbf{R}^{\mathbf{top}}$$

The function $\log^{\mathbf{R}}: \mathcal{SO}(3) \mapsto \mathbb{R}^3$ is given as follows

$$\mathbf{Log}^{\mathbf{R}}(\mathbf{R}) = \begin{cases} [0, 0, 0]^{\mathbf{top}}, & \mathbf{R} = \mathbf{I} \\ \omega = \theta \mathbf{n}, & \text{otherwise} \end{cases}$$

$$\theta = \arccos\left(\frac{\text{trace}(\mathbf{R}) - 1}{2}\right), \mathbf{n} = \frac{1}{2\sin(\theta)} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

The generated rotation matrices can be obtained by integrating (24) as follows

$$\mathbf{R}(t+\Delta t) = \mathbf{E} \times \mathbf{p}^{\mathbf{R}} \Big(\Delta t \frac{[\boldsymbol{\eta}]_{\times}}{\tau} \Big) \mathbf{R}(t)$$

The function $\exp^{\mathbf{R}}(\cdot): \mathbb{R}^3 \mapsto \mathcal{SO}(3)$ is given as follows

$$\begin{aligned} \mathbf{Exp}^{\mathbf{R}}([\boldsymbol{\omega}]_{\times}) &= \mathbf{I} + \sin(\theta) \frac{[\boldsymbol{\omega}]_{\times}}{\|\boldsymbol{\omega}\|} + (1 - \cos(\theta)) \frac{[\boldsymbol{\omega}]_{\times}^2}{\|\boldsymbol{\omega}\|^2} \end{aligned}$$

where $\theta(t) = t\|\boldsymbol{\omega}\|$ express the rotation angle within time t . An exemplar rotation matrix DMP is shown in Figure 5.

2.1.3. SPD matrices. Abu-Dakka and Kyrki (2020) generalized DMP formulation in order to encode robotic manipulation data profiles encapsulated in the form of SPD matrices. The importance of SPD matrices comes from the fact that many robotics data are encapsulated in such matrices, for example, full stiffness matrices, manipulability ellipsoids, and inertia matrices among others. By defining $\mathbf{X} \in \mathcal{S}_m^{++}$ as an arbitrary SPD matrix and $\mathbf{X}_i = [t_j, \mathbf{X}_{j-1}]_{j=1}^n$ as the set of SPD matrices in one demonstration, where $\mathcal{S}_m^{++}$ defines the set of $(m \times m)$ SPD matrices. Afterward, we can rewrite (1) and (2) as follows

$$\begin{aligned} \tau \dot{\boldsymbol{\sigma}} &= \mathbf{a}_z \Big(\beta_z \mathbf{v} \mathbf{e} \mathbf{c} \Big) \mathbf{B}_{\mathbf{X}_j} \mathbf{m}_g \Big(\mathbf{X}_j \Big)^+ \Big(\mathbf{X}_j \mathbf{g} \Big) \Big(\mathbf{B} \Big)^{-1} \boldsymbol{\sigma} \Big) \quad + \quad \mathcal{F}_+(x), \end{aligned}$$

$$\tau \dot{\mathbf{x}} = \boldsymbol{\sigma}$$

where $\boldsymbol{\sigma} = \text{vec}(\boldsymbol{\Sigma})$ is the Mandel representation of the symmetric matrix $\boldsymbol{\Sigma}$ and $\dot{\boldsymbol{\Sigma}}$ is the time derivative of $\mathbf{X}_i$ so that $\dot{\mathbf{X}}_i = (\log_{\mathbf{X}_{j-1}}(\mathbf{X}_j)) / \Delta t$. The function $\log_{\mathbf{X}_{j-1}}(\mathbf{X}_j) : \mathcal{M} \mapsto \mathcal{T}_{\mathbf{X}_{j-1}} \mathcal{M}$ maps a point $\mathbf{X}_j$ in the manifold $\mathcal{M}$ to a point in the tangent space $\Delta \in \mathcal{T}_{\mathbf{X}_{j-1}} \mathcal{M}$. $\text{vec}(\cdot)$ is a function that transforms a symmetric matrix into a vector using Mandel's notation, for example, a vectorization of a (2×2) symmetric matrix is as follows

$$\text{vec} \begin{bmatrix} a & b \\ b & d \end{bmatrix} = \begin{bmatrix} a \\ d \end{bmatrix} \sqrt{2} b$$

$\mathbf{x}_i$ is the vectorization of $\mathbf{X}_i$. $\mathbf{X}_g \in \mathcal{S}_m^{++}$ represents the goal of SPD matrix. $\mathbf{B}_{\mathbf{X}_j} \mathbf{m}_g(\mathbf{X}_j) = \log_{\mathbf{X}_{j-1}}(\mathbf{X}_j) \mathbf{X}_g$ is the vectorization of the transported symmetric matrix $\log_{\mathbf{X}_{j-1}}(\mathbf{X}_j) \mathbf{X}_g$ over the geodesic from $\mathbf{X}_{j-1}$ to $\mathbf{X}_j$. Then we integrate (30) as follows

$$\widehat{\mathbf{X}}(t+\Delta t) = \mathbf{E} \times \mathbf{p}^{\mathbf{X}} \Big(\mathbf{X}(t)^+ \left(\frac{\mathbf{B}_{\mathbf{X}_1} \mathbf{m}_g(\mathbf{X}_1)}{\tau \Delta t} \right) \Big)$$

where the function $\text{mat}(\cdot)$ is the inverse of $\text{vec}(\cdot)$ and denotes to the matricization using Mandel's notation. $\widehat{\mathbf{X}} \in \mathcal{S}_m^{++}$ represents the new SPD-matrices-based robot skills. The function $\text{Exp}_{\mathbf{X}_{j-1}}^{\mathbf{X}} \left(\Delta \right) : \mathcal{T}_{\mathbf{X}_{j-1}} \mathcal{M} \mapsto \mathcal{M}$ maps a point $\Delta \in \mathcal{T}_{\mathbf{X}_{j-1}} \mathcal{M}$ to a point $\mathbf{X}_j \in \mathcal{M}$, so that it lies on the

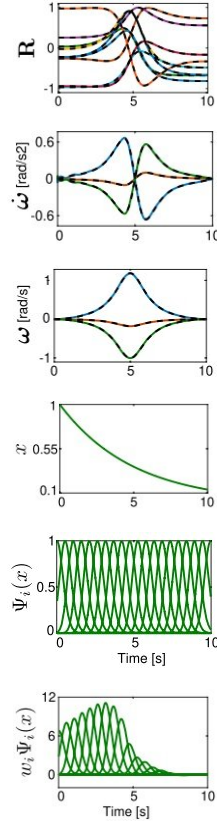


Figure 5. A rotation matrix DMP is used to generate a discrete motion connecting (R_1) and (R_g) . The training data (black-dashed lines) are obtained from a minimum jerk trajectory connecting (R_1) and (R_g) in $(T = 10s)$ and used to learn the weights (w_i) of 20 Gaussian basis functions equally distributed in time. The results of the parameter learning procedure are shown in the bottom right panel.

The exponentially decaying phase variable is used as shown in the middle right panel.

geodesic starting from $(\mathbf{X}_{j-1})$ in $(\mathcal{S}_{++}^m)$ in the direction of (Δ) . An exemplar SPD DMP is shown in Figure 6.

Moreover, Abu-Dakka and Kyrki (2020) rewrote (13) for smooth goal adaptation in case of sudden goal switching as follows

$$\begin{bmatrix} \tau \dot{\mathbf{g}}_+ = \mathbf{a}_g \log_{\mathbf{g}_-}(\mathbf{g}_+) \\ \mathbf{w}_+^+ \left(\mathbf{g}_+ \right) \end{bmatrix}$$

so $\mathbf{g}$ now is updated continually.

2.2. Periodic DMP

The periodic DMP (sometimes called rhythmic DMP) is used when the encoded motion follows a rhythmic pattern.

2.2.1. Classical DMP. The classical periodic (or rhythmic) DMPs were first introduced by Ijspeert et al. (2002b), where they redefined the second-order differential equation system described in (1) and (2) as follows

$$\ddot{z} = \Omega(\alpha(\beta(-y)z) + f(\phi)),$$

$$\dot{y} = \Omega z,$$

$$\tau \dot{\phi} = 1$$

where ω is the frequency and y is the desired periodic trajectory that we want to encode with a DMP. The main difference between periodic DMPs and point-to-point DMPs is that the time constant related to trajectory duration is replaced by the frequency of trajectory execution (refer to Ijspeert et al., 2013 and 2002b for details). In

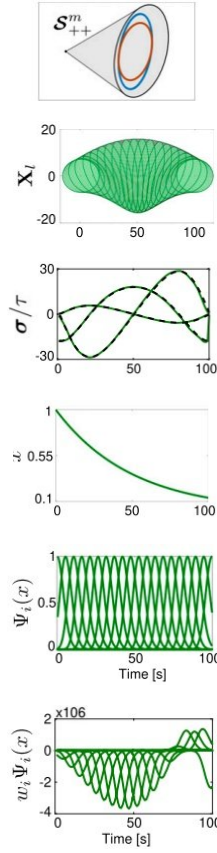


Figure 6. An SPD DMP is used to generate a discrete motion connecting $X_{1\backslash}$ and $X_{g\backslash}$. The training data (black-dashed lines) are obtained from a minimum jerk trajectory connecting $X_{1\backslash}$ and $X_{g\backslash}$ in $(T = 100s\backslash)$ and used to learn the weights $(w_{i\backslash})$ of 20 Gaussian basis functions equally distributed in time. The cone in the upper left corner represents the manifold of SPD data and includes the geodesic of the SPD profile. The results of the parameter learning procedure are shown in the bottom right panel. The exponentially decaying phase variable is used as shown in the middle right panel. Results are obtained with the open-source implementation available at <https://gitlab.com/dmp-codes-collection>.

addition, the periodic DMPs must ensure that the initial phase ($\phi = 0$) and the final one ($\phi = 2\pi$) coincide in order to achieve smooth transition during the repetitions.

Similar to (4), $f(\phi)$ is defined with N Gaussian kernels according to the following equation

$$f(\phi) = \frac{\sum_{i=1}^N \Psi_i(\phi) w_i}{\sum_{i=1}^N \Psi_i(\phi)}$$

$$\Psi_i(\phi) = \exp(h(\cos(\phi - c_i) - 1))$$

where the weights are uniformly distributed along the phase space and r is used to modulate the amplitude of the periodic signal (Gams et al., 2009; Ijspeert et al., 2002b) (if not used, it can be set to $r = 1$ [Peternel et al. 2016]).

Similar to discrete DMPs, LWR (Schaal and Atkeson, 1998) can be used to update the weight to learn a desired trajectory. In a standard periodic DMP setting (Gams et al., 2009; Ijspeert et al., 2002b), the desired shape f_d is approximated by solving

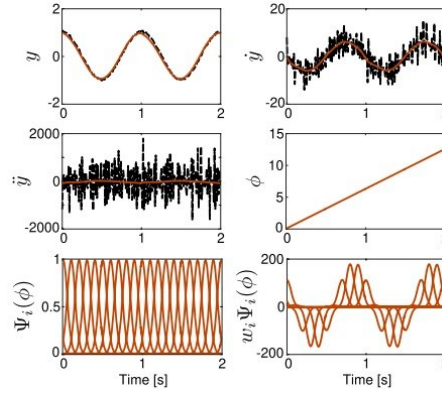


Figure 7. A classical DMP is used to reproduce a rhythmic motion (brown solid line in the top left panel). The desired trajectory is obtained by adding Gaussian noise to $y_d = \cos(2\pi t)$ with $t \in [0, 2]$ s and computing the numerical derivatives with $(\Delta t = 0.01 \text{ s})$ (black-dashed lines). The forcing term is obtained as the weighted summation of 20 Gaussian bases equally distributed in time (bottom left panel). The results of the parameter learning procedure are shown in the bottom right panel. Results are obtained with the open-source implementation available at <https://gitlab.com/dmp-codes-collection/dmp-classical>.

$$\begin{aligned} & \left[\begin{array}{l} \mathbf{r} \\ \frac{\sum_{i=1}^N \mathbf{w}_i \Psi_i(\phi)}{\sum_{i=1}^N \Psi_i(\phi)} \end{array} \right] \& \& \\ & \left\{ \mathbf{A}_r^{-1} \cdot \left(\mathbf{\Omega}^{-1} \cdot \boldsymbol{\omega} \right) - \left(\alpha_z \cdot \beta_z \right) \operatorname{L o} \right. \\ & \left. \mathbf{g}^q \cdot \overline{\mathbf{g}_q} \right\} \cdot \boldsymbol{\omega} \end{aligned}$$

where $(\mathbf{A}_r)$ is the (3×3) diagonal matrix of amplitude modulators.

The integration of (44) is done similarly as in (19), that is,

$$\mathbf{q}(t + \Delta t) = \exp\left\{ \frac{\Delta t}{2} \mathbf{\Omega} \boldsymbol{\eta}(t) \right\}$$

2.3. Formulation summary

A summary of the existence DMP formulations mentioned in the earlier sections is shown in Table 3. The table shows the variations of the formulation in its standard shape based on the space that they are applied to. However, the modifications of this standard shape (e.g., adding a coupling term) are discussed in the next section as an extension of the DMP formulations.

3. DMP extensions

3.1. Generalization

A desirable property of motion primitives is the ability to generalize to unforeseen situations. In this section, we present approaches that allow to adapt DMP motion trajectories to novel executive contexts.

Table 3. Summary of DMP basic formulations.

Type of movement	Space	System of equations	Reference	Short description
Discrete	$\mathbb{R}$	$\dot{z} = \alpha_z(\beta_z(g - y) - z) + f(x)$	Equations (1)-(3) (Ijspeert et al., 2002c)	A single DoF, discrete motion trajectory is encoded into a linear, second-order dynamical system with an additive, nonlinear forcing term. Convergence to the desired goal g is ensured by a vanishing phase variable x .
	$\mathcal{S}^3$	$\dot{\eta} = \alpha_z(\beta_z 2 \log^d \{ \mathbf{g}_q * \overline{\mathbf{q}} \} - \eta) + f_q(x)$	Equations (16) and (17) (Abu-Dakka et al., 2015a)	A quaternion-based orientation trajectory (3 DoFs) is encoded into a second-order dynamical system with an additive, nonlinear forcing term. The error definition complies with the geometry of the unit quaternion space.
	$\mathcal{SO}(3)$	$\dot{\eta} = \alpha_z(\beta_z \log^R \{ \mathbf{g}_R * \mathbf{R}^{\top} \} - \eta) + f_R(x)$	Equations (23) and (24) (Ude et al., 2014)	A rotation matrix-based orientation trajectory (3 DoFs) is encoded into a second-order dynamical system with an additive, nonlinear forcing term. The error definition complies with the geometry of the rotation matrices space.
	$\mathcal{S}_+^m$	$\dot{\sigma} = \alpha_z(\beta_z \mathbf{E}_X \mapsto \mathbf{X}_i \log \{ \mathbf{X}_i * \mathbf{g} \} - \sigma) + \mathcal{F}(x)$	Equations (29) and (30) (Abu-Dakka and Kyrki, 2020)	An SPD matrices trajectory, $\mathcal{M}^m$ ($(m+1)/2$ DoFs), is encoded into a second-order dynamical system with an additive, nonlinear forcing term. The error definition complies with the geometry of the SPD matrices space.
Periodic	$\mathbb{R}$	$\dot{z} = \Omega(\alpha(\beta(-y) - z) + f(\phi))$	Equations (34)-(36) (Ijspeert et al., 2002b)	A single DoF, periodic motion trajectory is encoded into a linear, second-order dynamical system with an additive, nonlinear forcing term. The resulting system generates a stable limit cycle.
	$\mathcal{S}^3$	$\dot{\eta} = \Omega(\alpha(\beta_z 2 \log^d \{ \mathbf{g}_q * \overline{\mathbf{q}} \} - \eta) + \mathbf{f}_q(\phi))$	Equations (43) and (44) (Abu-Dakka et al., 2021)	A quaternion-based orientation trajectory (3 DoFs) is encoded into a second-order dynamical system with an additive, nonlinear forcing term. The error definition complies with the geometry of the unit quaternion space.

3.1.1. Start, goal, and scaling. Classical DMPs are time-invariant, meaning that time scaling τ with $\tau > 0$ generates topologically equivalent trajectories (Ijspeert et al., 2013). Using a simple modification of the transformation system, namely, substituting (1) with

$$\dot{z} = \alpha_z(\beta_z(g - y) - z) + (g - y_0)f(x)$$

where y_0 is the initial value of y , Ijspeert et al. (2013) show that DMPs are also scale-invariant, meaning that the scaling of the movement amplitude $\sigma(g - y_0)$ with $\sigma > 0$ generates topologically equivalent trajectories. The purpose of the green color used in (48) is to highlight differences (w.r.t.) (1). Kulvicius et al. (2011) use a similar strategy and scale the forcing term by the ratio between the start- and end-points of the demonstrated and desired trajectories. Apart from generating scaled—in time and space—versions of the demonstrated motion trajectory, classical DMPs also generalize to different initial/target states. However, the classical formulation—and its extension in (48)—may exhibit dangerous behaviors like over-amplification of the trajectory when reaching a different target and high accelerations when switching to a different target on-line (Ijspeert et al. 2013; Pastor et al. 2009). To alleviate the second issue, Ijspeert et al. (2013) replaced hard goal switches with the smooth switching law as in (13). However, the over-amplification issue still remains. Moreover, a DMP that uses (48) fails to learn motions with the same initial and target states (i.e., $(g = y_0)$, $(z_0 = 0 \rightarrow y(t) = y_0 = g \nabla t)$).

In order to remedy those issues, Pastor et al. (2009) proposed to modify the transformation system as follows

$$\dot{\tau}_z = a_z(\beta_z(g - y - (g - y_0)x + f(x) - z))$$

where the green color is used to highlight differences between (49) and (1). The most important change in this formulation is the term $((g - y_0)x)$ that has several benefits. It prevents high accelerations at the beginning of the motion $((g - y - (g - y_0)x = 0)$ for $(t = 0)$) or when the goal is close to the initial state. It allows to reproduce motions with the same initial and target states and prevents over-amplifications and trajectory mirroring effects $(\wedge^1)$ when changing the goal. Hoffmann et al. (2009) derived a multidimensional representation of (49) from the behavior of the spinal force fields in frogs.

The goal can also change over time and, in this case, the tracking performance of the DMP mostly depends on the gains (α_z) and (β_z) . As proposed by Koutras and Dougeri (2020b), the tracking performance can be improved by adapting the temporal scaling (τ) .

Dragan et al. (2015) showed that DMPs solve a trajectory optimization problem in order to minimize a particular Hilbert norm between the demonstration and the new trajectory subject to start and goal constraints. In this light, DMP adaptation capabilities to different start and goals can be improved by choosing (or learning) a proper Hilbert norm that reduces the deformation in the retrieved trajectory.

3.1.2. Via-points. A via-point can be defined as a point in the state space where the trajectory has to pass. Failing to pass a via-point may cause the robot to fail the task execution. Therefore, having a motion primitive representation with the capability of modulating the via-points is important in robotic scenarios. It is not surprising that researchers have extended the DMP formulation to consider intermediate vi-points in the trajectory generation process.

Ning et al. (2011, 2012) extend the classical DMP to satisfy position and velocity constraints at the beginning and at the end of a sample trajectory. Their approach to traverse via-points consists of creating a sample trajectory by combining locally linear trajectories connecting the viapoints. This sample trajectory is used to fit a DMP that is constrained to pass the viapoints.

Weitschat and Aschemann (2018) considered each viapoint as an intermediate goal (via-goal) (g_v) for $(v = 1, \dots, V)$ to reach. The last via-goal (g_V) corresponded to the target state of the DMP. In their formulation, they defined a variable goal as follows

$$g_{via}(x) = \sum_{\nu=1}^V \Psi_{\nu}(x) g_{\nu}$$

where $(\Psi_{\nu}(x))$ are the Gaussian basis functions centered at the time corresponding to the (ν) – th via-goal. The effectiveness of the approach is demonstrated in a task where the robot has to reach a different target while preventing possible self-collisions of the end-effector with the robot body. To this end, the authors placed the via-goals along the trajectory used to learn the DMP, forcing the generated trajectory to stay close to the demonstration while reaching the new target.

The problem of generalizing to via-point close (interpolation) and far (extrapolation) from the demonstration is faced by Zhou et al. (2019). Their approach, namely, Viapoints Movement Primitives (VMPs), combines the benefits of DMP and Probabilistic Movement Primitives (ProMPs) (Paraschos et al., 2013). Authors assumed that the motion trajectory is generated as follows

$$y_{vmp}(x) = e(x) + f_{vmp}(x)$$

where (x) is the phase variable defined as in (3) and the elementary trajectory $(e(x))$ can be defined as the linear attractor $(e(x) = (y_0 - g)x)$. The shape modulation term $(f_{vmp}(x))$ is defined as follows

$$f_{vmp}(x) = \sum_{i=1}^N w_i \Psi_i(x) + \epsilon_f$$

where the Gaussian kernels $(\Psi_i(x))$ are defined as in (5), (w_i) are learnable weights, and (ϵ_f) is the Gaussian noise. As detailed in Paraschos et al. (2013), learning the shape modulation term $(f_{vmp}(x))$ means Learning from Demonstrations (LfDs) the prior probability distribution of the weights (w_i) . Having separated the generated trajectory into two parts like in (51) allows to adopt different strategies to pass a via-point (y_v) at (x_v) , Zhou et al. (2019) proposed to modify the shape modulation term for interpolation

cases—when the vi-point is “close” to the demonstrations. In extrapolation cases, instead, the elementary trajectory $\mathbf{e}(x)$ is rewritten as the polygonal line connecting $\mathbf{y}_0$, $\mathbf{y}_v$, and $\mathbf{g}$. This approach easily generalizes to the case of multiple via-points. VMPs are experimentally compared with ProMPs, showing better performance especially in extrapolation cases.

3.1.3. Task parameters. Reaching a different goal, or passing through via-points, may not be enough to successfully execute a task in a different context. Approaches presented in this section adapt the DMP motion to new situations by adjusting the weights $\{w_i\}$ of the forcing term (4), which modifies the entire DMP trajectory.

Weitschat et al. (2013) considered that L demonstrations are given, each encoded in a different DMP. In order to generalize, for instance, to a new goal $\mathbf{g}_{\text{new}}$, they proposed to interpolate the weights of nearby DMPs, that is., DMPs that reached points around $\mathbf{g}_{\text{new}}$. In formulas

$$\mathbf{w}_{n \in \mathcal{O}} = \frac{\sum_{o \in \mathcal{O}} \mathbf{w}_o}{|\mathcal{O}|}$$

where $\mathcal{O}$ represents the indices of the nearby DMPs for which it holds that $d_o < d_{\max}$, d_o is the distance (or, more

generally, a cost) between $\mathbf{g}_{\text{new}}$ and $\mathbf{g}_o$, $d_{\max}$ is the maximum distance to consider 2 DMPs close, and $\mathbf{w}_{\text{new}} = [w_{1,\text{new}}, \dots, w_{N,\text{new}}]^{\top}$ and $\mathbf{w}_o = [w_{1,o}, \dots, w_{N,o}]^{\top}$ are the new weights and the weight of nearby DMPs, respectively.

The approach by Forte et al. (2011, 2012) also assumes that L demonstrations are given and that each demonstration is encoded in a different DMP. Further, the authors exploited GP (Rasmussen and Williams, 2006) to learn a mapping between the query points $\mathbf{q}_l$ for $l = 1, \dots, L$ (e.g., the goal of each DMP) and the DMP parameters $[\mathbf{w}_l, \mathbf{g}_l, \tau_l]$. Given the new query point $\mathbf{q}_{\text{new}}$, Gaussian Process Regression (GPR) (Rasmussen and Williams, 2006) is used to retrieve the new set of parameters $[\mathbf{w}_{\text{new}}, \mathbf{g}_{\text{new}}, \tau_{\text{new}}]$, that can be used to generate a DMP motion. This approach builds on previous work (Gams and Ude, 2009; Ude et al., 2010) where raw data from the L demonstrations are stored in memory and LWR is used to generate new DMP weights. Alizadeh et al. (2016) extend the approach in Ude et al. (2010) to retrieve the DMP weights even when the task parameters are partially observable. Finally, Zhou and Asfour (2017) extend the approach in Ude et al. (2010) to consider task-specific costs while learning the mapping between query points and DMP weights.

The aforementioned approaches follow a 2-steps procedure where first the shape parameters $\mathbf{w}$ are estimated given new task parameters and then execute the DMP. Matsubara et al. (2010, 2011) augmented the forcing term with a style parameter used to capture human variability across multiple demonstrations. Stulp et al. (2013) proposed a 1-step procedure where the DMP forcing term (4) is reformulated to explicitly depend on the task parameters. Their experiments show that a 1-step approach gives more freedom w.r.t. the used regression technique and increases the generalization performance. Along the same line, Pervez and Lee (2018) embedded task parameters directly in the forcing term. The authors proposed to use a mixture of Gaussians (Cohn et al., 1996) to learn the mapping between the task parameters (e.g., new goal and the height of an obstacle) and the forcing term. Given a new query task parameter, regression over the mixture of Gaussians is used to retrieve the forcing term parameters and generate the DMP motion. The approach is tested on a variety of tasks including sweeping and striking and additionally compared with the approaches presented by Forte et al. (2012), Stulp et al. (2013), and Ude et al. (2010) showing better performance, especially in extrapolation.

A Mixture of Motor Primitives (MoMP) is proposed in Mülling et al. (2010, 2013) and used to generalize table tennis skills like hitting and batting a ball. MoMP uses an augmented state that contains robot position and velocity as well as the meta-parameters of the table tennis task like the expected hitting position and velocity. The adapted motion is generated by the weighted summation of L DMPs and the responsibility of each DMP, representing the probability that a particular DMP is the correct one for the sensed augmented state, is also learned from data.

In high DoF systems, like humanoid robots, it is non-trivial to find a relationship between the task and the DMP parameters. This is especially true when the DMPs are used to encode joint space trajectories. Bitzer and Vijayakumar (2009) showed that such a relationship is easier to find in a latent (lower-dimensional) space obtained from training data. Therefore, they used dimensionality reduction techniques to find the latent space where to fit a DMP and show that the interpolation of DMP weights in the latent space results in better generalization performance.

3.2. Joining multiple DMPs

An important and desired feature of any motion primitive representation is the possibility to combine basic movements to obtain more complex behaviors (Schaal, 1999). We review here three prominent approaches developed to smoothly join a sequence of DMPs. In this tutorial, we name the approach by Pastor et al. (2009) as velocity threshold, that in Kober et al. (2010b) as target crossing, and that in Kulvicius et al. (2011, 2012) as basis functions overlay. Some of the presented approaches modify the DMP formulations in Sections 2.1.1 and 2.1.2. The main differences are highlighted with green text. The 3 approaches have been implemented in MATLAB for both position (Section 2.1.1) and orientation (Section 2.1.2) DMPs. The source code is included in our public repository (see Table 4). Results on synthetic data are shown in Figures 8–11.

3.2.1. Velocity threshold. A properly designed DMP reaches the desired target with zero velocity and acceleration, that is, once a DMP is fully executed the robot comes to a full stop. This also implies that the velocity “close” to the target is continuously decreasing. Using this property, Pastor et al. (2009) propose to combine successive DMPs by simply terminating the current DMP when the velocity is below a certain threshold and then starting the following primitive. When executing a single DMP, it is a common practice to initialize its velocity to zero—the robot is assumed to be still. In principle, this initialization can be used to sequence multiple DMPs (Lioutikov et al., 2016; Xu and Wang, 2004), but it may generate discontinuities if the robot does not fully stop in between two consecutive primitives. To prevent these discontinuities, Pastor et al. initialized the state of the current DMP with that of the previous one.

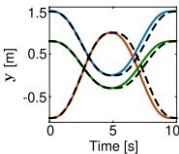
The velocity threshold approach is simple and effective since it directly applies to the DMP formulations in Sections 2.1.1 and 2.1.2. For instance, Saveriano et al. (2019) showed how to join multiple quaternion DMPs $\mathcal{Q}^2$ (see Section 2.1.2.1) with the velocity threshold approach.

Results in Figure 8 are obtained when the velocity threshold is applied to merge 2 DMPs separately trained to fit minimum jerk trajectories (black-dashed lines). Figures 8(a)–(e) show the position and Figures 8(f)–(j)

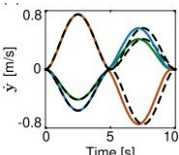
Table 4. Open-source implementations of DMP-based approaches that we have released to the community. The source code for each approach is available at <https://gitlab.com/dmp-codes-collection>.

Approach	Author	Language	Description
Discrete DMP	Fares J. Abu-Dakka	C++	An implementation for discrete DMP based on the work in Abu-Dakka et al. (2015a) and Ude et al. (2010, 2014).
Periodic DMP	Luka Peternel	Python	An implementation for periodic DMP based on the work in Peternel et al. (2016).
Unit quaternion	Fares J. Abu-Dakka	MATLAB and C++	An implementation for unit quaternion DMP and goal switching based on the work in Abu-Dakka et al. (2015a) and Ude et al. (2014).
SPD DMP	Fares J. Abu-Dakka	MATLAB	An implementation for SPD DMP and goal switching based on the work in Abu-Dakka and Kyrki (2020).
Joining DMPs	Matteo Saveriano	MATLAB	An implementation for joining multiple DMPs based on the work in Saveriano et al. (2019).
Coupling-force DMPs	Aljaz Kramberger	MATLAB	An implementation for discrete DMPs and force coupling terms based on the work in Kramberger et al. (2018).

(a)



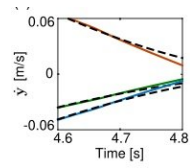
(b)



Position

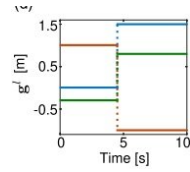
(c)

Linear Velocity



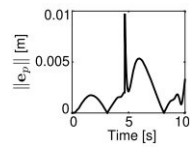
(d)

Linear Velocity



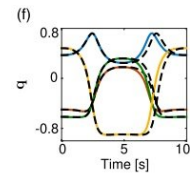
(e)

Goal Switch

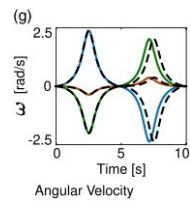


(g)

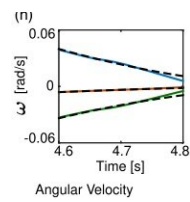
Position Error



(h)

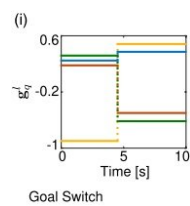


Angular Velocity

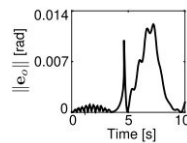


Angular Velocity

(j)



Goal Switch



Orientation Error

Figure 8. Results obtained by applying the zero velocity switch approach to join two DMPs trained on synthetic data. The training trajectory for the position and the orientation are shown as black-dashed lines in (a)–(b) and (f)–(g), respectively. Results are obtained with the open-source implementation available at <https://gitlab.com/dmp-codes-collection>.

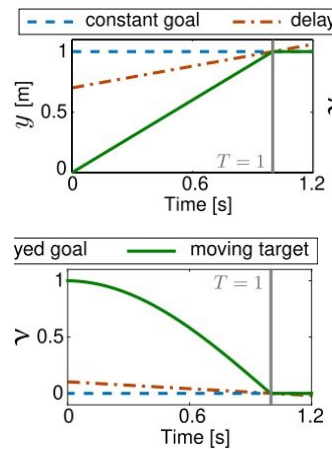
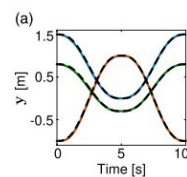
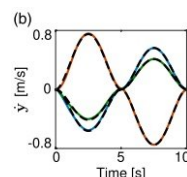


Figure 9. Constant goal, moving target, and delayed goal obtained with $\mathbf{y}(0)=0$ [m]; $\mathbf{g}=1$ [m]; $\hat{\mathbf{y}}=0.3$ [m/s] (left); and $\mathbf{q}(0)=1+[0,0,0]^{\top}$, $\mathbf{g}_-=0+[1,0,0]^{\top}$, and $\hat{\boldsymbol{\omega}}=[0.2,0.2,0.2]^{\top}$ [rad/s] (right). The sampling time is $\Delta t=0.01$ [s]. Only the scalar part [eqn7] of the quaternion is shown for a better visualization.

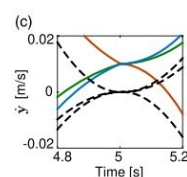
the orientation (unit quaternion) parts of the motion. The merged trajectory is generated by following the first DMP until the distance from the via-point is below 0.01 [m] and 0.01 [rad]. As shown in Figures 8(d) and (i), the switch occurs after about 4.7 [s]. Figures 8(e) and (j) show that the desired trajectory is accurately reproduced. More or less accurate trajectories can be obtained by tuning the distance from the via-point. However, the value of this



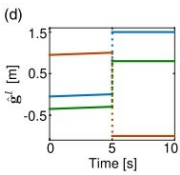
Position



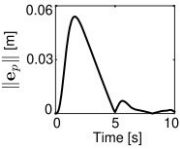
Linear Velocity



Linear Velocity



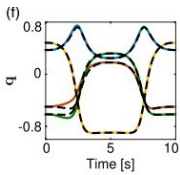
(f)



(e)

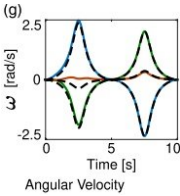
Goal Switch

Position Error

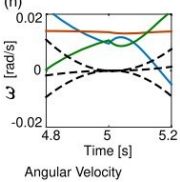


(g)

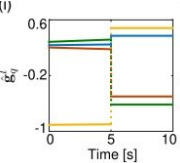
Quaternion



(h)

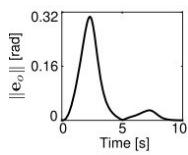


(i)



(j)

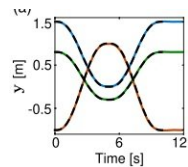
Goal Switch



Orientation Error

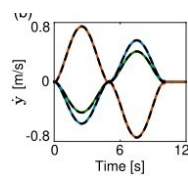
Figure 10. Results obtained by applying the target crossing approach to join two DMPs trained on synthetic data. The training trajectory for the position and the orientation are shown as black-dashed lines in (a)–(b) and (f)–(g), respectively. Results are obtained with the open-source implementation available at <https://gitlab.com/dmp-codes-collection>.

(a)

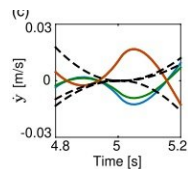


(b)

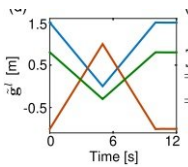
Position



(d)

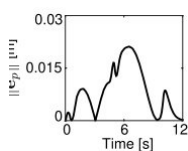


(e)



(f)

Goal Switch



(g)

Position Error

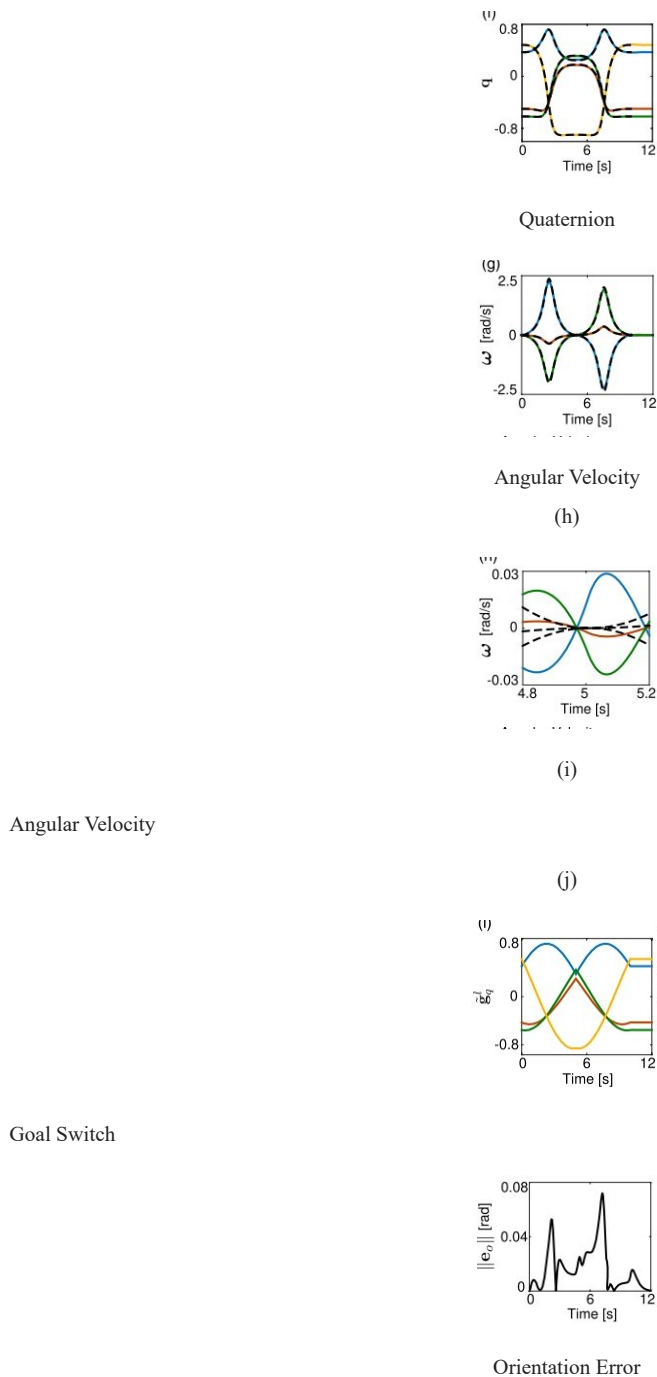


Figure 11. Results obtained by applying the basis functions overlay approach to join two DMPs trained on synthetic data. The training trajectory for the position and the orientation are shown as black dashed lines in (a)–(b) and (f)–(g), respectively. Results are obtained with the open-source implementation available at <https://gitlab.com/dmp-codes-collection>.

distance is the time duration of the generated trajectory—a bigger (smaller) distance results in a shorter (longer) trajectory. For instance, in the considered case, the total motion ends after 9.5 [s] while the demonstration lasts for 10 [s]. Depending on the application, the time difference may cause failures; therefore, it has to be taken into account. Finally, the velocity threshold approach may generate discontinuities if the target of the current DMP is far from the demonstrated initial point of the following primitive.

3.2.2. Target crossing. There exist movements like hitting or batting that are correctly executed only if the target is reached with a non-zero velocity. To this end, Kober et al. (2010b) extend the classical DMP formulation in Section 2.1.1 to let the DMP to track a target moving at a given velocity. In their approach, the DMP passes the target with a given velocity exactly after T seconds. To achieve this, the acceleration in (1) is re-written as follows

$$\dot{\tau}z = (1-x)a_z \text{Big}(\beta_z(\widehat{g}-y) + \tau \text{Big}(\dot{\widehat{y}} - \dot{y}) \text{Big}(\text{Big}) + f(x))$$

$$\widehat{y}_m$$

$$\widehat{g}$$

$$\widehat{g} = \widehat{g}(0) + \dot{\widehat{y}} \frac{\tau \text{ln}(x)}{a_x}$$

$$\widehat{g}(0) = g - \frac{T \dot{\widehat{y}}}{\tau}$$

By inspecting (55) and (56), and considering that the term $(-\tau \ln(x))^{\alpha_x}$ represents the elapsed time if (x) is the phase defined in (3), it is possible to show that the moving target $\widehat{g}$ is designed to reach the goal (g) after (T) seconds, that is, $(\widehat{g}(T) = g)$ (Figure 9-left). The initial position of the moving target $(\widehat{g}(0))$ is obtained by moving the goal position (g) for (T) seconds at constant velocity $(-\dot{y})$. High accelerations at the beginning of the movement are avoided by the pre-factor $((1-x))$ which is set to zero at the beginning of the motion $((x(0) = 1))$. The approach by Nemec and Ude (2012) combines a moving target and a particular initialization of the subsequent DMP to ensure continuity of the movement up to second-order derivatives.

Saveriano et al. (2019) extended this idea to quaternion DMP. The angular acceleration in (16) is modified as follows

$$\begin{aligned} \ddot{\eta} &= (1-x) \alpha_z \text{Big}(\beta_z \text{operatorname{Lo}} \\ &g \text{Big}(\widehat{\mathbf{g}}_q \overline{\mathbf{q}}) + \tau (\widehat{\boldsymbol{\omega}} - \boldsymbol{\omega})) + \\ &+ \mathbf{f}_q(x), \end{aligned}$$

where $(\widehat{\boldsymbol{\omega}})$ is the angular velocity of the moving quaternion target $(\widehat{\mathbf{g}}_q)$ and $(2 \log^q(\widehat{\mathbf{g}}_q \overline{\mathbf{q}}))$ measures the error between the current orientation $(\mathbf{q})$ and $(\widehat{\mathbf{g}}_q)$. The pre-factor $((1-x))$ is used to avoid high angular accelerations at the beginning of the motion. The moving target for the quaternion DMPs is defined as follows

$$\begin{aligned} \widehat{\mathbf{g}}_q &= \text{Exp}^{\mathbf{q}} \text{Big}(-\frac{\tau \text{ln}(x)}{2 \alpha_x}) \widehat{\boldsymbol{\omega}} * \widehat{\mathbf{g}}_q(0), \widehat{\mathbf{g}}_q(0) = \text{Exp}^{\mathbf{q}} \\ &\text{Big}(-\frac{T}{2}) \widehat{\boldsymbol{\omega}} * \mathbf{g}_q, \end{aligned}$$

where $(\mathbf{g}_q)$ is the goal quaternion, (T) is the time duration of the DMP, and the exponential map $(\text{Exp}^{\mathbf{q}})$ is defined in (20). As shown in Figure 9-right, the moving target $(\widehat{\mathbf{g}}_q)$ reaches the goal orientation after (T) seconds, that is, $(\widehat{\mathbf{g}}_q(T) = \mathbf{g}_q)$. This can be easily verified by considering that the initial value of the moving target $(\widehat{\mathbf{g}}_q(0))$ is computed by moving the goal orientation $(\mathbf{g}_q)$ for (T) seconds at the desired velocity $(-\widehat{\mathbf{w}})$.

The presented target crossing approach allows crossing the target after (T) seconds. Assuming to have two DMPs with time duration (T^1) and (T^2) , respectively, one can join them by running the first DMP for (T^1) seconds and then switching to the second one. As for the velocity threshold approach, possible discontinuities at the switching point are prevented by initializing the state of (DMP_2) with the final state of (DMP_1) . This procedure can be repeated to join $(L \geq 2)$ consecutive DMPs.

Results in Figure 10 are obtained when the velocity threshold is applied for merging 2 separately trained DMPs to fit the minimum jerk trajectories (black-dashed lines). Figures 10(a)–(e) show the position and Figures 10(f)–(j) the orientation (unit quaternion) parts of the motion. The merged trajectory is generated by following the first DMP for $(T^1 = 5\text{s})$ and then switching to the second one. The required

intermediate velocity is set to 0.01m/s (rad/s for the orientation) in each direction. The generated trajectory reaches the goal in 10s, that is, demonstration and execution times are the same. As required, the via-point is crossed at $(T = 5s)$ with the desired velocity (Figures 10(c) and (h)). However, the non-zero crossing velocity introduces a deformation in the first part of the trajectory (Figures 10(e) and (j)).

3.2.3. Basis functions overlay. The approach by Kulvicius et al. (2011, 2012) combines multiple DMPs into a complex one, guaranteeing a smooth transition between the primitives by ensuring that the basis functions composing $(f(x))$ in (4) overlap at the switching instances. First of all, Kulvicius et al. adopted a sigmoidal phase variable in (14) instead of the exponentially decaying one (3). As discussed in Section 2.1.1.3, the sigmoidal phase is (≈ 1) for the large part of the motion which makes it possible to use smaller forcing terms to reproduce the demonstrations. On the contrary, the exponential phase is close to zero already before T_s (Figure 3), which results in larger forcing terms.

The classical acceleration dynamics in (1) is modified as follows

$$\dot{z} = a_z (\beta_z (\tilde{g} - y) - z) + f(s)$$

Similar to target crossing, Kulvicius et al. used a moving target $(\tilde{g})$ in the acceleration dynamics, but called it the delayed goal function. The $(\tilde{g})$ term in (59) is obtained by integrating

$$\dot{\tilde{g}} = \begin{cases} \frac{\Delta t}{T} (g - y_0), & t \leq T \\ 0, & \text{otherwise} \end{cases}$$

with $(\tilde{g}(0) = y_0)$. The delayed goal function in Figure 9 moves linearly from (y_0) to (g) in (T) seconds and then remains constant, that is, $(\tilde{g}(t \geq T) = g)$.

The nonlinear forcing term $(f(s))$ is in green in (59) because it slightly differs from the classical one in (4). $(f(s))$ is defined as follows

$$f(s) = \frac{\sum_{i=1}^N w_i \Psi_i(t) \sum_{i=1}^N \Psi_i(t) s}{\sum_{i=1}^N \Psi_i(t) \exp\left(-\frac{\left(\frac{t}{\tau T} - c_i\right)^2}{2\sigma_i^2}\right)},$$

where (σ_i) is the width and (c_i) is the center of the (i) -th basis function, and (s) is obtained by integrating (14). The term $(t/\tau T)$ is used in (61) instead of the phase variable (x) . Being $(0 \leq t/\tau T)$

$(t/\tau T \leq 1)$, the basis functions are equally spaced between 0 and 1. Finally, (σ_i) are the widths of each kernel. They are constant and depend on the number of kernels.

Having presented the main differences with the canonical approach, it is possible to focus on how Kulvicius et al. (2012) solved the problem of joining $(L \geq 2)$ DMPs. In general, each of the L DMPs has a different time duration (T^k) , desired target (g^k) , and initial position (y_0^k) , from which it is possible to compute the delayed goal functions by integrating

$$\dot{\tilde{g}}^k = \begin{cases} \frac{\partial}{\partial t} \left(T^k (g^k - y_0^k) \right), & \sum_{k=1}^{L-1} T^k \leq t \leq \sum_{k=1}^L T^k \\ 0, & \text{otherwise} \end{cases}$$

Note that, being $(\tilde{g}^k(0) = y_0^k)$, the acceleration (59) is s . For this reason, the term $(1 - x)$ used in (54) is not needed in (59).

Assuming that L DMPs have been trained and that each DMP has N kernels, we can merge them into one DMP as follows. The centers of the joined DMP are computed as follows

$$c_i = \begin{cases} \frac{T^1 (i-1)}{T^1 + \dots + T^{L-1}}, & i = 1, \dots, N \\ \frac{T^L (i-1)}{T^L + \dots + T^{L-1}}, & \text{otherwise} \end{cases}$$

where T^l is the duration of the l -th DMP and $T_{\text{join}} = \sum_{l=1}^L T^l$ (duration of the joined motion). The widths of the joined DMP are computed as follows

$$\sigma_i = \frac{\sigma_i^L T^L}{T_{\text{join}}}$$

The centers and widths computed in (63) and (64), respectively, overlap at the transition points allowing for smooth transitions between consecutive DMPs. The weights of the joined DMP are obtained by stacking the N weights of the L DMPs. Therefore, the joined DMP has $N \cdot L$ kernels and $N \cdot L$ weights. The phase variable (14) is modified to run for the duration T_{join} of the joint motion.

Saveriano et al. (2019) extended the basis functions overlay approach to quaternion DMPs. Assuming that a sequence of L quaternion DMPs is given. The angular acceleration in (16) is reformulated for each DMP as follows

$$\ddot{\tau} = a_z \frac{1}{2} \log \left(\frac{\tilde{\mathbf{g}}_q}{\mathbf{g}_q} \right) \frac{\tilde{\mathbf{g}}_q}{\mathbf{g}_q} \frac{1}{\|\tilde{\mathbf{g}}_q\|} \quad (65)$$

where l indicates the l -th quaternion DMP and $\mathbf{f}_{q^l}(s)$ is defined as in (61). The term $\tilde{\mathbf{g}}_q$ is the quaternion delayed goal function, and it ranges from $\mathbf{q}^l(0)$ to $\mathbf{g}_{q^l}$ in T^l seconds (see Figure 9 [right]). To generate this moving target while preserving the geometry of $\mathbf{S}^3$, it is needed that $\tilde{\mathbf{g}}_q$ moves along the geodesic connecting $\mathbf{q}^l(0)$ to $\mathbf{g}_{q^l}$. Therefore, $\tilde{\mathbf{g}}_q$ is defined as follows

$$\tilde{\mathbf{g}}_q(t + \Delta t) = \exp \left(\frac{\tau}{2} \tilde{\mathbf{w}}_q \right) \mathbf{q}^l(t) \quad (66)$$

where

$$\tilde{\omega}_q(t) = \begin{bmatrix} \frac{2}{T^l} \log \left(\frac{\tilde{\mathbf{g}}_q}{\mathbf{q}^l(0)} \right) \\ \sum_{k=1}^{L-1} T^k \mathbf{w}_k \end{bmatrix} \quad (67)$$

The angular velocity in (67) is computed for each l . The term $2 \log \left(\frac{\tilde{\mathbf{g}}_q}{\mathbf{q}^l(0)} \right)$ represents the angular velocity that rotates $\mathbf{q}^l(0)$ into $\mathbf{g}_{q^l}$ in a unit time. Note, that the mappings $\log(\cdot)$ and $\exp(\cdot)$ are defined in (18) and (20), respectively. The delayed goal $\tilde{\mathbf{g}}_q$ crosses all the via-goals $\mathbf{g}_{q^l}$, $l=1, \dots, L-1$ and then reaches the goal $\mathbf{g}_{q^L}$.

Results in Figure 11 are obtained when the velocity threshold is applied to merge 2 DMPs separately trained to fit the minimum jerk trajectories (black-dashed lines). Figure 11(a)–(e) show the position and Figure 11(f)–(j) the orientation (unit quaternion) parts of the motion. This approach does not require a switching rule and automatically generates a smooth trajectory—with continuous velocity as shown in Figure 11(c) and (h)—that passes close to the via-point which favors the overall reproduction accuracy (Figure 11(e) and (j)). However, the distance from the via-point depends on the weights of the joined primitives and cannot be separately decided. The trajectory generated with this approach tends to last longer than the demonstrations. This is due to the sigmoidal phase that vanishes after $(T + \Delta t)$ s (Figure 3). Depending on the application, the time difference may cause failures and has to be taken into account.

3.3. Online adaptation

The standard periodic DMP learning approach approximates the shape $\mathcal{f}_d(t)$ of the input trajectory $\mathcal{y}_d$ in (39) by changing the weights of the Gaussian kernel functions (Ijspeert et al., 2013). Updating of the weights is performed in such a way that the difference between the reference trajectory and the DMP is reduced at every control step and gradually throughout the periodic repetitions. However, the DMP can also be reshaped by some external feedback function to achieve different functionalities for different applications, for instance, tasks that require trial-and-error approach (Kober et al., 2008), obstacle avoidance (Ginesi et al., 2021a; Hoffmann et al., 2009; Park et al., 2008; Tan et al., 2011), coaching (Gams et al., 2016; Petrić et al., 2014b) for robots, and adaptation of assistive exoskeleton behavior (Peternel et al., 2016). Alternatively, the frequency of the existing periodic DMPs can be modulated online (Gams et al., 2009; Petrić et al., 2011).

3.3.1. Robot obstacle avoidance and coaching. In Hoffmann et al. (2009), Park et al. (2008), and Tan et al. (2011), the detected obstacle was fitted with a potential field function to change the shape of the DMP to avoid it. More in detail, Tan et al. (2011) used the potential field to compute a time-varying goal and modified the resulting DMP trajectory, while Hoffmann et al. (2009), Park et al. (2008), and Zhai et al. (2022) added an extra forcing term to the DMP. Similarly in Gams et al. (2016), the human arm was fitted with a potential field function, which was used to reshape the DMP to perform coaching. The potential field was coupled to the position of the human hand to make pointing gestures and indicate the direction in which the robot arm position trajectory should change:

$$\dot{z} = \Omega(a(\beta(-y)-z) + C_{\mathcal{O}} + f)$$

The added coupling term $C_{\mathcal{O}}$ is the obstacle avoidance term that contains the potential field and is given in a simplified form for the sake of explanation as follows:

$$C_{\mathcal{O}} = d_s (\left| \mathcal{O} - y \right|) \exp(-\zeta(\mathcal{O} - y))$$

where $\mathcal{O}$ is the obstacle (or human pointing gesture) and y is the robot position. Exponential and ζ functions determine the potential field, while function d_s controls the distance at which the perturbation field should start affecting the DMP. For the full formulation of $C_{\mathcal{O}}$ and its parameters, see Gams et al. (2016). In Rai et al. (2017), the method was extended to include generalization of the obstacle avoidance formulation in (68). In a recent study, Ginesi et al. (2021a) proposed a novel approach for implementing volumetric obstacle avoidance by utilizing the theory of super quadratic potential functions.

Alternatively, the faulty segment of collision DMP trajectory can also be directly adjusted online by the human demonstrator (Karlsson et al., 2017b). On the other hand, the method in Kim et al. (2015) considers obstacle avoidance as a constraint of an optimization problem, which modifies the DMP trajectory to prevent collisions.

3.3.2. Robot adaptation based on force feedback. Similarly, as for obstacle avoidance, task dynamics can also be incorporated into DMP as coupling terms. In Gams et al. (2014), task dynamics were coupled on the acceleration and velocity level of the DMP. The presented method was utilized for interaction tasks, where the human changed the behavior of the robot based on the exerted dynamics on the manipulator.

$$\tau \dot{z} = a_z (\beta_z(g-y)-z) + \dot{C}_f + f(x),$$

$$\tau \dot{y} = z + C_f$$

whereas the force coupling term $C_f = \zeta F$ is defined as a virtual or measured force F and ζ is a scaling factor, which essentially changes the dynamic behavior of the DMP, enabling the motion primitive to instantly react to the coupled force. Later, Zhou et al. (2016b) introduced a PD controller-based coupling term formulation $C_{PD} = \zeta (K_P(F_d - F^e) - D^V \dot{F}^e)$ coupled to the velocity part of the DMP (71). In the formulation F_d represents the desired

force, (F^e) is the measured force, (ζ) is a scaling factor, and $(K^{\mathcal{P}})$ and $(D^{\mathcal{V}})$ are the proportional and derivative gains of the Proportional Derivative (PD) controller, respectively. The coupling term formulation allows for a controlled adaptation of robot motion to changes in the environment.

In Kramberger et al. (2018) this approach was extended, with a force feedback loop coupled to the velocity (2) and the goal g of the DMP. The outcome of this approach is a similar behavior as an admittance controller (Villani and De Schutter, 2008), with a difference that the execution is directly on the trajectory generation level.

$$\dot{\tau}_z = a_z (\beta_z ((g + C_a) - y) - z) + f(x),$$

$$\dot{\tau}_y = z + \dot{C}_a$$

Here $\dot{C}_a = \text{varsigma}(F_d - F^e)$ is the first time-derivative of the admittance coupling term, which changes the velocity and consequently the integrated coupling term, the position output of the DMP. The described approach can be used for Cartesian space motion, where the forces have to be substituted for desired and measured torques. This approach can be implemented in robot tasks involving contact with the environment as well as contact with humans. The constrained DMP formulation in Lu et al. (2021) unifies the formulations described in Sections 3.3.1 and 3.3.2 while ensuring stability of the generated motion.

3.3.3. Exoskeleton joint torque adaptation. In Peternel et al. (2016), human effort was used to provide information about the direction in which the assistive exoskeleton joint torque DMP should change in order to minimize it. The human was included into the robot control loop by replacing the error calculation in (41) with the human effort feedback term $(U(E))$:

$$w_i(t_{j+1}) = w_i(t_j) + \Psi_i P_i(t_{j+1}) U(E)$$

where $(E(t))$ is the current effort measured by human muscle activity through Electromyography (EMG) signals. $(\wedge^3)$ Equations (34)–(38) and (42) are used in the original form. Equations (39)–(41) are not used, since (74) is used to modulate the weights in (37) instead.

The effort feedback term $(U(E))$ closes the loop and acts as feedback for adapting the weights of Gaussian kernels that define the shape of the trajectory. A positive $(U(E))$ increases, while a negative $(U(E))$ decreases the values of weights at a given section of the periodic DMP that encodes joint torque. If the shape of the DMP does not provide enough assistive power, the human has to exert effort (i.e., muscle activity) to produce the rest of the power required to achieve the desired task under given dynamics. In turn, muscle activity feedback then increases the magnitude of the DMP until the human effort term $(U(E))$ is minimized. Note that each joint has its own torque DMP and $(U(E))$ term (Peternel et al., 2016). After that point, the DMPs do not change unless the task, dynamics, or conditions change. If they change, the

human has to compensate for the change by an additional muscle activity, which in turn adapts the DMPs to the new required joint torques.

3.3.4. Trajectory adaptation based on reference velocity. In many LfD scenarios, it is desired to modify both the spatial motion and the speed of the learned motion at any stage of the execution. Speed-scaled dynamic motion primitives first presented in Nemec et al. (2013a) are applied for the underlying task representation. The original DMP formulations from (1) and (2) were extended by adding a temporal scaling factor (ν) on the velocity level of the DMP.

$$\dot{\tau}_z = a_z (\beta_z (g - y) - z) + f(x),$$

$$\dot{\tau}_y = z \nu$$

From (75) and (76), it is evident that the velocity term is a function of phase and therefore encoded with a set of RBFs similarly as in (4). This method allows for modification of the spatial motion as well as the speed of the execution at any stage of the trajectory execution. The authors demonstrated the proposed method in a learning scenario, where after every learning cycle (using Iterative Learning Control [ILC]) a new velocity profile was encoded based on the wrench feedback and thus converged to an optimal velocity for the specific task. Vuga et al. (2016) extended the approach by incorporating a compact representation for non-uniformly accelerated motion as well as simple modulation of the movement parameters.

Later on, in Nemec et al. (2018) the authors extended the previous approach to also incorporate velocity scaling of the encoded orientation trajectories represented with unit quaternions. The outcome of the presented work is a unified approach to velocity scaling for tasks executed in Cartesian space. Furthermore, a reformulation of the velocity approach called Arc-Length-DMPs (AL-DMPs) was presented by Gašpar et al. (2018). In this work, they present a method, where the spatial and temporal components of the motion are separated, by means of the arc-length based on the time-parameterized trajectory. Arc-length, based on the differential geometry of curves, is related to the speed of the movement, given as the time derivative of the demonstrated trajectory. The approach is well suited when multiple demonstrations are compared for the extraction of relevant information for learning. Following the AL-DMPs idea, Simonič et al. (2021) introduced a constant speed DMPs to fully decouple the spatial and temporal part of the task. Pahic et al. (2021) used deep NN to map images into spatial paths represented by AL-DMPs. Weitschat and Aschemann (2018) added an extra forcing term to keep the velocity within a certain predefined limit. The aim of this work is to guarantee a safe execution of the robot task when interacting with humans, as well as provide a framework for safe interaction in a changing environment where the robot position and velocity have to change over time. For a full formulation of the coupling term, see Weitschat and Aschemann (2018).

To maintain consistency along the trajectory while ensuring a bounded velocity for each DoF, it is essential to decrease the velocities uniformly across all dimensions. This rationale prompts the consideration of temporal coupling, which involves modifying the DMP time constant, equivalent to the duration of path traversal, rather than spatial coupling. By increasing the time constant, the temporal evolution of all DoFs can be simultaneously reduced by the same factor. A temporal coupling approach based on tracking error was introduced in Ijspeert et al. (2013) and an enhanced version was proposed in Karlsson et al. (2017a). However, this method requires distorting the path in order to slow down the trajectory, and therefore, it cannot be directly utilized to limit the velocity or acceleration. One year later, the authors provided stability analysis of temporally coupled DMPs in Karlsson et al. (2018). Dahlin and Karayiannidis (2020) in their work proposed a temporal coupling based on a repulsive potential, keeping the DMP velocity within the predefined velocity limits while ensuring the path shape invariance. Subsequently, Dahlin and Karayiannidis (2021) introduced a temporal coupling method for DMPs that enables control over the velocity and acceleration constraints of the generated trajectory. This approach incorporates a filtering mechanism that proactively decelerates the trajectory as the acceleration constraints are approached, thereby mitigating the potential risk of infeasibility.

3.4. Robots with flexible joints

DMPs are commonly employed for generating trajectories in manipulators with noticeable elasticity. This choice is often justified by the relatively low execution speeds, resulting in lower acceleration and jerk levels. However, when rapid motions with high acceleration and jerk are required, combining DMPs for trajectory generation with inverse dynamics for feedforward controls can lead to oscillations in robots with flexible modes. In industrial settings, where execution speed is typically faster than the speed during demonstration due to cycle time requirements, this becomes a significant challenge. To overcome this issue, Wahrburg et al. (2021) proposed an extended framework for DMPs toward flexible joint robots, dominated as FlexDMP, by introducing a fourth-order system for generating trajectories as follows

$$\begin{array}{l} \tau \ddot{z} = a_1(a_2(a_3(a_4(g-y)-z)-\dot{z})-\ddot{z})+f(x), \end{array}$$

$$\tau \ddot{y} = \ddot{z},$$

$$\tau \ddot{y} = \dot{z},$$

$$\tau \dot{y} = z$$

where y , $\dot{y}$, $\ddot{y}$ are position, velocity, acceleration, and jerk, respectively. The canonical system and forcing terms are similar to the ones in (3) and (4). For critically damped system $\alpha_1 = 4\beta$, $\alpha_2 = 3/2\beta$, $\alpha_3 = 2/3\beta$, and $\alpha_4 = 1/4\beta$, where $\beta > 0$.

3.5. Alternative formulations

LfD is a wide research area, and many different approaches have been developed to reproduce human demonstrations (Billard et al., 2016). As already mentioned, the aim of this tutorial survey is to provide a comprehensive overview of DMP research, and we intentionally skip the rich literature in the field of LfD. However, we found some representations that are closely related to the DMP formulation. This section briefly reviews them.

Calinon et al. (2009) computed an acceleration command for the robot in a PD-like form

$$\ddot{\mathbf{y}} = \mathbf{K}^{\mathcal{P}}(\mathbf{y}_d - \mathbf{y}) + \mathbf{D}^{\mathcal{V}}(\dot{\mathbf{y}}_d - \dot{\mathbf{y}})$$

where $\mathbf{K}^{\mathcal{P}}$ is a stiffness and $\mathbf{D}^{\mathcal{V}}$ a damping gain, $\mathbf{y}$ is the measured state of the robot and $\dot{\mathbf{y}}$ its time derivative (velocity), and $\mathbf{y}_d$ and $\dot{\mathbf{y}}_d$ are desired position and velocity retrieved with GMR, respectively. Authors then showed that the acceleration command $\ddot{\mathbf{y}}$ can be seen as a mixture of linear dynamics, each converging to a certain attractor. Despite later work like Kormushev et al. (2010) referred to this representation as “a modified version” of DMPs, there are significant differences with the DMP formulation properly highlighted by Calinon et al. (2012).

Herzog et al. (2016) computed an acceleration command for the robot from the linear system

$$\ddot{\mathbf{y}} = \mathbf{u} = \mathbf{K}^{\mathcal{P}}(\mathbf{y}_d - \mathbf{y})$$

where $\mathbf{y}$ is the measured state of the robot, $\mathbf{y}_d$ is a human demonstration, and $\mathbf{K}^{\mathcal{P}}$ is a control gain computed using the linear-quadratic regulator method. Then, a compact representation of the control input trajectory $\mathbf{u}$ is computed by means of Chebyshev polynomials. This representation does not require a vanishing phase variable to ensure convergence, but the generalization to different start/goal positions requires the application of the linear-quadratic regulator method to find a new sequence of control inputs.

Regarding periodic motions, Ajallooeian et al. (2013) proposed a dynamical system-based framework to learn rhythmic movements with an arbitrary shape and basin of attraction. They exploit phase-based scaling functions to represent the mapping between a known, base limit cycle and a desired periodic orbit. The basic limit cycle can be, for example, the one generated by periodic DMPs, which makes the approach of Ajallooeian et al. (2013) a more general formulation of periodic primitives.

4. DMPs integration in complex frameworks

This section reviews approaches where DMPs have been integrated into bigger executive frameworks. We categorize these approaches into five main research areas, namely, grasping and manipulation, impedance learning, reinforcement learning, deep learning, and incremental and life-long learning.

4.1. Manipulation tasks

4.1.1. Grasping and tool usage. Successfully grasping an object is the first step toward robotic manipulation. Grasping necessitates the perception of the environment, particularly visual perception, to locate the object and determine suitable grasping points based on its geometric characteristics. In this context, even slight uncertainties can lead to object slippage and failed grasps. To improve the robustness

of vision-driven grasping, Krömer et al. (2010a) augmented DMPs with a potential field based on visual descriptors to adapt hand and finger trajectories according to the local geometry of the object. This grasping strategy was integrated within a hierarchical control architecture where the upper level determines the object's grasp location and the lower level locally adjusts the motion to achieve a robust grasp of the object (Krömer et al., 2010b). Stein et al. (2014) proposed a point cloud segmentation approach based on the convexity and concavity of surfaces. This approach is particularly well-suited for recognizing object handles, and DMPs are employed to execute grasping with a real robot.

The ability to grasp and use tools is also desirable to perform daily-life manipulation. In this respect, Guerin et al. (2014) proposed the so-called tool movement primitives that transform the demonstrations in a tool affordance frame. The result is a motion that generalizes to different tool poses and to tools that share the same affordance(s). Li and Fritz (2015) considered tool usage with low-cost, non-dexterous grippers and propose a framework to learn bi-manual strategies for tool usage and compensate for the lack of dexterity. Bi-manual robotic manipulation is a challenging task that requires precise coordination between hand movements and adherence to spatial constraints. Thota et al. (2016) developed a DMP-based control framework for bi-manual manipulation that ensures time synchronization of the two hands while being robust to spatial perturbations and goal changes.

4.1.2. Motion primitives sequencing. Beyond object grasping, everyday manipulation requires a precise execution of complex movements. Often such complex movements are hard to encode into a single motion primitive, but they can be conveniently split into simpler motions (e.g., reach and grasp) that can be properly sequenced and executed (Figure 12).

The possibility of exploiting DMPs as the building blocks of complex tasks was investigated in Caccavale et al. (2018, 2019) and Ramirez-Amaro et al. (2015). In these works, a human teacher demonstrated a relatively complex task consisting of several actions performed on different objects. The demonstration was then automatically segmented into M basic motions used to fit M DMPs. While Ramirez-Amaro et al. (2015) exploit semantic rules (e.g., reach an object with a knife means cut) to infer high-level human activities, Caccavale et al. built a hierarchical structure to schedule the execution of the complex task by selecting the proper DMP for the current executive context. They used kinesthetic teaching and verbal cues (open/close gripper commands) to provide task demonstrations. Schwaner et al. (2021) used DMPs to build a

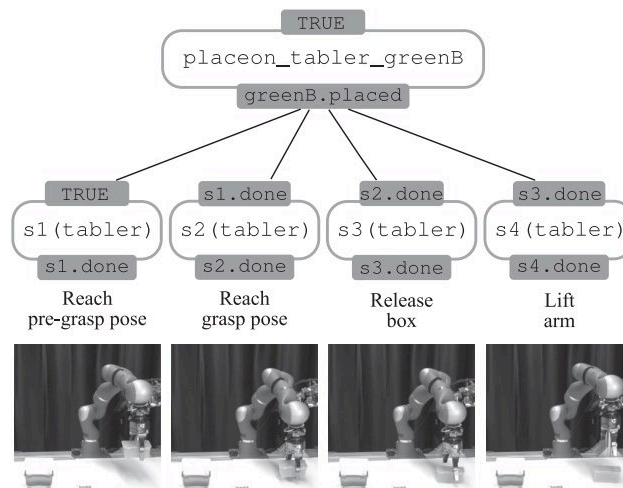


Figure 12. An example of hierarchical task decomposition and motion primitives sequencing from Agostini et al. (2020).

library of surgical skills which used to autonomously execute part of a suturing task. Lemme et al. (2014) organize segmented task demonstrations into a motion primitives' library learned from self-generated trajectory patches. They also introduced a mechanism to remove unused skills and update the library. Kinesthetic teaching and haptic feedback were also used by Eiband et al. (2019) to segment and recognize basic motions or skills and to build a tree describing geometric relationships—like reference frames and goal poses—between consecutive skills. At run time, the robot performed haptic exploration to locate objects in the scene and update the skill tree. The transformations in the skill tree were then used to define the initial and goal pose of the DMPs and execute the task. Finally, Wu et al. (2018) integrated DMPs into a dialogue system with speech and ontology to learn or re-learn a task using natural interaction modalities.

4.1.3. Data collection. Collecting demonstrations becomes an issue of kinesthetic teaching or marker-based motion trackers cannot be used. The latter requires an expensive sensor infrastructure that is hard to build in real-world scenarios like factory floors. Kinesthetic teaching needs torque-controlled/collaborative robots that are still uncommon in industrial scenarios. To remedy this issue, Mao et al. (2015) exploited a low-cost RGB-D camera and track the human hand using the markerless approach proposed by Oikonomidis et al. (2011). Collected data were then segmented into basic motions and used to fit DMPs. Also, the approach in Yang et al. (2022) does not require tracking markers or manual annotations. Instead, authors exploit videos of random unpaired interactions with objects by the robot and human for unsupervised learning of a keypoint model of visual correspondences. Bayesian optimization is then used to find the parameters of rhythmic DMPs from a single human video demonstration within a few robot trials.

The described approaches assume that human teachers always provide consistent and noiseless task demonstrations. Ghalamzan E. et al. (2015) encoded noisy demonstrations into a GMM and computed a noise-free trajectory using GMR. The noise-free trajectory was then used to fit a DMP that generalized to different start, goal, and obstacle configurations. Dong et al. (2023) propose to fit a DMP from correct (positive) and incorrect (negative) demonstrations to increase the representation and generalization capabilities of the model. Niekum et al. (2012, 2015) designed a framework that learns from unstructured demonstrations by segmenting the task demonstrations, recognizing similar skills, and generalizing the task execution. Interestingly, a user study on 10 volunteers conducted by Gutzeit et al. (2018) showed that existing strategies for segmentation and learning are sufficiently robust to enable automatic transfer of manipulation skills from humans to robots in a reasonable time.

4.1.4. Task learning and execution. Some work (Deniša and Ude, 2013a, 2013b, 2015; Denisa et al., 2021) exploited transition graphs and trees to embed parts of a trajectory and search algorithms to discover a sequence of partial parts and generate motions that have not been demonstrated. Approaches that rely on a hierarchical, tree-like structure to represent the task have limited task generalization capabilities. Lee and Suh (2013) used probabilistic inference and object affordances to infer the adequate skill that can handle uncertainties in the executive context. Beetz et al. (2010) learned stereotypical task solutions from observation

and used task planning and symbolic reasoning to execute novel mobile manipulation tasks. A generative learning framework was proposed by Wörgötter et al. (2015) to augment the robot's knowledge base with missing information at different levels of the cognitive architecture, including symbolic planning as well as object and action properties. Paxton et al. (2016) used task and motion planning to generalize the execution of complex assembly tasks and proposed a learning by demonstration approach to ground symbolic actions. Agostini et al. (2020) performed task and motion planning by combining an object-centric description of geometric relations between objects in the scene, a symbol to motion hierarchical decomposition depending on three consecutive actions in the plan, and the LfD approach developed in Caccavale et al. (2019) (Figure 12). A manipulation task was described at three different levels by Aein et al. (2013). The top level provides symbolic descriptions of actions, objects, and their relationships. The mid-level uses a finite state machine to generate a sequence of action primitives grounded by the lower level. A common point among these approaches is that they use DMP to execute the task on real robots.

4.2. Variable impedance learning control

Impedance control can be used to achieve compliant motions, in which the controller resembles a virtual spring–damper system between the environment and robot end-effector (Hogan, 1985). Such an approach permits smooth, safe, and energy-efficient interaction between robots and environments (possibly humans). A standard model for such interaction is defined as follows

$$\begin{aligned} \frac{d}{dt} \mathbf{M} \ddot{\mathbf{y}} + \mathbf{K} \mathbf{y} + \mathbf{D} \dot{\mathbf{y}} &= \mathbf{F}_g - \mathbf{F}_e \\ \mathbf{M} &= \mathbf{M}_0 + \mathbf{M}_1 \dot{\mathbf{y}} + \mathbf{M}_2 \ddot{\mathbf{y}} \\ \mathbf{K} &= \mathbf{K}_0 + \mathbf{K}_1 \dot{\mathbf{y}} + \mathbf{K}_2 \ddot{\mathbf{y}} \\ \mathbf{D} &= \mathbf{D}_0 + \mathbf{D}_1 \dot{\mathbf{y}} + \mathbf{D}_2 \ddot{\mathbf{y}} \end{aligned}$$

where (81) and (82) correspond to translational and rotational cases, respectively, $\mathbf{M}$, $\mathbf{K}_t^P$, and $\mathbf{D}_t^V$ are the mass, stiffness, and damping matrices, respectively, for translational motion, while $\mathbf{I}$, $\mathbf{K}_t^O$, and $\mathbf{D}_t^V$ are the moment of inertia, stiffness, and damping matrices, respectively, for rotational motion. $\widehat{\mathbf{R}}$, $\mathbf{R}_t \in \text{SO}(3)$ are rotation matrices and correspond to desired rotation goal and actual orientation profile of the end-effector, respectively. $\mathbf{f}_t^e$ and $\mathbf{r}_t^e$ represent the external force and torque applied to the robot end-effector, respectively.

In fact, Variable Impedance Control (VIC) plays an important role when a robot needs to interact with any environment in order to avoid high impact forces and damage for the environment or the robot (i.e., change to low stiffness) (Ajoudani et al., 2012; Abu-Dakka et al., 2018; Peternel et al., 2018a). On the other hand, it is important in rejecting unexpected and unpredictable perturbations from the environment to achieve the desired position tracking precision (i.e., change to high stiffness) (Yang et al., 2011). In addition, it is also important in the coordination of human–robot collaborative movements (Petemel et al., 2017b). However, a robotic system still needs to learn how to adapt such VIC to unseen situations while avoiding hard-coding. Such a paradigm of learning is called Variable Impedance Learning Control (VILC). Interested readers can refer to our recent survey on VILC (Abu-Dakka and Saveriano, 2020) or teleimpedance (Petemel and Ajoudani, 2022).

In this review, we will mention some of the works that integrate DMP with VIC in a VILC framework. Figure 13 shows a simple generic example where DMP is integrated in a VIC control scheme.

Buchli et al. (2011a) proposed one of the earliest approaches that integrates DMP with Policy Improvement with Path Integrals (PI²) algorithm (Theodorou et al., 2010) to learn movements (position and velocity presented by DMP) while optimizing impedance parameters. Later, the authors exploited a diagonal stiffness matrix and expressed the variation (time derivative) of each diagonal entry as follows

$$\dot{k}_{\vartheta_j,t} = a_j \bigwedge_{\text{top}}(\vartheta_j + \epsilon_{j,t}) - k_{\vartheta_j,t} \quad j=1, \dots, J$$

where j indicates the j -th joint, k_{ϑ_j} , i is the stiffness of joint j , $\epsilon_{j,t}$ is a time-dependent exploration noise, each λ_j is a vector of N Gaussian basis functions, and ϑ_j are the learnable parameters for joint j . The stiffness parameterization in (83) is also linear in the parameters, and PI^2 can be applied to find the optimal policy. Later, authors used PI^2 to learn VIC in deterministic and stochastic force fields (Stulp et al., 2012a). Nakanishi et al. (2011) proposed a method that optimizes a periodic motion along with a time-varying joint stiffness.

Basa and Schneider (2015) introduced an extension to DMP formulation by adding a second nonlinear function to cope with elastic robots as follows

$$\tau\dot{z} = a_z(\beta_z(g-y)-z) + f(x) + f_2$$

where f_2 is defined as (4) but without the phase variable x . The main purpose of f_2 is to compensate for the gravitational influence on the moved DoF at the end of the movement time and beyond. Differently, Haddadin et al. (2016) used optimal control to execute near-optimal motion of elastic robots.

Nemec et al. (2016) proposed a cooperative control scheme that enables a dual-arm robot to adapt its stiffness online along with the executed trajectory in order to provide accurate evolution. Umlauf et al. (2017) used GP along with DMPs (as proposed in Fanger et al. (2016)) to predict the trajectories. During the execution, their admittance controller adapts both stiffness and damping online. The energy-tanks passivity-based control method has been integrated with DMPs to enforce passivity in order to stably adapt to contacts in unknown environments by adapting the stiffness online (Shahriari et al., 2017; Kramberger et al., 2018; Kastritsi et al., 2018).

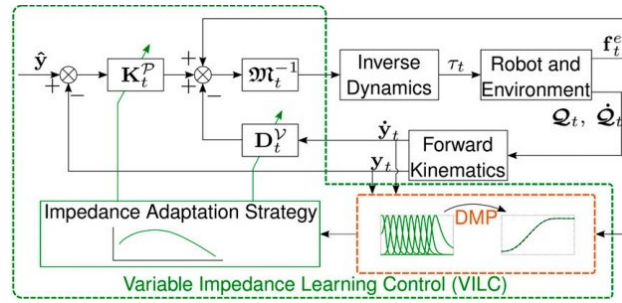


Figure 13. General control scheme of VIC and DMP.

Methods in Bian et al. (2019), Peternel et al. (2014, 2018b, 2018a), and Yang et al. (2018, 2019) designed different multi-modal interfaces to let the human to explicitly teach an impedance behavior to the robot. Most of them combined EMG-based variable impedance skill transfer with DMP-based motion sequence planning, inheriting the merits of these two aspects for robotic skill acquisition. Hu et al. (2018) used Covariance Matrix Adaptation-Evolution Strategies (CMA-ES) to update the parameters of DMPs and variable impedance controller in order to reduce the impact during the robot motion in noisy environments. Dometios et al. (2018) integrated a Coordinate Change-DMPs (CC-DMP) with a vision-based motion planning method to adapt the reference path of a robot's end-effector and allow the execution of washing actions.

Travers et al. (2016, 2018) proposed a shape-based compliance controller for the first time in locomotion, by implementing amplitude compliance on a snake robot moving in a complex environment with obstacles. Their approaches allow snake-like robots to blindly adapt to such complex unstructured terrains, thanks to their proprioceptive gait compliance techniques.

Recently, an adaptive admittance controller is proposed (Wang et al., 2020) which integrates GMR for the extraction of human motion characteristics, DMP to encode a generalizable robot motion, and an RBF-NN-based controller for trajectory tracking during the reproduction phase. In their work, Spector and Zacksenhouse (2021) introduced a residual admittance policy within the framework of DMPs. This policy explicitly learned full asymmetric stiffness matrices and aimed to correct the movements generated by a baseline policy. The effectiveness of the learned policy was demonstrated through successful peg insertion tasks involving various shapes and sizes of pegs. Additionally, the policy exhibited robustness in handling uncertainties in hole location and peg orientation and showed good generalization to new shapes. Moreover, the learned policy demonstrated successful transferability from simulations to real-world scenarios.

Novel LfD approaches explicitly take into account that training data are possibly generated by certain Riemannian manifolds with associated metrics. Abu-Dakka and Kyrki (2020) reformulated DMPs based on Riemannian metrics, such that the resulting formulation can operate with SPD data in the SPD manifold. Their formulation is capable to adapt to a new goal-SPD-point.

Recently, biomimetic controller has been integrated with DMPs (Zeng et al., 2021) in order to learn and adapt compliance skills.

4.3. Reinforcement Learning (RL)

In RL, an agent tries to improve its behavior via trial-and-error by exploring different strategies (actions) and receiving a feedback (reward) on the outcome of its actions. Actions a are drawn from a policy $\pi(s, a)$ that represents a mapping between states s and actions a . The goal of RL is to find an optimal policy π^* that maximizes the cumulative expected reward, that is, the sum of expected rewards over a possibly infinite time interval. When the agent is a robot performing tasks in the real world, the state and action spaces are inherently continuous. Moreover, the robotic agent is affected by imperfect (e.g., noisy) perception and inaccurate models (e.g., contacts). Finally, performing a large amount of interactions with the real world (rollouts) is expensive and possibly dangerous. As discussed by Kober et al. (2013), robotic-specific challenges require specific solutions to make the RL problem feasible.

4.3.1. DMPs as control policies. One possibility is to use parameterized policy and use RL to search for an optimal, finite set of policy parameters. In this respect, DMPs have been widely used as policy parameterization. The general idea is shown in Figure 14. More in detail, (Peters and Schaal, 2008a; Schaal, 2006) showed that various policy gradient and actor-critic RL approaches can be effectively applied to improve robotic skills parameterized as DMPs. Other research focused on developing policy search algorithms specifically for parameterized policies. Inspired by stochastic optimal control, Theodorou et al. (2010)

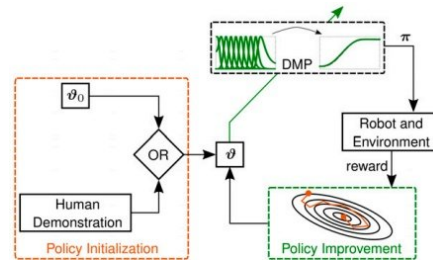


Figure 14. General block scheme of DMP-based policy improvement.

proposed $\backslash(\text{PI}^{\wedge}\{2\})\backslash$ which is an application of path integral optimal control to DMPs. $\backslash(\text{PI}^{\wedge}\{2\})\backslash$ and DMPs have been successfully applied in several domains including VILC (Buchli et al., 2011a, 2011b) and in-contact tasks (Hazara and Kyrki, 2016), grasping under-state estimation uncertainties (Stulp et al., 2011a), bi-manual manipulation (Zhao et al., 2020), nonprehensile manipulation (Sun et al., 2022), and robot-assisted endovascular intervention (Chi et al., 2018). Kober and Peters (2011) derived from expectation-maximization the so-called Policy Learning by Weighting Exploration with the Returns (PoWER). PoWER and DMPs have been successfully applied to perform highly dynamic tasks including ball-in-a-cup (Kober and Peters, 2011) and pancake flipping (Kormushev et al., 2010).

4.3.2. Limit the search space. Even with parameterized policies the number of rollouts needs to search for optimal policy parameters may become large, especially for robots with many DoFs. Dimensionality reduction techniques can be exploited to perform policy search in a reduced space (Colomé and Torras, 2014). The effectiveness of this approach was demonstrated in the challenging task of clothes (i.e., soft tissues) manipulation (Colomé and Torras, 2018). IL arises as an effective approach to policy initialization and to speed up policy search by reducing the number of rollouts (Kober and Peters, 2010). In this respect, Kober et al. (2008, 2010a) augmented DMPs with a perceptual coupling term and proposed to initialize the DMP via human imitation and to refine the motor skill via RL. IL can be eventually combined with dimensionality reduction (Tan and Kawamura, 2011), and several rollouts can be performed firstly in simulation (Cohen and Berman, 2014) to further speed up the policy search. When multiple demonstrations are given, one can learn a mapping between policy parameters and query points (e.g., goal positions) and use the mapping to generalize to new situations (Section 3.1.3). This strategy was used by Nemec et al. (2011, 2012, 2013b) to provide a good initial policy for a new situation which is then further refined using RL. Being the mapping estimated using example query points, the search space can be effectively constrained within query points making the policy search more efficient. Vuga et al. (2015a, 2015b) combined this approach with a different DMP formulation to optimize the velocity of execution. The approach was tested on diverse tasks including pouring water into a cup, where it prevented the water to split from the cup during the motion. Schroeder et al. (2016) provided demonstrations in the form of soft via-points (Section 3.1.2) which reduce the search space to the neighborhood of the taught via-points. Multiple demonstrations were used by Reinhart and Steil (2014, 2015) to build a parameterized skill memory that connects low-dimensional skill parameterization to motion primitive parameters. This low-dimensional embedding is then leveraged for efficient policy search. Instead of learning a mapping from task to policy parameters, Queißer et al. (2016) used data from the rollouts to incrementally learn a parametric skill (bootstrapping) and used it to generate a good initial policy for a new task.

4.3.3. DMP generalization and sequencing. Instead of using generalization to provide a better initial policy, some researchers exploit RL to improve and generalize the motion primitive. André et al. (2015) adapted DMP policies to walk on sloped terrains. Mülling et al. (2010) generalized to new situations using a mixture of DMPs. In their approach, RL was used to estimate the shape parameters as well as to estimate the optimal responsibility of each DMP. Mülling et al. (2013) used episodic RL to estimate meta-parameters like the temporal and spatial interception point of the ball and the racket typical of table tennis tasks. Lundell et al. (2017) used parameterized kernel weights and RL to search for optimal parameters, while Forte et al. (2015) augmented the given demonstration using RL-based state space exploration to autonomously expand the robot's task knowledge. Metric RL was exploited by Hangl et al. (2015) to smoothly switch between learned DMP policies and execute a task in new situations.

RL can be also applied to sequence multiple motion primitives and perform more complex tasks; a successful strategy when the robot has to perform, for instance, a manipulation task (Section 4.1). To sequence multiple primitives, it is also of importance to learn the goal of each motion. Tamosiunaite et al. (2011) used continuous value function approximation to optimize the goal parameters of a DMP used to perform a pouring task. Kober et al. (2011, 2012) learned a meta-parameter function that maps the current state to a set of meta-parameters including the goal and duration of the movement. Instead of separating shape and goal learning into different processes, Stulp et al., 2011b,

2012) extended PI^2 to simultaneously learn the shape and goal of a sequence of DMPs. Wang et al. (2022) describe both complex manipulation tasks and user execution preferences as logic and temporal constraints and use RL to find a set of DMP parameters that fulfill the constraints.

4.3.4. Skills transfer. Learned skills can be potentially transferred across different tasks to speed up the learning process and increase robot autonomy. To this end, Fabisch and

Metzen (2014) considered the case where the robot can actively choose which task to learn to make the best progress in learning. The process of actively selecting the task was considered as a non-stationary bandit problem for which a suitable algorithmic solution exists while intrinsic motivation heuristics were exploited to reward the agent after the selection. Cho et al. (2019) defined the complexity of a motor skill based on the temporal and spatial entropy of multiple demonstrations and used the measured complexity to generate an order for learning and transferring motor skills. Their experimental findings provided useful guidelines for skill learning and transfer. In short, humans have to demonstrate, when possible, the most complex task and then the robot is able to transfer the motor skills. Vice versa, if demonstrations are not given, it is more effective to start learning simple skills first and then transfer the simpler skills to more complex tasks.

4.3.5. Learning hierarchical skills. RL often lacks scalability to high-dimensional continuous state and action spaces. To remedy this issue, hierarchical RL exploits a divide et impera approach by decomposing an RL problem into a hierarchy of sub-tasks in order to reduce the search space. Different levels in the hierarchy represent information at different times and/or spatial scales.

Stulp and Schaal (2011) proposed to represent different options as DMPs to sequence. PI^2 was extended to optimize shape and (sub-)goal of each DMP at different levels of temporal abstraction. In particular, the shape was adjusted based on the cost up to the next primitive in the sequence, while the sub-goal considers the cost of the entire sequence of two DMPs. Layered direct policy search in End et al. (2017) did not rely on a set of predefined sub-policies and/or sub-goals but instead used information theoretic principles to uncover a set of diverse sub-policies and sub-goals.

Reducing the number of rollouts required to discover optimal policies is also important in Hierarchical RL (HRL). As already mentioned, IL is a valuable option to find good initial policies. However, there are applications like manipulation with multi-fingered robotics hands for which it is hard or impossible to provide expert demonstrations. To make policy search more efficient, Ojer De Andres et al. (2018) used HRL where the upper level considers discrete action and state spaces to search for optimal finger gaiting and synchronization among the fingers. This information was passed to the lower level where rhythmic DMPs and PI^2 generated continuous commands for the fingers. Another possibility to increase data-efficiency is to use model-based approaches for RL. Colome et al. (2015) exploited a friction model to improve a DMP policy and manipulate soft tissues (a scarf). A model-based HRL approach was proposed by Kupcsik et al. (2017) for data-efficient learning of upper level policies that generalize well across different executive contexts. Li et al. (2018) proposed a hybrid hierarchical framework where the higher level computes optimal plans in Cartesian space and converts them to desired joint targets using an efficient solver. The lower level is then responsible to learn joint space trajectories under uncertainties using RL and DMPs. Recently, Davchev et al. (2022) proposed residual LfD, a framework that combines DMPs and RL to learn a residual correction policy for assembly tasks. In the paper, they show that applying residual learning directly in task space and operating on the full pose of the robot can significantly improve the overall performance of the DMPs.

4.4. Deep learning

A popular method of machine learning is NNs. Due to their non-parametric nature, they can effectively represent non-linear mappings. A major drawback of NNs in the past was their computational complexity of learning. In recent years, there is a renewed interest in NNs. New deep learning approaches were successfully (LeCun et al., 2015) applied in machine vision and language processing.

In recent years, deep learning has been applied also in robotics to learn task dynamics (Yang et al., 2016) and movement dimensionality reduction (Chen et al., 2015). The authors (Chen et al., 2015, 2016) introduced a framework called AutoEncoded DMP (AEDMP) which uses deep auto-encoders to find movements represented in latent feature space. In this space DMPs can optimally be generalized to new tasks, as well as the architecture enables the DMPs to be trained as a unit. Pervez et al. (2017b) in their work coupled the vision perception data for object calcification with task-specific movement definitions represented with DMPs. The data was modeled with

Convolutional Neural Networks (CNNs), where the images and the associated movements were directly processed by the deep NN, thus preserving the associated DMP properties and eliminating the need for extracting the task parameters during motion reproduction. Later on, Kim et al. (2018b) combined deep RL with DMPs to learn and generalize robotic skills from demonstration. The framework builds on an RL approach to learn and optimize a new DMP skill based on a demonstration. The RL approach is backed up with a hierarchical search strategy, reducing the search space for the robot, which allows for more efficient learning of complex tasks. Furthermore, Pan and Manocha (2018) presented a deep learning approach for motion planning of high-dimensional deformable robots in complex environments. The locomotion skills are encoded with DMPs, and an NN is trained for obstacle avoidance and navigation. The data is further optimized with deep Q-Learning showing that the learned planner can efficiently plan and navigate tasks for high-dimensional robots in real-time.

Pahic et al. (2018) proposed a deep learning approach for perception-action couplings, demonstrating the coupling between the vision-based images and associated movement trajectories. Later on, they extended the approach to incorporate CNNs and give a distinguishing property formulation for the approach (Pahič et al., 2020), which utilizes a loss function to measure the physical distance between the

movement trajectories as opposed to measuring the distance between the DMPs parameters which have no physical meaning, leading to better performance of the algorithm. Recently, they extended the usage of GPR to create a database needed to train autoencoder NNs for dimensionality reduction (Lončarević et al., 2021). Mavsar et al. (2022) in their work presented a recurrent neural architecture, capable of transforming variable-length input motion videos into a set of parameters describing a robot trajectory which is later encoded with DMP, where predictions can be made after receiving only a few frames, in addition, a simulation environment is utilized to expand the training database and to improve the generalization capability of the network, which is used for handover robotic tasks. Furthermore, Jaques et al. (2021) in their study introduced the Newtonian Variational Autoencoder (Newtonian VAE), a framework for learning latent dynamics. Drawing inspiration from Newton's second law, they define a linear dynamic system in a hidden space. This system is based on a rigid-body model with J DoFs. The Newtonian VAE framework enables the formation of a PD-controllable state space, which facilitates robust path following based on visual demonstrations using DMPs within the learned latent space.

4.5. Lifelong/Incremental learning

Lifelong (incremental) learning is a framework which provides continuous learning of tasks arriving sequentially (Chen and Liu, 2018; Fei et al., 2016; Thrun, 1996). The essential component of this framework is a database that maintains the knowledge acquired from previously learned tasks $\{TSK_1\}$, $\{TSK_2\}$, ..., $\{TSK_{N-1}\}$. Incremental learning starts from the task manager assigning a new task $\{TSK_N\}$ to a learning agent. In this case, the agent exploits the knowledge in the database as prior data for enhancing the generalization performance of its model on the new task. After the new task $\{TSK_N\}$ is learned, the database is updated with the knowledge obtained from learning $\{TSK_N\}$. In fact, the incremental learning framework provides an agent with three capabilities: (i) continuous learning, (ii) knowledge accumulation, and (iii) re-using previous knowledge for future learning enhancements. Figure 15 shows a general structure of DMP integrated with a lifelong framework.

Churchill and Fernando (2014) proposed a cognitive architecture capable of accumulating adaptations and skills over multiple tasks in a manner which allows recombination and reuse of task-specific competences. Lemme et al. (2014) segmented demonstrations based on geometric similarities and subsequently created a motion primitives library. The library is updated by removing unused skills and including new ones. Multiple demonstrations are used by Reinhart and Steil (2014, 2015) to build a parameterized skill memory that connects low-dimensional skill parameterization to motion primitive parameters. This low-dimensional embedding is then leveraged for efficient policy search. Piece-wise linear phase is used to improve incremental learning performance (Samant et al. 2016).

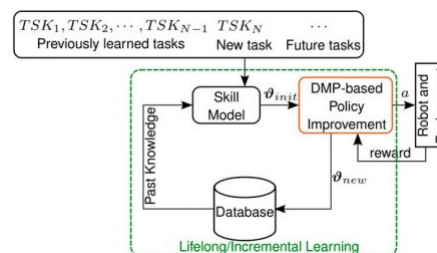


Figure 15. General framework of lifelong/incremental learning approach.

Duminy et al. (2017) designed a framework for learning which data collection strategy is most efficient for acquiring motor skills to achieve multiple outcomes and generalize over its experience to achieve new outcomes for cumulative learning.

A generative learning framework is proposed to augment the robot's knowledge-base with missing information at different levels of the cognitive architecture including symbolic planning as well as object and action properties (Wörgötter et al., 2015). Aforementioned approaches use DMPs to represent the learned skills and execute them on real robots.

Wang et al. (2016) proposed a modified formulation of DMPs called DMP+ which is capable of efficiently modifying learned trajectories by improving the usability of existing primitives and reducing user fatigue during IL. Later, DMP+ had been integrated into a dialogue system with speech and ontology to learn or re-learn a task using natural interaction modalities (Wu et al., 2018).

In the literature, it has been shown that incremental learning provides better generalization than the isolated learning approaches in terms of interpolation, extrapolation, and the speed of learning (Hazara and Kyrki, 2017). Hazara and Kyrki (2018) improved their Global Parametric Dynamic Movement Primitive (GPDMP) (Lundell et al., 2017) in order to construct, incrementally, a database of motion primitives, which aims to improve the generalization to new tasks. Furthermore, it has been transferred incrementally from simulation to the real world (Hazara and Kyrki, 2019). Moreover, authors endow incremental learning with a task manager, which is capable of selecting a new task by maximizing future learning while considering the current task performance (Hazara et al., 2019).

5. DMPs in application scenarios

We categorize the applications into several subsections based on different topics. We first separate the use of DMPs for robot interaction with the passive environment (e.g., tools, objects, and surfaces) and for interaction with an agent that involves co-manipulation (e.g., human and another robot). Additionally, we examine several other major application areas, such as human body augmentation/rehabilitation with exoskeletons, teleoperation, motion



Figure 16. Human operators teach the robot how to perform different tasks. Left scenarios use robots' gravity compensation mode to enable kinesthetic guiding, where a human operator guides the robot's tool center point along the desired trajectory in such a way that the desired task is successfully executed (Abu-Dakka et al., 2015a, 2018; Caccavale et al., 2019; Sloth et al., 2020). Right scenarios use a teleoperation system to demonstrate appropriate robot movements either through a haptic interface (Peternel et al., 2018a) or magnetic trackers (Abu-Dakka et al., 2015a).

analysis/recognition, high DoF robots, and autonomous driving and field robotics.

5.1. Robots in contact with passive environment

Most of the daily tasks that the robots perform involve some kind of physical interaction with the environment that requires control of forces or positions. Nevertheless, simultaneous control of force and position in the same axis is not possible (Stramigioli, 2001), and thus the control approaches have to make a compromise between prioritizing position control or force control (Schindlbeck and Haddadin, 2015). The key to such control is for the robot to learn appropriate force or position reference trajectories that can lead to the

desired task performance in interaction with the environment. Factors such as positional information, muscle stiffness of the human arm, and contact force with the environment are crucial in comprehending and generating robot manipulation behaviors that resemble those of humans. In Wang et al. (2021), both positional and contact force profiles are represented using DMPs to facilitate the transfer of human–robot skills.

5.1.1. Demonstration of interaction tasks. A common approach to teaching robot motion trajectories is kinesthetic guidance (Figure 16-Left), where the human operator holds the robot arm and shows the appropriate movements to be encoded by DMPs (Abu-Dakka et al., 2015a; Joshi et al., 2017; Kormushev et al., 2011; Papageorgiou et al., 2020; Schaal, 2006). Recently, the technology is protruding into high-risk fields such as invasive surgery, where high-dimensional fine human-like manipulation skills are being demonstrated (Su et al., 2021) and executed with robots (Ginesi et al., 2019; Su et al., 2020). In Kormushev et al. (2011), the human held the robot arm and used kinesthetic guidance to teach the position and orientation trajectories necessary to perform ironing and door-opening tasks. In the second stage, the corresponding forces and torques were recorded with a haptic device in a teleoperation setup. For setups where the robot arm is equipped with multiple force/torque sensors, the two demonstration steps with additional control policies can be combined into one (Montebelli et al., 2015; Steinmetz et al., 2015).

An alternative to learning force trajectories is to learn the impedance of the robot by learning the desired stiffness trajectories. The ability to change the impedance of the arm is crucial to simplify the physical interaction in unpredictable and unstructured environments (Burdet et al., 2001; Hogan, 1984). In Peternel et al. (2018a), teleoperation was used with a push-button interface to command the robot impedance, which was learned by DMPs that enabled the robot to perform various collaborative assembly tasks. For example, the learned position and stiffness DMPs were used to insert a peg in a groove to bind the two parts or to screw a bolt (Peternel et al., 2018a). A similar approach was used in Yang et al. (2018) to learn DMPs used for a vegetable cutting task.

While teleoperation-based methods are very effective to teach the robot DMPs for interaction tasks, it usually involves a complex and expensive system. The method in Abu-Dakka et al. (2018) enabled the robot to learn stiffness profiles through measurement of interaction force with the environment to perform a valve turning task. The method in Peternel

et al. (2017a) used human demonstration and EMG to learn stiffness DMPs from human muscle activity measurements in order to perform sawing and wiping (Figure 17) tasks.

Nevertheless, adaptation of a single trajectory is unlikely to generate an appropriate solution for more general cases, where the task execution needs to change significantly. After learning the initial DMP motion trajectories through kinesthetic guidance, the robot can then adapt them based on the measured force of interaction while performing the task. Pastor et al. (2011) introduced a method for real-time adaptation of demonstrated DMP trajectories depending on the measured sensory data. They developed an adaptive regulator for trajectory adaptation based on estimated and actual force data. Prakash et al. (2020) extended the real-time adaptation approach incorporating a fuzzy fractional-order sliding mode controller in order to efficiently and stably adapt the demonstrated DMP trajectory to fast movements, such as a ping pong swing. Recently, Cui et al. (2022) presented a method for coupling multiple DMPs for modeling robot tasks for transportation tasks of deformable objects.

Sutanto et al. (2018) presented a data-driven framework for learning a feedback model from demonstrations. They

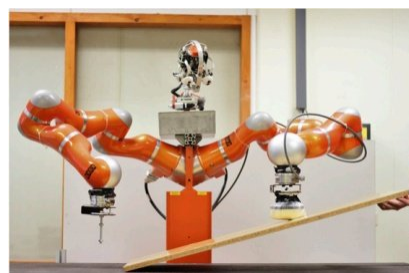


Figure 17. Using DMPs for adapting to changing surfaces (e.g., wiping task) (Kramberger et al., 2018).

used an RBF-NN to represent the feedback model for the movement primitive. Similar to this research, Gams et al. (2010) proposed a method for adaptation of demonstrated movements depending on the desired force, with which the robot should act on the environment. Thus, they ensured the adaptation of the learned movements to different surfaces. This approach was later expanded (Pastor et al., 2011) to provide the statistically most likely force–torque profile (Pastor et al., 2012), and furthermore, force–torque data was used for training a classifier (Straizys et al., 2020) in order to modulate the demonstrated trajectory for the use with delicate tasks such as tissue or fruit cutting.

Moving onward from policy learning, Do et al. (2014) presented an adaptation framework, where not only the desired adaptation force or trajectory but also the entire skill can be learned. They demonstrated the method with a wiping task under different environmental conditions.

5.1.2. Assembly tasks. Assembly presents one of the more challenging tasks to automate, where not only position trajectories but also task dynamics have to be taken into account. To deal with this challenge, various methods were proposed. Abu-Dakka et al. (2015a) proposed a method that can learn the orientation aspect of the complex physical interaction, like the peg-in-the-hole assembly tasks (Figure 18). The proposed method was integrated into an industrial assembly framework where the key challenge was to adapt to uncertainties presented by the assembly task (Abu-Dakka et al., 2014; Krüger et al., 2014).

Complex assembly tasks that are subject to change cannot be demonstrated and executed on the fly; therefore, adaptation methods are required for ensuring a successful execution. Nemec et al. (2020) used exception strategies, modeled as DMPs, for dealing with complex assembly cases. Sloth et al. (2020) presented an exception strategy framework, combining discrete and periodic DMP, coupled with force control to learn an assembly task under tight tolerances. Angelov et al. (2020) incorporated several different control policies by taking into account the

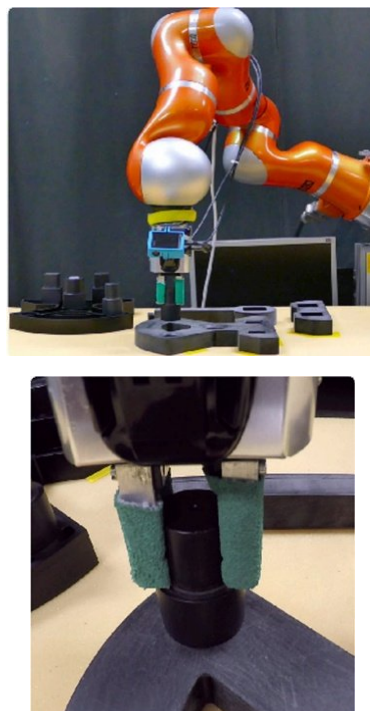


Figure 18. An example of using DMPs in assembly tasks (e.g., peg-in-the-hole) (Kramberger et al., 2016b).

dynamics and sequencing of the task. The approach uses DMPs to generate free motions and convolutional neural networks for the assembly.

In some cases, active exploration and autonomous database expansion can be used for learning assembly policies automatically. In Petric et al. (2015), the proposed algorithm can build and combine CMP motion knowledge from a database in an autonomous manner.

Complementary to assembly tasks, disassembly is also challenging by solely using the demonstrated trajectories. As described in Ijspeert et al. (2013), DMPs have a unique point attractor in the specified goal parameter of the movement, essentially repelling the idea of reversibility. Therefore, Nemec et al. (2018) proposed a framework, where the disassembly challenge was tackled by learning two separate

DMPs from a single demonstrated motion: one forward and one backward. Iturrate et al. (2019) took the idea further and reformulated the DMP phase system with a logistic differential equation to obtain two stable point attractors. This Reversible Dynamic Movement Primitive (RevDMP) approach provided a reversibility formulation of the dynamical system and demonstrated the effectiveness of the algorithm on a peg-in-hole assembly task.

5.1.3. Learning methods for contact adaptation. Desired force–torque profiles can be tracked using ILC (Gams et al., 2014, 2015b). In repetitive robotic tasks, iterative learning has been gaining increased popularity (Bristow et al., 2006) due to its effectiveness and robustness. However, in order to achieve effective results, a careful tuning of learning parameters is required. Norrlöf (1991) and Tayebi (2004) presented an adaptive learning approach for automated tuning of learning parameters.

Another approach is to use RL to adapt DMPs. For example, in Buchli et al. (2011b, 2011a), stiffness parameters were adjusted during the task execution by RL.

Alternatives to the feedback-based adaptation of DMPs and RL are scalability and generalization approaches. Matsubara et al. (2011) proposed an algorithm for the generation of new control policies from existing knowledge, thereby achieving an extended scalability of DMPs, while a mixture of motor primitives was used for generation of table tennis swings (Mülling et al., 2010). On the other hand, generalization of DMPs was combined with model predictive control by Krug and Dimitrov (2015) or applied to DMP coupling terms by Gams et al. (2015a), which were learned and later added to a demonstrated trajectory to generate new joint space trajectories.

Stulp et al. (2013) proposed to learn a function approximator with one regression in the full space of phase and task parameters, bypassing the need for two consecutive regressions. Forte et al. (2012) performed a comparison study of LWR and GPR for trajectory generalization. This work shows that higher accuracy can be achieved with LWR trajectory approximation. Koropouli et al. (2015) presented a generalization approach for force control policies. By learning both the policy and the policy difference data using LWR, they could estimate the policy at new inputs through superposition of the training data.

Deniša et al. (2016a) used GPR-based generalization over combined joint position trajectories and torque commands in the framework of CMPs. To showcase the versatility of the approach, Petric et al. (2018) applied it for robot-based assembly tasks. Finally, Kramberger et al. (2017) extended the approach to account for variations of the desired tasks, for example, assembly of similar objects. This enables the robot movements to be automatically generated with the use of LWR from a demonstrated database of successful task executions, which include kinematic and dynamic demonstrated trajectories encoded with DMPs. The newly obtained data is used to account for the changes in the work-space. Nevertheless, a major problem in statistical learning is how to efficiently deal with singularity-free representations of orientation trajectories. To resolve this issue, Kramberger et al. (2016a) proposed a formulation for Cartesian space DMPs where orientations are represented with unit quaternion.

5.2. Human–robot co-manipulation

While control of robot interaction with the passive environment can solve the majority of the tasks, in some cases the robot needs to interact with an active agent (e.g., human and another robot). Human–robot collaboration is becoming one of the key fields in robotics (Ajoudani et al., 2018). To perform a successful physical human–robot collaboration, the robot must be able to control complex movements in coordination with the human partner. In this direction, the ability to modulate the impedance is important to coordinate the physical interaction during human–robot com-manipulation of tools (Peternel et al., 2017b). DMPs offer an elegant solution to encode such coordinated dynamic movements.

In Peternel et al. (2014), the collaborative robot was taught online through teleoperation how to perform collaborative saving with a human co-worker. The impedance was commanded to the robot through the muscle activity measurement using EMG. DMPs were used to encode coordinated phase-dependent motion and impedance as demonstrated by the human teleoperator. Teaching through teleoperation is an effective way to convey the physical interaction skill to the collaborative robot; however, the setup can be expensive and is not widely available.

An intuitive alternative to teleoperation is for the robot to learn the skill directly through physical interaction with the human partner while they are collaborating. Numerous methods have focused on learning the synchronized motion between collaborative partners (Gams et al., 2014; Kulvicius et al., 2013; Lu et al., 2022; Peternel et al., 2018b; Prada et al., 2013; Sidiropoulos et al., 2019, 2021; Ugur and

Girgin, 2020; Umlauft et al., 2014; Wu et al., 2022; Zhou et al., 2016a). For example, in Kulvicius et al. (2013), the interactive movements were encoded with DMPs and adapted based on the measured force arising from the disagreements.

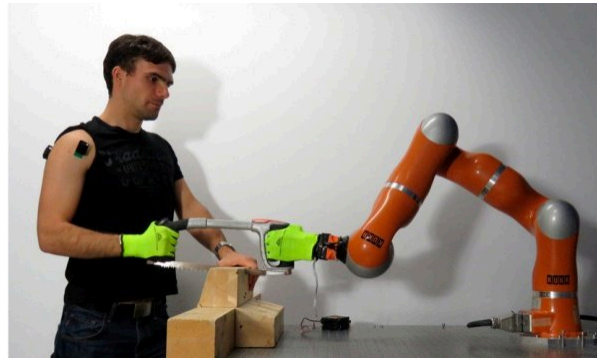


Figure 19. An example of using DMPs for collaborative human–robot sawing from Peternel et al. (2018b).

between agents during co-manipulation. Similarly, in Gams et al. (2014), the collaborative movements were encoded with DMPs and adapted using force feedback and ILC. The approach in Zhou et al. (2016a) combines two DMPs to encode the movements of each partner's arm, which are coupled in a leader-follower manner.

Besides adapting the collaborative movements, in Peternel et al. (2018b), the robot used DMPs to also learn the impedance online directly from the co-manipulation with the human (Figure 19). The robot started with a basic skill set that enabled it to collaborate with the human in a pure follower role. Through the collaborative task execution, the robot then learned the motion and impedance trajectories online and encoded them with DMPs. When the human became fatigued, the robot used the learned advanced skill to take over the majority of the task execution.

The method in Ben Amor et al. (2014) proposed an upgraded version of standard DMPs called Interaction Primitives that can account for a probabilistic nature of collaborative movements. Rather than having a single value of weights, the DMP includes weight distributions. This distribution enabled the robot to learn the inherent correlations of cooperative actions and infer the behavior of the human partner during the cooperation. Cui et al. (2016, 2019) used visual information to extract context-related parameters that augment the interaction primitives to increase the robustness during the task execution.

There are also other types of co-manipulation scenarios, such as within-hand bi-manipulation or human–robot object handover. For example, in Amadio et al. (2022), Gao et al. (2019), and Koene et al. (2014) DMPs were used to perform bi-manipulation, while in Abdelrahman et al. (2020), Iori et al. (2023), Lafleche et al. (2019), Prada et al. (2014), and Solak and Jamone (2019) DMPs were used for human–robot object handover.

When the environment is hazardous for human workers or when there are too many robots compared to the number of human workers, the obvious solution is to make robots collaborate between themselves. The method in Peternel and Ajoudani (2017) used DMPs to make novice robots learn from the expert robot through co-manipulation. Initially, the novice robot remained compliant to let the expert robot lead the task execution. In the first stage, the novice robot learned the reference motion through DMPs. In the second stage, it became stiff to perform the newly learned motion, while the expert robot initiated stiff/compliant phases expected in the collaborative task execution. Finally, the novice robot then learned in which phases of the task to increase or decrease the impedance and encoded this impedance behavior with DMPs.

5.3. Human assistance, augmentation, and rehabilitation

The most common type of co-manipulation is the classic human–robot collaboration, where a human and a robotic agent are physically performing industrial or daily tasks. Another type of co-manipulation occurs when a human is using a wearable robot such as an exoskeleton. There are different types of functions that the exoskeleton can be used for. One function is augmentation where the current human motion is amplified to augment existing (healthy) human capabilities such as in tasks involving heavy loads. When human capabilities are impaired, the exoskeleton has to act in an assistance function. If human capabilities are impaired to a larger degree, the

exoskeleton can be employed in a rehabilitation function to perform physical therapy. In the augmentation function, DMPs can be employed on the robot to offload a hard and/or repetitive motion of healthy human workers, while in the assistance and rehabilitation functions, they can be used to assist impaired humans in their daily tasks or perform repetitive physical therapy on patients that would lead to recovery.

Besides the type of exoskeleton function, another important aspect is the shared load between the exoskeleton and the human during physical human–robot interaction. For example, in the case of highly impaired patients in

early-stage rehabilitation, DMPs generated by the exoskeleton may take almost all the load of the movement and assume the role of the leader to perform passive physiotherapy. As the recovery progresses and more active exercise is preferred, the majority of the load can be shifted to the human who leads the movements, while the DMP system can act in a support capacity as a follower. When full recovery is not possible, DMP can provide assistance in daily tasks where the load can be partially shared. In case of augmentation of existing (healthy) human capabilities, the DMP system can adapt to human movement and add extra power on the top of the human effort that is needed to perform a specific heavy-load task. In some cases, the exoskeleton can learn DMP from human movements to take over a repetitive action completely.

The methods in Lauretti et al. (2017, 2018) obtained DMPs offline by learning by demonstration, which were then used by an arm exoskeleton to support human movements. In Peternel et al. (2016), the control method employed DMPs to interactively adapt the joint torques required to perform the arm exoskeleton assistance and compensate all the underlying dynamics for periodic movements (Figure 21-Left). The phase-dependent torque trajectory was updated online in real-time in order to minimize the muscle activity feedback measured by EMG. A similar adaptation capability was achieved for discrete exoskeleton movements by the DMP-based method proposed in Lanotte et al. (2021). In Petrič et al. (2016), the robot encoded the assistive motion with DMPs and then adapted it by taking into account aspects of human motor control through the Fitts' law. Li et al. (2021a) introduced a hierarchical control strategy that enables the learning of human–machine cooperative operations by integrating DMPs and GMM. The proposed approach utilizes well-defined primitives, which offer comprehensive mathematical formulations and rigorous analyses of stability and convergence for control methods based on DMPs.

Gait-related assistance and rehabilitation with exoskeletons is a very common application of DMPs, and there are numerous examples (Abu-Dakka et al., 2015; Amatya et al., 2020; Escarabajal et al., 2023; Huang et al., 2016a; Hwang et al., 2019, 2021; Schaal, 2006; Yuan et al., 2020; Zou et al., 2021). In Abu-Dakka et al., 2015, 2020), a parallel robot was used for ankle rehabilitation, where the movements were generated by DMPs (Figure 20). A similar parallel robot was applied in Escarabajal et al. (2023) for knee rehabilitation, where RevDMPs was used to enable a patient to reverse the movement in order to maintain their own desired pace. In Huang et al. (2016a), DMPs were used to learn the gait motion trajectories for a lower body exoskeleton. This approach was then extended with an RL method to adapt a force coupling term (similar to earlier approaches presented in Section 3.3.2) to enable online adaptation of motion trajectories (Huang et al., 2016b). In Luo et al. (2022), DMPs were adapted to the different starting and ending locations of the foot in the swing phase of gait. The method in Xu et al. (2023) used DMPs for leg



Figure 20. An example of using DMPs for teaching passive exercises for ankle rehabilitation (Abu-Dakka et al., 2015, 2020).

exoskeleton movements in a mirror therapy concept where the motion of the healthy limb is transferred to the impaired limb. Hong et al. (2023) used DMPs to plan obstacle-avoidance leg movements during walking with a prosthesis.

Besides normal gait, DMPs were also applied for stair-ascend (Xu et al., 2020) and sit-to-stand (Kamali et al., 2016) assistive movements of lower body exoskeletons. In Joshi et al. (2019), a robotic arm was used to assist humans with putting the clothes on their body, where the movements were generated by DMPs. Ding et al. (2022) developed a framework for assistance of older adults combining DMPs' admittance control for mobility assistance and manipulation support. The framework was implemented on a mobile platform with a robotic arm utilized for LfD.

Besides assistive body movement and rehabilitation, DMPs were also applied for relaxation purposes. For example, in Li et al. (2020), a robotic arm provided massage movement through DMPs.

5.4. Teleoperation

Teleoperation is one of the major fields of robotics and enables a human to have a direct and real-time control over a (remote) robot. Typically, the control is done through interfaces that can capture the human commands to be sent to the robot and that can provide haptic feedback from the robot. While teleoperation focuses on giving the human operator a full or shared control over the robot, DMPs are used to encode autonomous robot behaviors. Therefore, here we mostly examine cases where teleoperation is used to teach the robot new autonomous behavior encoded by DMPs.

In Kormushev et al. (2011), a combination of kinesthetic teaching and teleoperation was employed to form the DMP-based robot skill for ironing. After the motion trajectories were learned through kinesthetic guidance, the corresponding forces were recorded by using a haptic device and a teleoperation system. In Peternel et al. (2014),

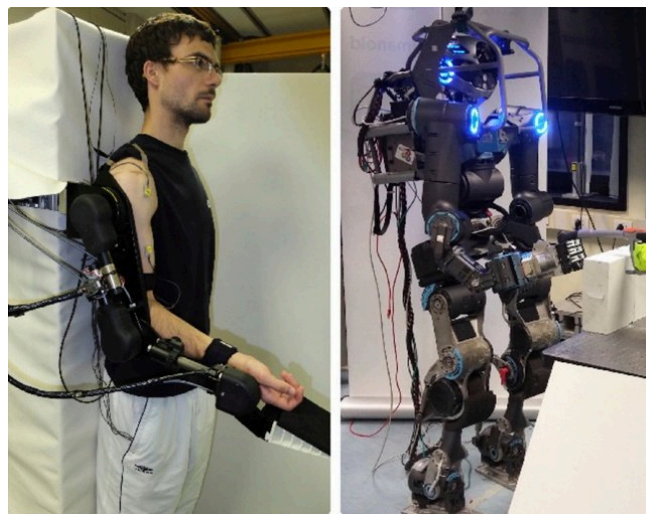


Figure 21. Left photo shows arm exoskeleton application from Peternel et al. (2016). The right photo shows high-DoF humanoid robot Walk-man (Tsagarakis et al., 2017) performing sawing in Peternel and Ajoudani (2017).

teleoperation was used to teach the robot how to physically collaborate with another human. Since there was no haptic feedback, the teleoperation setup was unilateral, but the human was able to teach also the impedance of the robot in addition to motion. The former was commanded by muscle activity measurement through EMG, while the latter was commanded by the movement of the human operator's arm as measured by an optical motion capture system. Similar teleoperation-based DMP learning approaches were used in Lentini et al. (2020) and Yang et al. (2018).

In Peternel et al. (2018a), the human operator taught the robot through teleoperation how to perform autonomous assembly actions (Figure 16-right). DMPs were used to encode the commanded impedance and motion; however, a more practical push-button impedance command interface was employed instead of EMG. More importantly, the teleoperation setup was bilateral, and the haptic interface provided the human operator with feedback about the forces the robot felt. Learned DMPs through teleoperation can also be generalized to improve adaptability (Yang et al., 2018). In Luo et al. (2023), DMPs were employed to encode skilled operator movements, which were then used for training novice operators. The method in Zhang et al. (2023) used DMPs to partially automate the surgical process of appendectomy.

A real robot is not always necessary to acquire new skills. In Beik-Mohammadi et al. (2020), the robot and the environment were simulated and the human operator used a virtual reality system. A combination of DMPs and RL was used to form an adaptive skill. The scenario proposed in Abu-Dakka et al. (2015a) was teleoperation in its basis; however, the human demonstrator did not just pretend that he/she is embodied in the robot, but the robot task environment was cloned at the human side (Figure 16-right). This removed the need for force feedback and a haptic device, since the human felt the real environment on his/her side, while the motion was captured by a non-contact-based sensory system (i.e., magnetic trackers) and then mirrored on the robot.

Multiple demonstrations through teleoperation can be inconsistent, especially if done in a multi-agent shared-control setting. The method proposed in Pervez et al. (2019) can synchronize inconsistent demonstrations through shared-control teleoperation and encode them with DMPs. Maeda (2022) investigated the possibility of using DMPs to implicitly blend human and robot policies without requiring the design of task-specific arbitration functions or the need to provide multiple (possibly inconsistent) demonstrations.

5.5. High DoF robots

DMPs provide an elegant and fast way to deal with systems with high-dimensional space by sharing one canonical system (3) among all DoFs and maintain only a separate set of transformation systems. By high-dimensional space, we are referring to systems with 10 or more DoFs (i.e., Walkman humanoid robot in Figure 21-right). In this section, we will quickly mention some of the potential works with a high number of DoFs.

Ijspeert et al. (2002b, 2002a) used DMPs in an IL framework to learn tennis forehand, a tennis backhand, and rhythmic drumming using 30-DoFs humanoid robot. Pastor et al. (2009) used DMPs to encode a 10-DoFs exoskeleton

robot arm. Luo et al. (2015) integrated DMPs with stochastic policy gradient RL and GPR in order to design an online adaptive push recovery control strategy. The approach had been applied to PKU-HR5 humanoid robot with 20-DoFs. André et al. (2015, 2016) implemented a predictive model of sensor traces that enable early failure detection for humanoids based on an associative skill memory to periodic movements and DMPs. They applied their algorithm on DARwIn-OP with 20-DoFs in simulation. Pfeiffer and Angulo (2015) represented gestures by applying DMPs on REEM robotic platform with 23-DoFs. Nah et al. (2020) proposed an approach to optimize DMP parameters in order to deal with the complexity of high DoF system like a whip. They tested their approach in simulation for 10-, 15-, 20-, and 25-DoF systems. In order to reduce the number of required rollouts for adaptation to new task conditions, Queißer and Steil (2018) used CMA-ES to optimize DMP parameters. In addition, they introduced a hybrid optimization method that combines a fast coarse optimization on a manifold of policy parameters with a fine-grained parameter search in the unrestricted space of actions. The approach was successfully illustrated in simulation using a 10-DoF robot arm. Liu et al. (2020) proposed DMP-based trajectory generation to enable a full-body humanoid robot with 10-DoFs (for the two legs) to realize adaptive walking. Liang et al. (2021) developed an efficient approach to enable service robots with 26 DoFs with the skill of performing sign language motions.

Travers et al. (2016, 2018) proposed a framework that integrates DMP with Gaussian-shaped spatial activation windows in order to plan the motion for high DoF robotic systems (e.g., snake-like robot) in complex environments (with obstacles) by linking low-level controllers to high-level planners. DMPs can also be applied to soft continuum robots where the separation between DoFs is more blurred (Salem et al., 2023).

5.6. Motion analysis and recognition

DMPs tend to fit topologically similar trajectories with similar shape parameters $\{w_i\}$ (Ijspeert et al., 2013). This behavior, due to the temporal and spatial invariance of DMPs, makes the shape parameters a useful descriptor to recognize similar motions. Indeed, Strachan et al. (2004) have shown that the shape parameters computed for 5 repetitions of 4 classes of discrete hand gestures—measured with a 3 DoF accelerometer—are linearly separable, that is, easy to classify. Lantz and Murray-Smith (2004) draw similar conclusions for 10 classes of periodic hand gestures. Xu et al. (2005) used the correlation between the parameter vectors of two DMPs to measure the similarity between the original motion and recognize gait patterns. Similarly, Ijspeert et al. (2013) used the correlation between parameter vectors to recognize the 26 letters of the Graffiti alphabet.

The shape parameters $\mathbf{w}_i$ are also suitable to fit more sophisticated classifiers like support vector machines. This strategy was used to successfully classify gestures observed with a monocular (Liu et al., 2014) or a binocular (Wang and Payandeh, 2015) camera. Instead of considering a fixed number of basis functions (number of shape parameters), Zhang et al. (2017) used fast dynamic time warping (Salvador and Chan, 2007) to align parameter vectors of different lengths and then used K-nearest neighbors to classify different motions.

Motion recognition can also be used to determine whether the robot is correctly executing a task by comparing sensed data with a movement template. In this respect, André et al. (2016) used an associative skill memory, like the one in Pastor et al. (2011), as a predictive model of sensor traces that enables early failure detection. In this work, DMPs were used to compactly encode the associative skill memory and speed up the failure detection. Described approaches demonstrate that DMPs are a valuable option for gesture recognition especially for systems with limited computational power. Liang et al. (2021) presented a solution to the motion retargeting problem for generating dual-arm sign language motions. Their approach involves an offline-constrained optimization technique that minimizes the deviation from trajectories generated by DMPs, which encodes the human demonstrations. It should be noted that their approach was exclusively applied and tested in an offline setting.

Humans tend to perform the same task in slightly different manners. Sometimes, differences in the execution style contain useful information to adapt the motion to different executive contexts. This is the case, for instance, of a reaching motion with and without an obstacle on the way. To capture the execution style, Matsubara et al. (2010) augmented the forcing term of the DMP with a style parameter learned from multiple demonstrations. At run time, different style parameters can be used to smoothly interpolate between demonstrated behaviors. Zhao et al. (2014) not only employed movements with different styles but also learned a smooth mapping between style parameters and goal to improve the generalization.

When humans provide seamless demonstrations, DMPs can be used for online segmentation and recognition. To this end, Meier et al. (2011) assumed that a library of DMPs is given and used it to recognize motion segments during a task demonstration. Instead of using exemplar templates for each class of primitives, Chang and Kulić (2013) segmented a video stream using motion to non-motion transitions, fitted DMPs on segmented data, and performed clustering to group similar motion segments in an unsupervised fashion. Song et al. (2020) performed unsupervised trajectory segmentation using the concept of key points, that is, shared features across different task demonstrations. Mandery et al. (2016) segmented whole-body motions by detecting contacts with the environment and used them to build a probabilistic language model where words represent the poses and sentence sequences of poses. The learned language model was used to plan whole-body motion trajectories executed by joining multiple DMPs (see Section 3.2). Kordia

and Melo (2021) introduced a method for recognizing an observed trajectory from a library of pre-learned motions and predicting its target position. They leveraged distinctive features of the observed trajectory to aid in the recognition and utilized DMPs for representing the movement.

DMPs have been developed as a computational model of the neurobiological motor primitives (Schaal et al., 2007). Experimental findings from neurophysiology related to the spinal force fields in frogs have inspired the modification of DMP formulation in Hoffmann et al. (2009). As discussed in Section 3.1.1, this multidimensional representation overcomes limitations of classical DMPs like trajectory overshooting and dependence of the trajectory from the reference frame used to describe the motion. Hoffmann et al. (2009) also derived a collision avoidance strategy for DMPs, inspired by the way humans avoid collisions during arm motion. DeWolf et al. (2016) investigated the human ability to cope with changes in the arm dynamics and kinematic structure during motion control. They proposed a spiking neuron model of the motor control system that uses DMPs to implement the preparation and planning functionalities of the premotor cortex. The effects of changes in the robot's dynamic parameters on the tracking performance of a DMP trajectory were studied in Kuppuswamy and Alessandro (2011). Their findings suggest that the change in the body parameters should be explicitly considered in the DMP learning process. Hotson et al. (2016) augmented a brain-machine interface that captures neural signals with a DMP model of the endpoint trajectories executed by a non-human primate. The system was used to decode real trajectories from a primate manipulating four different objects.

5.7. Autonomous driving and field robotics

DMPs can be utilized in various autonomous non-stationary fields of robotics. Perk and Slotine (2006) utilized DMPs for defining flight paths and obstacle avoidance for Unmanned Aerial Vehicles (UAVs), where the trajectories were generated based on the joystick movements controlling the throttle of the UAV motors. Later, Fang et al. (2014) extended the approach to encode user-demonstrated UAV data, extracting and encoding the rhythmic and linear segments of the flight trajectory, and combining them into a flight control skill. Furthermore, Tomić et al. (2014) formulated the UAV movements as an optimal control problem. The output of the optimal control solver was encoded with DMPs, enabling them to generalize and apply in-flight modifications to the UAV flight trajectories in real-time. Similarly, Lee et al. (2018) and Kim et al. (2018a) presented a framework for UAV cooperative areal manipulation tasks, based on an adaptive controller which adapts the movement of the UAV in relation to the mass and inertial properties of the payload. In addition, DMPs were incorporated in the control scheme to modify the flight trajectories and avoid obstacles on the fly. The approach was later extended to incorporate path optimization, where DMPs play a significant role in real-time obstacle avoidance (Lee et al., 2020).

As mentioned before, DMPs represent a versatile movement representation, which can be implemented in various tasks and scenarios. One of the recent applications in this field is also Autonomous Underwater Vehicles (AUVs). Carrera et al. (2015) integrated the DMPs in a learning by demonstration scenario for an AUV. The demonstrated data consisted of the manipulator and vehicle sensory outputs, which were efficiently used to demonstrate an underwater valve-turning task.

6. Discussion

DMPs are also represented in the autonomous driving domain. In the recent work of Wang et al. (2018, 2019), the authors propose a framework which decomposes the complex driving data into a more elementary composition of driving skills represented as motion primitives. In the proposed framework, DMPs are utilized to represent the driver's trajectory with acceptable accuracy and can be generalized to different situations.

This section provides guidelines to choose, among the several discussed in this work, the most appropriate approach for a given application. A useful criterion to decide whether to use a particular approach is the availability of code that greatly simplifies the implementation. We have searched for open-source DMP implementations and listed them in a Git repository (see Section 6.2). To further contribute to the community, we have also released the implementations listed in Table 4. This section ends with a discussion on the limitations inherent to the DMP formulation, the open issues, and the possible research directions. These are summarized in Table 5.

6.1. Guidelines for different applications

Previous sections present different DMP formulations and extensions together with possible application scenarios. As usual, there is not a single formulation that serves all the scopes and purposes, and the suitable approach to use depends on the goal to achieve and the conditions of application. For this reason, we present some guidelines to guide the user in the process of selecting the formulation to use.

6.1.1. Discrete versus periodic. For a task with distinct starting and ending points, discrete DMPs are a logical option to encode the movement trajectories between them. Examples of these tasks include reaching and pick-and-place (Caccavale et al., 2019; Deniša et al., 2016a; Forte et al., 2012; Stulp et al., 2009), specific actions of assembly (Krüger et al., 2014; Abu-Dakka et al., 2014; Nemec et al., 2020; Angelov et al., 2020), and cutting (Straizys et al., 2020; Yang et al., 2018).

When the starting and ending points coincide, periodic DMPs are the logical option, since the encoded movements can be repeated over and over again. Good examples of their application are repetitive tasks such as locomotion (Nakanishi et al., 2004; Rückert and d'Avella, 2013; M. Wensing and Slotine, 2017), human body augmentation/

Table 5. A summary of DMP features and limitations that have been solved (✓) or partially solved (□).

Limitation	Related work	Status
Via-points	Ning et al. (2011, 2012), Weitschat and Aschemann (2018), Saveriano et al. (2019), and Zhou et al. (2019)	✓
Start-point	Hoffmann et al. (2009), Ijspeert et al. (2013), Weitschat et al. (2013), and Dragan et al. (2015)	✓
Goal-point	Ijspeert et al. (2013), Weitschat et al. (2013), Abu-Dakka and Kyrki (2020), Dragan et al. (2015), and Weitschat and Aschemann (2018)	✓
Obstacle avoidance	Park et al. (2008), Hoffmann et al. (2009), Tan et al. (2011), Kim et al. (2015), and Rai et al. (2017)	✓
Geometry-constrained data	Pastor et al. (2009), Abu-Dakka et al. (2015a), Ude et al. (2014), Saveriano et al. (2019), and Abu-Dakka and Kyrki (2020)	$\setminus$ ($I^{\wedge}\{5\}$)
Probabilistic	Ben Amor et al. (2014)	I
Extrapolation	Pervez and Lee (2018) and Zhou et al. (2019)	I
High-dim input	Pervez et al. (2017a) and Pahić et al. (2020)	I
Closed-loop	Peternel et al. (2016) and Kramberger et al. (2018)	I
Multi-attractor	Nemec et al. (2018) and Iturrate et al. (2019)	I

rehabilitation (Petenerl et al., 2016), wiping a surface (Gams et al., 2016; Kramberger et al., 2018; Petenerl et al., 2017a), and sawing (Petenerl et al., 2018b). Nevertheless, even typically non-repetitive tasks that are executed just once every now and then can still be encoded with periodic DMPs when the starting and ending points coincide (Petenerl et al., 2018a).

There are cases where it is not possible to clearly distinguish if the motion is periodic or discrete. For instance, Ernesti et al. (2012) have shown that the first step in a gait of a humanoid robot is a transient toward a periodic motion. Their representation is a good candidate to encode transients converging to limit cycle trajectories. Finally, in some cases like in complex assembly, the task requires a combination of discrete and periodic DMPs (Sloth et al., 2020).

6.1.2. Space representation. The original formulations of DMPs were and are still successfully applied to multidimensional independent data with each DoF $\in \mathbb{R}$ (Sections 2.1.1 and 2.2.1). These data can be joint or Cartesian positions, forces, torques, etc., where every DoF of the data can be evolved independently from the rest. However, such formulation is not enough to successfully encode data with specific geometry constraints without pre- and/or post-processing the data. Examples of such data are i) orientation, where data are tight up by additional constraints (i.e., the orthogonality in case of rotation matrix representation or the unit norm of the quaternion representation); ii) full stiffness/damping matrices and manipulability matrices are encapsulated in SPD matrices.

In many early works, orientation trajectories were learned and adapted without considering their geometry constraints (Pastor et al., 2009), leading to improper orientation and hence requiring an additional re-normalization. In a different example, Umlauf et al. (2017) used eigendecomposition for impedance adaptation.

In order to comply with such geometry constraints, researchers provided a new formulation of DMPs that ensures proper unit quaternions or rotation matrices over the course of orientation adaptation (Abu-Dakka et al., 2015a; Koutras and Doulgeri, 2020a; Saveriano et al., 2019; Ude et al., 2014 and proper SPD matrices over the course of the adaptation of SPD profiles (e.g., stiffness or manipulability ellipsoids) (Abu-Dakka and Kyrki, 2020). We believe that using these geometry-aware DMPs is preferable to encode data with underlying geometry constraints.

6.1.3. Weights learning method. DMPs represent motion trajectories as stable dynamical systems with learnable weights that define the shape of the motion. In the LfD paradigm, DMP weights are usually learned in a supervised manner using human demonstrations. The procedure used to transform human demonstrations into training data for the DMP forcing term is highlighted in Section 2.1.1.1. The number of weights, which corresponds to the number of RBFs used to approximate the forcing term, is a hyperparameter that is typically provided by the user. As practical tuning guidelines, one has to consider that the number of RBFs increases with the length of the trajectory and with its complexity, which depends on changes in concavity and frequency and magnitude of picks. Given the training data, and the number of RBFs, different techniques can be used to fit the weights.

LWR is widely used when the forcing term is a combination of RBFs as in (4). However, in the literature one can use RBF-NN as in Si et al. (2021) or if multiple demonstrations are given, one can exploit GMM/GMR as in Li et al. (2021b) and Pervez et al. (2017a) or GPR as in Fanger et al. (2016) to represent the forcing term and use expectation-maximization to fit the (hyper-)parameters. Deep NNs, typically trained via back-propagation, seem an appealing possibility to map input images into forcing terms (Pervez et al., 2017b), mimicking the human perception-action loop. Although appealing, the possibility of exploiting deep learning techniques as motion primitives requires further investigations.

In real applications, there can be a misplacement between the DMP trajectory and the robot motion. Typical examples include assembly or other tasks that require physical interaction with the environment (see Section 5.1). In this

situation, the DMP motion can be incrementally adjusted to improve the robot's performance. ILC arises as an interesting approach to iteratively update the DMP weights as it ensures a rapid convergence to the desired performance (Abu-Dakka et al., 2015a; Gams et al., 2014; Kramberger et al., 2018). However, ILC assumes that a target behavior to reproduce is given. When the target behavior cannot be easily specified and the robot performance is not satisfactory, RL solutions have to be adopted. As detailed in Section 4.3, DMPs are effective control policies and, combined with policy search algorithms like π^2 or PoWER, are able to solve complex and highly dynamic tasks.

6.1.4. Online adaptation. Performing robotic tasks in the real world requires adaptation capabilities. When adaptation of DMPs based on some feedback is required, one of the extension methods should be applied. For example, to change the existing movement based a detected obstacle, the method in Gams et al. (2016), Hoffmann et al. (2009), Park et al. (2008), and Tan et al. (2011) can be used (see Section 3.3.1). If it is necessary to adaptively learn the movement dynamics based on real-time effort feedback, the method in Peternel et al. (2016) can be employed (see Section 3.3).

Furthermore, for industrial tasks, such as assembly or polishing, adaptation strategies combining force control with demonstrated trajectories can be applied (Abu-Dakka et al., 2015a; Gams et al., 2010; Kramberger et al., 2016), ensuring the system will follow the predefined trajectory and adapt to the environmental uncertainties. For online adaptation DMPs can be used as a trajectory generator, which output represents an input to the force control algorithm, on the other hand, force feedback can directly be incorporated as a coupling term in the DMP formulation (see Section 3.3.2), eliminating the need for an additional force controller. A similar approach can also be utilized for velocity-based adaptation of the movements (see Section 3.3.4).

6.1.5. Impedance versus force. In physical interaction tasks, DMPs can be used to either learn force or impedance (Petenerel et al., 2017a). If the task requires position control, then the impedance should be learned with DMPs in combination with the reference position. If the task requires to control a specific force, for example, pushing on a surface during the wiping and drilling, either force or impedance is feasible. However, if safety is the most critical aspect, the DMPs should be used to learn impedance control so that the robot can be made soft.

Furthermore, to overcome any undesirable movements, the control policy can be augmented with a tank-based passivity approach (Shahriari et al., 2017). This approach monitors the energy flow between the modeled sub-systems, for example, DMP trajectory generation, impedance control, and environment. In an event of an energy violation, the system will first try to passively compensate for the violation and subsequently, if the violation cannot be compensated, for example, the energy tank is depleted, stop the system. In cases, where the task characteristics are not fully known, a learning policy can be added on the top of the passivity approach (Kramberger et al., 2018) in order to learn the overall energy requirements for the task.

6.2. Resources and codes

The availability of code and datasets is useful to speed up the setup of novel applications without the need of re-implementing a promising approach from scratch. We have searched for available DMP implementation and found out that several researchers published their DMP codes in various open-source repositories. We decided to list the available implementations on the Git repository that accompanies this paper (<https://gitlab.com/dmp-codes-collection/third-party-dmp>). For each implementation, we mention the type of DMP, the author, the URL to download the code, and the used programming language. We also provide a short description of the key features.

Apart from listing existing approaches, the Git repository that accompanies this paper contains an implementation that we decided to release to the community. The list of provided implementations is given in Table 4.

6.3. Limitations and open issues

As any motion primitive representation, DMPs have strengths but also inherent limitations. The advantages of the DMPs have been widely discussed in previous sections. Here, we present the main limitations of the DMPs and discuss open issues that require further investigation. A summary of these limitations is presented in Table 5.

6.3.1. Implicit time dependency. The phase variable used to suppress the nonlinear forcing term and ensure convergence to a given goal introduces an implicit time dependency in the DMP formulation. The reason for representing the time dependency implicitly as a dynamical system is that such a phase variable can be conveniently manipulated. For example, in Section 2.1.1.2, we have seen how to manipulate the phase variable to slow down (or even stop) the execution. A drawback of the time dependency is that the shape of the DMP motion is significantly affected by the time evolution of the phase variable. If the phase vanishes too early, the last part of the trajectory is executed with a linear dynamics converging to the goal. If the phase lasts too long, the trajectory may overshoot and fail to reach the goal within the desired time. In both cases, the DMP motion may significantly deviate from the demonstration. A properly designed phase-stopping mechanism can remedy the issue, but the proper phase-stopping to adopt depends on the specific application.

In order to overcome this limitation, several authors focused on learning stable and time-independent (or autonomous) dynamical systems from demonstrations. A globally stable and autonomous system generates a vector field that converges to the given goal from any initial state. Without the need for a phase variable, the generated motion

depends only on the current state of the system. Notable approaches to learn stable and autonomous systems exploit Lyapunov theory (Khansari-Zadeh and Billard, 2011, 2014), contraction theory (Blocher et al., 2017; Ravichandar and Dani, 2015), diffeomorphic transformations (Perrin and Schlehuber-Caissier, 2016; Neumann and Steil, 2015), and passivity considerations (Kronander and Billard, 2015). These approaches have been effectively used to learn complex movements from demonstrations.

In general, autonomous systems have the potential to represent much more complex movements than DMPs. For example, autonomous systems can encode different motions in different regions of the state-space. In this respect, DMPs can only generate a stereotypical trajectory connecting the start to the goal, regardless of where the initial state is placed in the state-space. However, the stereotypical motion generation is also an advantage of DMPs since it makes it easier to predict the generated motion in regions of the state-space poorly covered by training data. On the contrary, it is hard to predict how an autonomous system generalizes where only a few or no training data are available. DMPs are known to scale well in high-dimensional spaces since the learned forcing term always depends on a shared, scalar phase variable. Autonomous systems perform learning directly on the high-dimensional state-space, which poses numerical challenges and requires much more training data. In synthesis, each representation has its own advantages and disadvantages and the choice between time-dependent and autonomous motion primitives depends on the specific application.

6.3.2. Stochastic information. Representing the demonstrated motion as a probability distribution has several advantages. For example, in a probabilistic framework, the generalization to new a goal (or a via-point) is achieved using conditioning on the new goal (via-point), while the covariance computed from the probability distribution can represent couplings between different DoFs (Paraschos et al., 2013). As a matter of fact, classical DMPs are deterministic and lack stochastic information on the modelled motion.

Ben Amor et al. (2014) proposed an approach to estimate the predictive distribution $\mathcal{P}(\mathbf{w} | \mathbf{y}_{1:\mathcal{T}})$ that relates the DMP weights $\mathbf{w}$ and a partial trajectory $\mathbf{y}_{1:\mathcal{T}}$ observed for $\mathcal{T}$ time instants. $\mathcal{P}(\mathbf{w} | \mathbf{y}_{1:\mathcal{T}})$ is used to estimate the most likely weights given a partial movement and to reconstruct the missing part of the trajectory. However, a full probabilistic characterization of DMPs is still missing.

The ProMP framework (Paraschos et al., 2013) proposed an alternative movement primitive representation that contains information about the variability across different demonstrations as well as different DoFs in the form of a covariance matrix. This enables to explicitly encode the couplings between different directions and to increase the generalization by conditioning on a desired goal, via-point, or intermediate velocity. The covariance computed by ProMPs represents the variability and the correlation in the demonstrations. In other representations, like GPR, the covariance is a measure of the model uncertainty due to the lack of training data. An attempt to unify ProMP

and DMP formulations was made in Li et al. (2023). Kernelized Movement Primitives (KMPs) (Huang et al., 2019; Silvério et al., 2019) offer the possibility of modelling variability, correlation, and uncertainty in the same framework. However, KMP's computational cost can be elevated compared to DMP in longer trajectories due to the computation of the inverse of the kernel matrix.

6.3.3. Closed-loop implementation and issues. A vast majority of methods employ DMPs only as a reference trajectory generator for the closed-loop controller, which then actually executes it. However, the DMPs can also be used as a part of the closed-loop controller itself where the sensor measurements, for example, forces and torques, are used as a coupling term in the DMP for changing its behavior. In other words, in the open-loop case, the DMP serves as the plan and does not change online during the execution (perhaps iteratively after each execution), while in the closed-loop case, DMP serves as the action generator and changes online during the execution. Until now, only a few methods explored the closed-loop concept. For example, in Peternel et al. (2016), the DMPs are directly torque generators for exoskeleton actuators in the control loop, which is closed by feedback from the human user's muscle activity. Nevertheless, in such scenarios, the closed-loop stability and passivity become crucial considerations that have to be addressed and resolved before the widespread application (Kramberger et al., 2018).

6.3.4. Coping with high-dimensional inputs. One of the main limitations of DMP is that it encodes human and robot trajectories explicitly with the time (i.e., 1-D input) which may lead to synchronization issues since human motions in the new evaluations could be significantly different (e.g., faster/slower velocity) from the demonstrated ones. In order to avoid synchronization problem, Ben Amor et al. (2014) designed a time-alignment strategy, while Pervez et al. (2017a) estimated the phase signal during the training using expectation-maximization (Bishop, 2006).

As the DMP models trajectories using basis functions, this works effectively when learning time-driven trajectories (i.e., 1-D input). However, when demonstrations comprise high-dimensional inputs, specifying the center vectors and widths of basis functions becomes quite cumbersome. Specifically, as discussed in Bishop (2006), the number of basis functions often increases exponentially when the dimension of inputs increases. To alleviate this limitation, some approaches investigated modern deep-learning techniques. Pahič et al. (2020) used a deep NN to synthesize DMP weights from an input image. The classical DMP formulation is then used to generate motion trajectories. Pervez et al. (2017b) used a CNN (LeCun et al., 2015) to predict 2-D task parameters (e.g., the position of a target) from an input image and a fully

connected NN to retrieve the forcing term from the 2-D parameters and the phase variable. The CNN and the fully connected NN are trained in two separate stages. The approach is promising, but the separate training of the two networks increases the pre-processing and complicates the learning process.

Alternative approaches in the literature, such as GMM/GMR (Calinon, 2016), Task-Parameterized GMM (TP-GMM) (Calinon, 2016), and KMP (Huang et al., 2019, 2021), can be directly applied for learning demonstrations comprising high-dimensional inputs.

6.3.5. Multi-attractor systems. The well-known second-order dynamic properties of the DMPs strive toward a single attractor system (Ijspeert et al., 2002a). The properties, for example, convergence and modulation of the motion, are well studied and implementations can be found in many research papers. Because of the second-order dynamics, the system becomes unstable if, for example, the motion is reversed during the execution. In the past years, two main approaches describing the reversibility problem have been introduced. In the first approach (Nemec et al., 2018), reversibility is considered as learning two separate primitives, one for each direction of the motion. The approach is promising, but does not reflect true reversibility, because it uses one attractor point for each primitive.

On the other hand, Iturrate et al. (2019) introduced an alternative formulation with two stable attractor systems. The first attractor is defined at the starting point $y_{\{0\}}$ of the trajectory and the subsequent one at the goal g , and the dynamical system between them guarantees a stable convergence depending on the selected attractor. The approach demonstrated true reversibility, while keeping all the DMP properties. Nevertheless, all questions have not been resolved yet, and the approach was evaluated on tasks and joint space position trajectories. A proper formulation for dealing with orientations, for example, quaternions in task space, is still missing.

7. Concluding remarks

Since their introduction in the early 2000s, DMPs have established as one of the most used and popular approaches for motor command generator systems in robotics. Several authors have exploited and extended the classical formulation to overcome some limitations and fulfill different requirements. Their research resulted in a large number of papers published over the last two decades.

One of the aims of this paper is to categorize and review the vast literature on DMPs. We took a systematic review approach and automatically searched for DMP-related papers in a popular database. A manual inspection of the resulting papers, guided by clear and unbiased criteria, led to the papers included in this tutorial survey.

Another aim of our work is to provide a tutorial on DMPs that presents the classical formulation and the key extensions in rigorous mathematical terms. We made an effort to unify the notation among different approaches in order to make them easier to understand. Moreover, we provide useful guidelines that guide the reader to select the right approach for a given application. In the tutorial vein, we have also searched for open-source implementations of the described approaches and released to the community several implementations of DMP-based approaches.

Advantages of DMPs have been discussed as well as their limitations and the open issues. We have summarized them in Table 5 where we also indicate the solved issues and the ones that require further investigation. In this respect, as research on DMP is still very active, we provide a comprehensive discussion that will help the readers to understand what has been done in the field and where they can put their research focus.

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ORCID iDs

Matteo Saveriano  <https://orcid.org/0000-0002-9784-3973>

Fares J Abu-Dakka  <https://orcid.org/0000-0001-9062-9416>

Aljaž Kramberger  <https://orcid.org/0000-0002-4830-4885>

Luka Peternel  <https://orcid.org/0000-0002-8696-3689>

Notes

1. As discussed by Pastor et al. (2009), a transformation system that uses (48) generates a mirrored trajectory while reaching a new goal $\mathbf{g}_{\text{new}}$ every time the signs of $(\mathbf{g}_{\text{new}} - \mathbf{y}_0)$ and $(\mathbf{g} - \mathbf{y}_0)$ differ.
2. Saveriano et al. used the multi-dimensional DMP formulation developed in Hoffmann et al. (2009) for both position and quaternion DMPs. In this review paper, we reformulate the merging approaches in Saveriano et al. (2019) to comply with the formulations in Section 2.1.1 and 2.1.2.1.
3. Note that other feedback that measures human effort can be used instead of EMG, such as joint torque or limb forces.
4. There is a duality in impedance-admittance, that is, the force produces motion and motion produces force; therefore if one is the input, the other can only be the output of the control system (Peternel et al. 2017a).
5. The referred work extended the classical DMP to different space like $\mathcal{SO}(3)$ or $\mathcal{S}^{m}_{++}$. Although formally similar, the extension to other Riemannian manifolds like the Grassmannian or the Hyperbolic manifolds is non-trivial and still not fully addressed.

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机器人学中的动态运动基元：一篇教程式综述



Matteo Saveriano ¹, Fares J Abu-Dakka ², Aljaž Kramberger ³ and Luka Peternel ⁴

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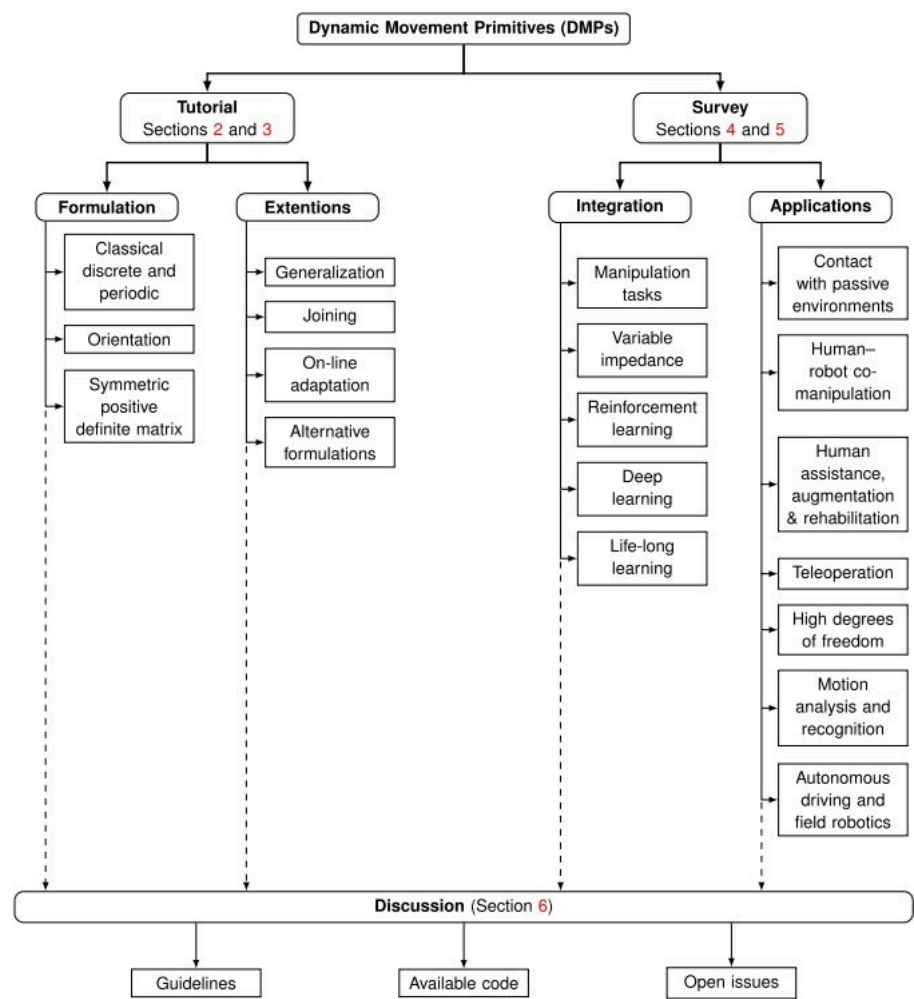


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摘要

包括人类在内的生物系统，天生具备以多功能且敏捷的方式执行复杂任务的能力。感觉运动控制领域的研究人员一直致力于理解并正式定义这一先天特性。多项实验证据支持生物系统能够将基本运动单元组合并适应于复杂任务，这一观点最终促成了运动基元理论的提出。在这方面，动态运动基元（DMPs）将运动基元优雅地数学表述为稳定的动力系统，非常适合为机器人等人工系统生成运动指令。过去几十年中，DMPs激发了不同机器人领域研究者的灵感，包括模仿学习与强化学习、最优控制、物理交互以及人机协作，从而产生了大量已发表的论文。本教程式综述的目标有二。一方面，我们以严谨的数学语言呈现现有的DMP表述，讨论每种方法的优势与局限性以及实际实现细节。秉承教程风格，我们还检索了所介绍方法的现有实现，并发布了多个其他实现。另一方面，我们对现有文献进行了系统而全面的回顾，并对DMP方面最先进的工作进行了分类。本文最后讨论了DMP的局限性，并概述了可能的研究方向。

关键词



人工系统的运动控制，运动基元理论，动态运动基元，从示教中学习

1. 引言

生物系统（如人类和动物）如何以多功能且富有创造性的方式执行复杂运动？

在过去几十年中，神经生物学和运动控制领域的研究人员为回答这一问题付出了巨大努力，其实验发现促成了运动基元或动作基元理论的提出。运动基元理论将复杂运动的执行解释为生物系统对动作单元（即所谓的运动基元）进行排序和适应的能力（Flash and Hohnner, 2005; Mussa-Ivaldi, 1999）。

动态运动基元（DMPs）根植于生物系统的运动控制，可被视为将运动基元严谨数学表述为稳定的非线性动力系统（Schaal, 2006a, 2006b）。就此而言，DMPs是回答下列研究问题的早期尝试之一：

人工系统（如（类人）机器人）如何以多功能且富有创造性的方式执行复杂运动？

除了其生物学动机外，DMPs具有简洁优雅的表述形式、保证收敛到给定目标、具备足够灵活性以创建复杂行为、能够实时响应外部扰动，并且可以利用高效算法从数据中学习。这些特性解释了DMPs在机器人应用中的"成功"，它们已成为学习和生成运动指令的突出工具。自从在Ijspeert等人（2002c; Schaal, 2006）的开创性工作中提出以来，DMPs已被成功应用于多种场景，实际上已成为模仿学习（IL）领域初学者在其机器人上首先采用的方法。

1.1. 现有综述与教程

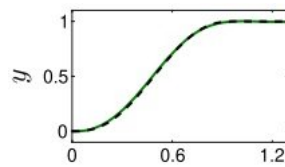
DMP的普及催生了大量工作，这些工作使用、修改或扩展了Ijspeert及其同事提出的原始公式。开创性工作（Ijspeert等人，2001）将期望位置和速度计算为多个具有不同时间尺度的动力系统的混合。一年后，（Ijspeert等人，2002c; Schaal, 2006）提出了另一种公式，该公式变得流行，并在本文中被称为经典DMP。如表1所示，一些教程和综述已尝试对DMP的现有工作进行分类和回顾。

Schaal等人（2007）将经典DMP描述为统一非线性动力系统与最优控制理论的一种尝试，即用于推导神经生物运动理论计算模型的两个主要框架（Flash和Hochner, 2005; Mussa-Ivaldi, 1999）。Ijspeert等人（2013）提出了节律性和离散性DMP的统一公式，并附带了一些扩展，包括耦合项、对不同目标的一般化以及用于碰撞避免的在线自适应。他们还描述了在模仿学习和运动识别方法中的可能应用。同年，Pastor等人（2013）提出了经典DMP的扩展，特别侧重于通过将感知信息整合到动作生成过程中来在线自适应DMP吸引子景观。随后，Deniša等人（2016b）回顾了所谓的顺从运动基元，该概念最早由Petrić等人（2014a）提出。CMP将经典DMP用于生成期望的运动学路径与力矩基元（高斯基函数的加权求和）相结合，以生成任务特定的动力学。如Deniša等人的综述所示，CMP能够以顺从方式精确跟踪运动学路径，因此非常适用于需要机器人与环境交互的任务。

然而，上述论文、综述和教程主要关注各自研究团队内部的方法和进展，或聚焦于特定问题或应用领域。另一方面，与DMP相关的文献广泛而丰富，来自众多研究团队的贡献在多个重要应用领域取得了进展。因此，本文提出的DMP综述与教程旨在扫描更广泛的范围，并提供一份包含统一结构化公式的教程，涵盖迄今为止的各种DMP方法与进展。这将有助于用户更清晰地理解不同方法之间的差异与联系，并促进更简便的应用。此外，我们还对机器人学中所有主要DMP应用领域进行了更全面、分类更细致的综述，有助于启发读者在不同领域应用DMP。

在教程部分，我们从（Ijspeert等人，2002c; Schaal, 2006）中提出的经典DMP开始，直至DMP最近在黎曼几何和对称正定矩阵方面的扩展（Abu-Dakka和Kyrki, 2020），介绍了现有DMP公式的数学表述、实现细节及潜在问题。在综述部分，我们以全面且系统的方式细致回顾了DMP的现有文献，重点关注其贡献的质量与重要性，而不对任何特定研究团队有所偏倚。系统性综述过程的细节如下所述。

1.2. 系统性综述过程



我们于2020年11月25日在Scopus中对包含字符串“Dynamic Movement Primitive”的文献进行了自动检索，共返回1223篇论文。我们从2001年DMP初步工作发表时开始检索，并于2023年6月20日进一步优化了检索，以纳入最新论文。

我们对所有论文进行了手动检查，剔除了那些未明确使用DMP，或仅在文献综述中与DMP进行对比的论文。首要的筛选标准是工作的技术质量，以及相对于特定论文发表前DMP领域现有技术水平的贡献重要性。换言之，我们提出的问题是：“该论文是否在该领域实现了重大的阶段性突破？”因此，我们排除了多次提出相似（或相同）想法，或对现有技术水平改进甚微的论文。若多篇论文提出相同/相似想法，则保留其中技术质量最全面的；若质量相近，则进一步考虑是否发表于更权威的期刊/会议或被引次数更高。这一手动筛选过程最终确定了本文分析的321篇关于DMP的论文（共373篇参考文献）。

1.3. DMP相关研究的分类体系

对DMP文献的系统性综述形成了如图1所示的分类体系，该图也描述了本文的结构。DMP位于树状结构的根部，分支为两个节点，即教程部分与综述部分。在教程部分，我们以严谨的数学语言呈现了不同的DMP公式及其扩展。

教程部分涵盖第2节和第3节。第2节包含离散运动与周期运动、方向轨迹以及SPD矩阵的DMP公式。第3节讨论DMP形式化的扩展，以涵盖技能泛化、多个基元的连接、基于力反馈的在线自适应或参考速度。本节最后简要描述了与DMP相关的形式化方法。

综述部分涵盖第4节和第5节。第4节介绍了DMP在操作与变阻抗任务、强化学习、深度以及

表1. 现有关于DMP的论文、综述和教程与本文教程调查的比较。

论文/综述/教程	主题	描述		
Schaal et al. (2007)	· 经典 DMPs · 在线自适应 · 优化	该教程提供了用于发展计算运动控制理论的两大主要方法（即微分方程和最优控制）的统一视角。本文将离散和节律性DMP（Ijspeert等，2002c; Schaal, 2006）作为运动基元理论（Mussa-Ivaldi 1999）的计算模型，该模型统一了非线性微分方程和最优控制。教程中有一节专门讨论超越模仿学习的DMP参数优化。Schaal等人展示了如何优化DMP参数以最小化各种代价函数，例如轨迹的总加加速度或末端方差。		
Ijspeert et al. (2013)	· 经典 DMPs · 在线自适应 · 耦合项 · 泛化	一篇关于经典DMP的论文，除其科学贡献外，还介绍了离散和节律性两种公式，这些公式主要来自（Ijspeert等，2002c, 2002; Schaal, 2006），以及它们在模仿学习和运动识别中的应用。该论文还提出了经典DMP公式的扩展，以防止运动开始时的高加速度，避免与不可预见障碍物的碰撞（Pastor等，2009），并在空间（例如，达到不同目标）和时间（例如，生成更长/更短的轨迹）上进行泛化。		
Pastor et al. (2013)	· 经典 DMPs · 在线自适应 · 耦合项 · 阻抗学习	一篇关于经典DMP的论文，除其科学贡献外，还介绍了离散和节律性两种公式，这些公式主要来自（Ijspeert等，2002c, 2002; Schaal, 2006）。该论文还提出了经典DMP公式的扩展，以避免与不可预见障碍物的碰撞（Pastor等，2009）并通过强化学习（RL）学习阻抗控制策略（Buchli等，2011b）。Ijspeert等（2013）与Pastor等（2013）的关键区别在于，后者有一个专门讨论感官关联以及利用Pastor等（2011）和Pastor等（2011a）开发的关联技能记忆框架对DMP轨迹进行在线、情境感知自适应的部分。		
Deniša et al. (2016b)	· 经典 DMPs · 顺应运动基元（CMPs）			
N	$\backslash$ (triangle q)	非线性基函数的数量	i	$\backslash$ (triangleq) 索引: $\backslash$ (i = 1, 2, ..., N)
J	$\backslash$ (triangle q)	关节数量或自由度 (DoFs)	j	$\backslash$ (triangleq) 索引: $\backslash$ (j = 1, 2, ..., J)
L	$\backslash$ (triangle q)	演示或DMP的数量	l	$\backslash$ (triangleq) 索引: $\backslash$ (l = 1, 2, ..., L)
V	$\backslash$ (triangle q)	途经点或途经目标的数量	v	$\backslash$ (triangleq) 索引: $\backslash$ (v = 1, 2, ..., V)
$\backslash$ (mathcal{T})	$\backslash$ (triangle q)	数据点的数量	j	$\backslash$ (triangleq) 索引: $\backslash$ (j = 1, 2, ..., \mathcal{T})
m	$\backslash$ (triangle q)	$\backslash$ (mathcal{S}_{-}^{++\wedge m}) 的维度	n	$\backslash$ (triangleq) $\backslash$ (mathbb{R}^{\wedge n}) 的维度
$\backslash$ ({}) d	$\backslash$ (triangle q)	期望值的下标	$\backslash$ ({}) q 或 $\backslash$ ({}) q	$\backslash$ (triangleq) 与四元数相关的变量

$\mathbf{R}$ 或 $\mathbf{R}$	$\backslash$ (triangle q)	与旋转矩阵相关的变量	$\mathbf{R}++$, $\mathbf{R}+$ 或 $\mathbf{R}+$	$\mathbf{R}$ 与 SPD 相关的变量
$\mathbf{g}$	$\backslash$ (triangle q)	目标值的下标	$\mathbf{a}_z$, $\mathbf{a}_{\beta_z}$, $\mathbf{a}_x$, $\mathbf{a}_{\beta_x}$, $\mathbf{a}_{y_x}$, $\mathbf{a}_{q_g}$	$\mathbf{R}$ 正增益
τ	$\backslash$ (triangle q)	时间调制参数	$\mathbf{c}_i$, $\mathbf{h}_i$	$\mathbf{R}$ 高斯基函数的中心和宽度
T	$\backslash$ (triangle q)	时间持续时间	t	$\mathbf{R}$ 连续时间
λ	$\backslash$ (triangle q)	遗忘因子	r	$\mathbf{R}$ 幅度调制参数
$\mathbf{x}$	$\backslash$ (triangle q)	相位变量	$\mathbf{y}$, $\dot{\mathbf{y}}$	$\mathbf{R}$ 轨迹数据及其一阶导数
$\mathbf{s}$	$\backslash$ (triangle q)	S形衰减相位	$\mathbf{z}$, $\dot{\mathbf{z}}$	$\mathbf{R}$ 缩放后的速度和加速度
$\mathbf{p}$	$\backslash$ (triangle q)	分段线性相位	$\mathbf{g}$, $\mathbf{g}_{\mathbf{q}}$, $\mathbf{g}_{\mathbf{q}^+}$	$\mathbf{R}$ 不同空间中的吸引点（目标）
ω	$\backslash$ (triangle q)	角速度	$\widehat{\mathbf{g}}$, $\widehat{\mathbf{g}_{\mathbf{q}}}$ 和 $\widetilde{\mathbf{g}}$, $\widetilde{\mathbf{g}_{\mathbf{q}}}$	$\mathbf{R}$ 不同空间中的移动目标和延迟目标函数
$\mathbf{Q}$, $\dot{\mathbf{Q}}_t$	$\backslash$ (triangle q)	关节位置及其一阶时间导数	$\mathbf{g}_v$	$\mathbf{R}$ 中间吸引子（途经目标）
$\mathbf{q}$, $\dot{\mathbf{q}}$	$\backslash$ (triangle q)	单位四元数及其一阶时间导数	$\mathbf{R}$, $\dot{\mathbf{R}}$	$\mathbf{R}$ 旋转矩阵及其一阶时间导数
$\mathbf{f}$, $\mathbf{f}_{\mathbf{q}}$, $\mathbf{f}_{\mathbf{R}}$, $\mathbf{f}_{\mathbf{q}^+}$, $\mathcal{F}_{\mathbf{q}^+}$	$\backslash$ (triangle q)	不同空间的强迫项	$\mathbf{w}_i$	$\mathbf{R}$ 可调权重

Ψ_i	$\triangleleft$	基函数	θ 和 ϑ	$\triangleleft$ 一个角度和可学习参数
$\mathcal{S}_{++}^n$	$\triangleleft$	$(m \times m)$ SPD流形	$\text{Sym } \mathcal{M}$	$\triangleleft (m \times m)$ 对称矩阵空间
M	$\triangleleft$	一个黎曼流形	X	$\triangleleft$ 任意一个SPD矩阵
T_Λ	$\triangleleft$	在任意点 Λ 处 M 的切空间	M	$\triangleleft \{\mathbf{X}_t\}_{t=1}^\Sigma$ 的均值
$\log_{\Lambda} \mathbf{Y}$	$\triangleleft$	$M \mapsto T_\Lambda M$, 将任意点 $\mathbf{Y} \in \mathcal{M}$ 映射到 $\mathbf{q} \in \mathcal{T}_\Lambda \mathcal{M}$ 中	$\mathbf{Y} = \text{Exp}_\Lambda(\mathbf{q})$	$\triangleleft T_\Lambda M \mapsto \mathbf{q} \in \mathcal{T}_\Lambda \mathcal{M}$ 映射到 $\mathbf{Y} \in \mathcal{M}$ 中
$\text{vec}(\cdot)$	$\triangleleft$	使用Mandel记号将 $\text{Sym } \mathcal{M}$ 变换为 $\mathbb{R}^n$ 的函数。	$\text{mat}(\cdot)$	$\triangleleft$ 使用Mandel记号将 $\mathbb{R}^n$ 变换为 $\text{Sym } \mathcal{M}$ 的函数。
$\mathbf{K}, \mathbf{K}^P, \mathbf{K}^O$	$\triangleleft$	不同形式的刚度增益	$\mathbf{D}, \mathbf{D}^V, \mathbf{D}^W$	$\triangleleft$ 不同形式的阻尼增益
π 和 $\mathbf{I}$	$\triangleleft$	质量和惯性矩阵	$\mathbf{F}, \mathbf{f}^e, \mathbf{f}^r$	$\triangleleft$ 力以及外力和扭矩
DMP	动态运动基元	IL	模仿学习	
CMP	顺应性运动基元	UAV	无人驾驶飞行器	
RL	强化学习			
运动类型	空间	方程组	参考文献	简要描述
离散	$\mathbb{R}$	$\dot{z} = \alpha_z(\beta_z(g - y) - z) + f(x)$	公式(1)-(3) (Ijspeert et al., 2002c)	一个单DoF的离散运动轨迹被编码到一个带有加性非线性强迫项的线性二阶动力系统中。通过一个逐渐消失的相位

				变量 $\mathbf{x}$ 确保收敛到期望目标 $\mathbf{g}$ 。
	$\backslash$ (\mathcal{I}\{S\}^3)	$\backslash(\tau \dot{\eta} = \alpha_z(\beta_z)^2 \log^d(\mathbf{g}_q) * \overline{\mathbf{q}}) - \eta + \mathbf{f}_q(\mathbf{x}))$	公式(16)和 (17) (Abu-Dakka et al., 2015a)	一个基于四元数的方向轨迹 (3 DoFs) 被编码到一个带有加性非线性强迫项的二阶动力系统中。误差定义符合单位四元数空间的几何结构。
	$\backslash$ (\mathcal{I}\{SO\}(3))	$\backslash(\tau \dot{\eta} = \alpha_z(\beta_z) \log^R(\mathbf{R}_g) * \mathbf{R}^{\text{top}}) - \eta + \mathbf{f}_R(\mathbf{x}))$	公式(23)和 (24) (Ude et al., 2014)	一个基于旋转矩阵的方向轨迹 (3 DoFs) 被编码到一个带有加性非线性强迫项的二阶动力系统中。误差定义符合旋转矩阵空间的几何结构。
	$\backslash$ (\mathcal{I}\{S\}_{++}^m)	$\backslash(\tau \dot{\sigma} = \alpha_z(\beta_z) \mathbf{vec}(\mathbf{E}_{\mathbf{X}_i} \mapsto \mathbf{X}_i) (\mathbf{Log}_{\mathbf{X}_i}^{++}(\mathbf{X}_g))) - \sigma + \mathcal{F}(\mathbf{x}))$	公式(29)和 (30) (Abu-Dakka and Kyrki, 2020)	一个SPD矩阵轨迹 ($\mathbf{m}$) ($\mathbf{m} + 1$)/2 DoFs) 被编码到一个带有加性非线性强迫项的二阶动力系统中。误差定义符合SPD矩阵空间的几何结构。
周期	$\backslash$ (\mathbb{R})	$\backslash(\dot{z} = \Omega(\alpha(\beta(-y) - z) + \mathbf{f}(\phi)))$	公式(34)-(36) (Ijspeert et al., 2002b)	一个单DoF的周期运动轨迹被编码到一个带有加性非线性强迫项的线性二阶动力系统中。生成的系统产生一个稳定的极限环。
	$\backslash$ (\mathcal{I}\{S\}^3)	$\backslash(\dot{\eta} = \Omega(\alpha_z(\beta_z)^2 \log^q(\mathbf{g}_q) * \overline{\mathbf{q}}) - \eta + \mathbf{f}_q(\phi)))$	公式(43)和 (44) (Abu-Dakka et al., 2021)	一个基于四元数的方向轨迹 (3 DoFs) 被编码到一个带有加性非线性强迫项的二阶动力系统中。误差定义符合单位四元数空间的几何结构。

3.1.1.1. 起始点、目标点与缩放。经典DMPs是时间不变的，这意味着时间缩放 $\backslash(\varsigma \tau)$ 与 $\backslash(\varsigma > 0)$ 生成拓扑等价的轨迹 (Ijspeert et al., 2013)。通过对变换系统进行简单修改，即将(1)替换为

$$\backslash[\tau \dot{z} = \alpha_z(\beta_z)(g - y - z) + (g - y_0)f(x)]$$

其中 $\backslash(y_0)$ 是 $\backslash(y)$ 的初始值, Ijspeert 等人 (2013) 证明 DMP 还具有尺度不变性, 即用 $\backslash(\varsigma > 0)$ 缩放运动幅度 $\backslash(\varsigma(g - y_0))$ 会产生拓扑等价的轨迹。(48)中使用绿色的目的是突出显示与 (1) 的差异 $\backslash(\text{w.r.t.})$ 。Kulvicius 等人 (2011) 采用了类似的策略, 通过演示轨迹和期望轨迹起点与终点之间的比率来缩放强迫项。除了生成演示运动轨迹在时间和空间上的缩放版本外, 经典 DMP 还能泛化到不同的初始/目标状态。然而, 经典公式及其在 (48) 中的扩展可能会表现出危险行为, 例如在到达不同目标时轨迹过度放大, 以及在线切换到不同目标时产生高加速度 (Ijspeert 等人, 2013; Pastor 等人, 2009)。为了解释第二个问题, Ijspeert 等人 (2013) 用 (13) 中的平滑切换律替代了硬目标切换。然而, 过度放大问题仍然存在。此外, 使用 (48) 的 DMP 无法学习具有相同初始状态和目标状态的运动 (即 $\backslash(g = y_0)$, $\backslash(z_0 = 0 \rightarrow y(t) = y_0 = g \nabla t)$)。

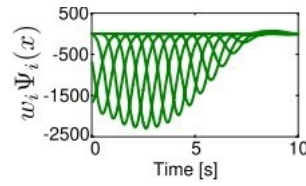
为了解决这些问题, Pastor 等人 (2009) 提出了如下修改变换系统的方法:

$$\dot{\tau} = a_z(\beta_z(g - y - (g - y_0)x + f(x)) - z)$$

其中使用绿色来突出显示 (49) 与 (1) 之间的差异。该公式中最重要的变化是项 $((g - y_0)x)$ ，它具有若干优点。它防止了运动刚开始时 $((g - y - (g - y_0)x = 0)$ (对于 $(t = 0)$) 或当目标接近初始状态时的高加速度。它允许复现具有相同初始状态和目标状态的运动，并在改变目标时防止过度放大和轨迹镜像效应 $(^1)$ 。Hoffmann 等人 (2009) 从青蛙脊髓力场的行为推导出了 (49) 的多维表示。

目标也可以随时间变化，在这种情况下，DMP 的跟踪性能主要取决于增益 (α_z) 和 (β_z) 。正如 Koutras 和 Doulgeri (2020b) 所提出的，可以通过调整时间缩放 (τ) 来提高跟踪性能。

Dragan 等人 (2015) 表明，DMP 通过求解轨迹优化问题，在起始和目标约束下最小化演示轨迹与新轨迹之间的特定希尔伯特范数。基于此，可以通过选择 (或学习) 适当的希尔伯特范数来减少检索轨迹中的变形，从而提高 DMP 对不同起始和目标的适应能力。



3.1.2. 路径点。路径点可以定义为状态空间中轨迹必须经过的点。未能经过路径点可能导致机器人任务执行失败。因此，在机器人场景中，具有调制路径点能力的运动基元表示是重要的。研究人员扩展了 DMP 公式以考虑轨迹生成过程中的中间路径点，这并不意外。

Ning 等人 (2011, 2012) 扩展了经典 DMP，以满足样本轨迹起始和结束时的位置与速度约束。他们遍历路径点的方法是通过组合连接路径点的局部线性轨迹来创建样本轨迹，然后利用该样本轨迹拟合一个被约束经过路径点的 DMP。

Weitschat 和 Aschemann (2018) 将每个路径点视为一个中间目标 (路径目标) (g_v) ，供 $(v = 1, \dots, V)$ 到达。最后一个路径目标 (g_V) 对应于 DMP 的目标状态。在他们的公式中，他们定义了一个可变目标如下：

$$g_v(x) = \sum_{n=1}^V \Psi_n(x) g_n$$

其中 $(\Psi_v(x))$ 是高斯基函数，其中心位于对应第 (n) 个路径目标的时间点。该方法的有效性在一个任务中得到验证，该任务要求机器人到达不同目标，同时防止末端执行器与机器人本体发生自碰撞。为此，作者沿着用于学习 DMP 的轨迹放置路径目标，迫使生成的轨迹在到达新目标的同时保持接近演示轨迹。

Zhou 等人 (2019) 面临了从演示中泛化到接近 (内插) 和远离 (外推) 途经点的问题。他们的方法，即途经点运动基元 (VMPs)，结合了 DMP 和概率运动基元 (ProMPs) (Paraschos 等人, 2013) 的优点。作者假设运动轨迹按如下方式生成

$$y_{vmp}(x) = e(x) + f_{vmp}(x)$$

其中 (x) 是按 (3) 定义的相位变量，基本轨迹 $(e(x))$ 可以定义为线性吸引子 $(e(x) = (y_0 - g)x)$ 。形状调制项 $(f_{vmp}(x))$ 定义如下

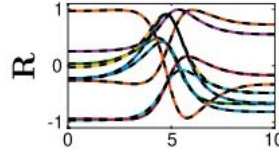
$$f_{vmp}(x) = \sum_{i=1}^N w_i \Psi_i(x) + \epsilon_f$$

其中高斯核 $(\Psi_i(x))$ 按 (5) 定义， (w_i) 是可学习权重， (ϵ_f) 是高斯噪声。如 Paraschos 等人 (2013) 所述，学习形状调制项 $(f_{vmp}(x))$ 意味着从演示中学习 (LFDs) 权重 (w_i) 的先验概率分布。由于将生成的轨迹像 (51) 那样分成两部分，可以采用不同的策略来通过 (x_v) 处的途经点 (y_v) ，Zhou 等人 (2019) 提出在内插情况下——当途经点“接近”演示时——修改形状调制项。在外推情况下，基本轨迹 $(e(x))$ 被重写为连接 (y_0) 、 (y_v) 和 g 的折线。该方法很容易推广到多个途经点的情况。VMPs 与 ProMPs 进行了实验比较，显示出更好的性能，尤其是在外推情况下。

3.1.3. 任务参数。到达不同的目标或通过途经点可能不足以在不同情境下成功执行任务。本节介绍的方法通过调整强迫项 (4) 的权重 w_i 来使 DMP 运动适应新情况，这会修改整个 DMP 轨迹。

Weitschat 等人 (2013) 考虑了给定 L 个演示，每个演示编码在不同的 DMP 中。为了泛化到，例如，一个新目标 g_{new} ，他们提出内插附近 DMP 的权重，即到达 g_{new} 附近点的 DMP。用公式表示为

$$\mathbf{w}_{new} = \frac{\sum_{o: d_o} \{o\}}{\sum_{o: d_o} 1}$$



其中 $\{o\}$ 表示附近 DMP 的索引, 满足 $d_o < d_{max}$, d_o 是 g_{new} 和 g_o 之间的距离 (或更一般地, 成本), d_{max} 是认为两个 DMP 接近的最大距离, $\mathbf{w}_{new} = [w_{1,new}, \dots, w_{N,new}]^{\top}$ 和 $\mathbf{w}_o = [w_{1,o}, \dots, w_{N,o}]^{\top}$ 分别是新权重和附近 DMP 的权重。

Forte 等人 (2011, 2012) 的方法也假设给定 L 个演示, 并且每个演示被编码在不同的 DMP 中。此外, 作者利用 GP (Rasmussen and Williams, 2006) 来学习查询点 q_l (对于 $l = 1, \dots, L$), 例如每个 DMP 的目标与 DMP 参数 $[w_l, g_l, \tau_l]$ 之间的映射。给定新的查询点 q_{new} , 使用高斯过程回归 (GPR) (Rasmussen and Williams, 2006) 来获取新参数集 $[w_{new}, g_{new}, \tau_{new}]$, 该参数集可用于生成 DMP 运动。该方法建立在先前工作 (Gams and Ude, 2009; Ude et al., 2010) 的基础上, 其中来自 L 个演示的原始数据存储在内存中, 并使用 LWR 生成新的 DMP 权重。Alizadeh 等人 (2016) 扩展了 Ude 等人 (2010) 的方法, 即使在任务参数部分可观测的情况下也能检索 DMP 权重。最后, Zhou 和 Asfour (2017) 扩展了 Ude 等人 (2010) 的方法, 在学习查询点与 DMP 权重之间的映射时考虑任务特定的成本。

前述方法遵循两步流程: 首先根据新任务参数估计形状参数 w , 然后执行 DMP。Matsubara 等人 (2010, 2011) 在强迫项中增加了风格参数, 用于捕捉多次演示中的人类变异性。Stulp 等人 (2013) 提出了一步法流程, 将 DMP 强迫项 (4) 重新表述为显式依赖于任务参数的形式。他们的实验表明, 一步法相较于所使用的回归技术具有更多自由度, 并提升了泛化性能。沿着同样的思路, Pervez 和 Lee (2018) 将任务参数直接嵌入强迫项中。作者提出使用高斯混合模型 (Cohn 等人, 1996) 来学习任务参数 (例如新目标及障碍物高度) 与强迫项之间的映射。给定新的查询任务参数, 通过对高斯混合模型进行回归来获取强迫项参数, 并生成 DMP 运动。该方法在包括扫掠和击打在内的多种任务上进行了测试, 并与 Forte 等人 (2012)、Stulp 等人 (2013) 以及 Ude 等人 (2010) 提出的方法进行了比较, 显示出更好的性能, 尤其是在外推方面。

Mülling 等人 (2010, 2013) 提出了运动基元混合模型 (MoMP), 并将其用于泛化乒乓球技能, 例如击球和挥拍。MoMP 使用包含机器人位置和速度以及乒乓球任务元参数 (如预期击球位置和速度) 的增广状态。自适应运动由 L 个 DMP 的加权求和生成, 每个 DMP 的责任度 (表示该特定 DMP 是感知到的增广状态的正确解的概率) 也从数据中学习得到。

在高自由度系统 (如人形机器人) 中, 寻找任务参数与 DMP 参数之间的关系并非易事。当 DMP 用于编码关节空间轨迹时尤其如此。Bitzer 和 Vijayakumar (2009) 指出, 这种关系在从训练数据中获得的潜空间 (低维空间) 中更容易找到。因此, 他们使用降维技术来寻找潜空间, 并在该空间中拟合 DMP, 结果表明在潜空间中对 DMP 权重进行插值可获得更好的泛化性能。

3.2. 连接多个 DMP

任何运动基元表示的一个重要且期望的特性是能够组合基本运动以获取更复杂的行为 (Schaal, 1999)。我们在此回顾三种用于平滑连接一系列 DMP 的主流方法。在本教程中, 我们将 Pastor 等人 (2009) 的方法称为速度阈值法, 将 Kober 等人 (2010b) 的方法称为目标穿越法, 将 Kulvicius 等人 (2011, 2012) 的方法称为基函数叠加法。其中某些方法修改了第 2.1.1 节和第 2.1.2 节中的 DMP 公式。主要差异以绿色文本突出显示。这三种方法已针对位置 DMP (第 2.1.1 节) 和姿态 DMP (第 2.1.2 节) 在 MATLAB 中实现。源代码包含在我们的公开仓库中 (见表 4)。合成数据的实验结果如图 8–11 所示。

3.2.1. 速度阈值法。设计得当的 DMP 会在速度和加速度为零时达到目标, 即一旦 DMP 完全执行完毕, 机器人将完全停止。这也意味着“接近”目标的速度会持续下降。利用这一特性, Pastor 等人 (2009) 提出通过当前 DMP 的速度低于某一阈值时直接终止该 DMP, 然后启动下一个基元来组合连续的 DMP。在执行单个 DMP 时, 通常将其速度初始化为零——假设机器人处于静止状态。原则上, 这种初始化可用于串联多个 DMP (Lioutikov 等人, 2016; Xu 和 Wang, 2004), 但如果机器人在两个连续基元之间未完全停止, 则可能产生不连续性。为避免这些不连续性, Pastor 等人使用前一个 DMP 的状态来初始化当前 DMP 的状态。

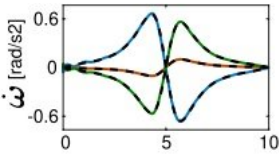
速度阈值方法简单有效, 因为它直接适用于第 2.1.1 节和第 2.1.2 节中的 DMP 公式。例如, Saveriano 等人 (2019) 展示了如何使用速度阈值方法连接多个四元数 DMP (见第 2.1.2.1 节)。

图8中的结果是在应用速度阈值合并两个分别训练以拟合最小加加速度轨迹（黑色虚线）的DMP时获得的。图8(a)–(e)显示了位置，图8(f)–(j)

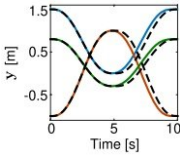
表4. 我们向社区发布的基于DMP方法的开源实现。每种方法的源代码可在 <https://gitlab.com/dmp-codes-collection> 获取。

方法	作者	编程语言	描述
离散DMP	Fares J. Abu-Dakka	C++	基于Abu-Dakka等人（2015a）以及Ude等人（2010，2014）工作的离散DMP实现。
周期性DMP	Luka Peternel	Python	基于Peternel等人（2016）工作的周期性DMP实现。
单位四元数	Fares J. Abu-Dakka	MATLAB and C++	基于Abu-Dakka等人（2015a）和Ude等人（2014）工作的单位四元数DMP及目标切换实现。
SPD DMP	Fares J. Abu-Dakka	MATLAB	基于Abu-Dakka和Kyrki（2020）工作的SPD DMP及目标切换实现。
DMP连接	Matteo Saveriano	MATLAB	基于Saveriano等人（2019）工作的多个DMP连接实现。
耦合力DMP	Aljaz Kramberger	MATLAB	基于Kramberger等人（2018）工作的离散DMP及力耦合项实现。

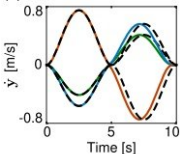
(a)



(b)

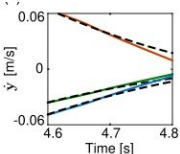


(c)



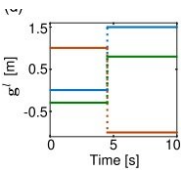
位置

(d)



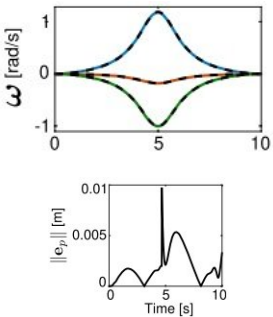
线速度

线速度



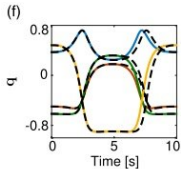
(e)

目标切换

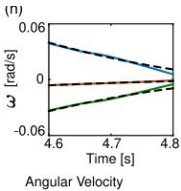
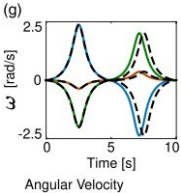


(g)

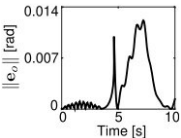
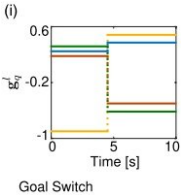
位置误差



(h)



(j)



方向误差

图8. 应用零速度切换方法连接两个基于合成数据训练的DMP所获得的结果。位置和方向的训练轨迹分别以黑色虚线显示在(a)–(b)和(f)–(g)中。结果来源于开源实现，地址为 <https://gitlab.com/dmp-codes-collection>。

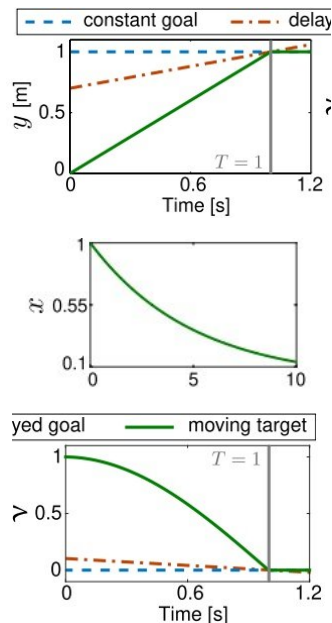
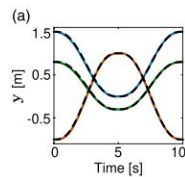
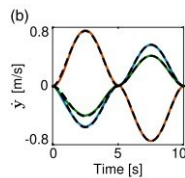


图9. 恒定目标、移动目标及延迟目标的结果，分别对应 $y(0)=0$ [m]、 $g=1$ [m]、 $\hat{y}=0.3$ [m/s]（左图）以及 $\mathbf{q}(0)=1+[0,0,0]^{\top}$ 、 $\mathbf{g}_q=0+[1,0,0]^{\top}$ 、 $\widehat{\boldsymbol{\omega}}=[0.2,0.2,0.2]^{\top}$ [rad/s]（右图）。采样时间为 $\Delta t=0.01$ [s]。为便于可视化，仅显示四元数的标量部分[eqn7]。

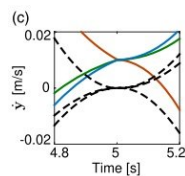
运动的（单位四元数）方向部分。合并轨迹通过跟随第一个DMP生成，直到与途径点的距离低于0.01 [m]和0.01 [rad]。如图8(d)和(i)所示，切换发生在大约4.7 [s]处。图8(e)和(j)表明期望轨迹被精确复现。通过调整与途径点的距离，可以获得精度不同的轨迹。然而，该值的



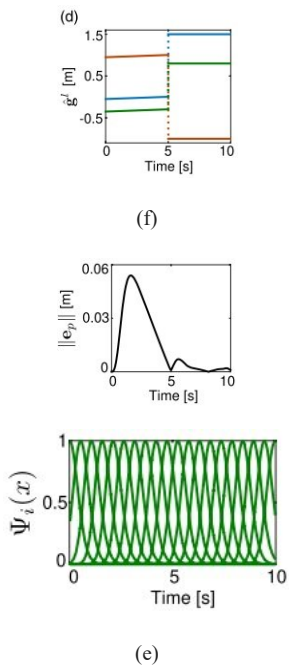
位置



线速度

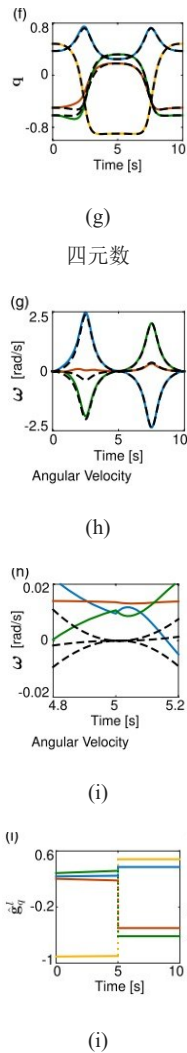


线速度

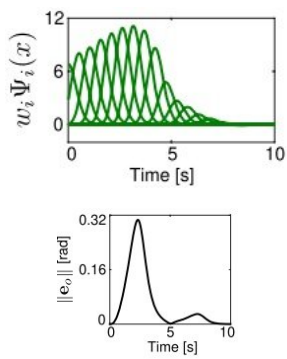


目标切换

位置误差



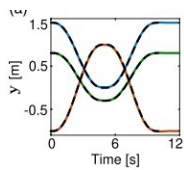
目标切换



朝向误差

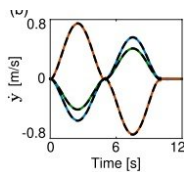
图10. 将目标交叉方法应用于连接两个在合成数据上训练的DMP所获得的结果。位置和朝向的训练轨迹分别以黑色虚线显示在 (a)–(b)和(f)–(g)中。结果通过开源实现获得，该实现可在<https://gitlab.com/dmp-codes-collection>获取。

(a)



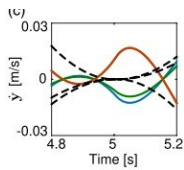
(b)

位置

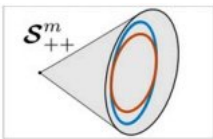


线速度

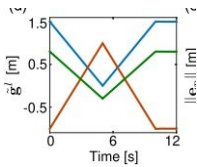
(c)



(d)

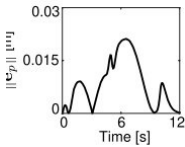


线速度



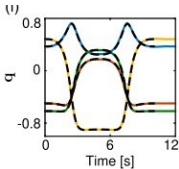
(e)
(f)

目标切换

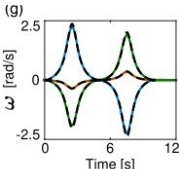


位置误差

(g)

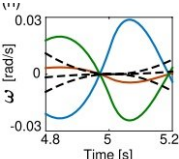
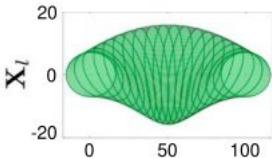


四元数



角速度

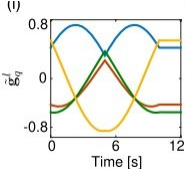
(h)



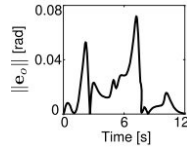
(i)

角速度

(j)



目标切换



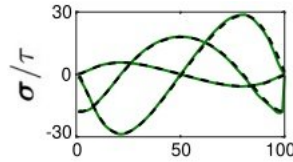
朝向误差

图11. 将基函数叠加方法应用于连接两个在合成数据上训练的DMP所获得的结果。位置和朝向的训练轨迹分别以黑色虚线显示在(a)–(b)和(f)–(g)中。结果通过开源实现获得，该实现可在<https://gitlab.com/dmp-codes-collection>获取。

距离是生成轨迹的时间持续时间——距离越大（越小）导致轨迹越短（越长）。例如，在所考虑的情况下，整个运动在9.5秒后结束，而示范持续10秒。根据具体应用，时间差异可能导致失败；因此必须加以考虑。最后，如果当前DMP的目标距离后续基元的演示起始点较远，速度阈值方法可能会产生不连续性。

3.2.2. 目标穿越。存在一些动作，如击打或挥棒，只有在非零速度下到达目标时才能正确执行。为此，Kober等人(2010b)对2.1.1节中的经典DMP公式进行了扩展，使得DMP能够跟踪以给定速度移动的目标。在他们的方法中，DMP在恰好T秒后以给定速度穿过目标。为了实现这一点，(1)中的加速度被重写如下

$$\ddot{\tau} \dot{z} = (1-x) a_z \Big(\beta_z (\widehat{g} - y) + \tau \Big(\dot{\widehat{y}} - \dot{y} \Big) \Big) + f(x)$$

 $\widehat{y}_m$ $\widehat{g}$

$$\widehat{g} = \widehat{g}(0) - \dot{\widehat{y}} \frac{\tau \ln(x)}{a_z},$$

$$\widehat{g}(0) = g - \frac{T \dot{\widehat{y}}}{\tau}$$

通过检查(55)和(56)，并考虑到项 $(-\tau \ln(x))^{\alpha_x}$ 表示如果 x 是(3)中定义的相位，则已经过去的时间，可以证明移动目标 $\widehat{g}$ 被设计为在 (T) 秒后到达目标 (g) ，即 $\widehat{g}(T) = g$ （图9左）。移动目标 $\widehat{g}(0)$ 的初始位置是通过将目标位置 (g) 以恒定速度 $(-\dot{\widehat{y}})$ 移动 (T) 秒得到的。运动开始时的较高加速度通过预因子 $(1-x)$ 避免，该因子在运动开始时 $(x(0) = 1)$ 被设为零。Nemec和Ude(2012)的方法结合了移动目标和后续DMP的特定初始化，以确保运动直到二阶导数的连续性。

Saveriano等人(2019)将此思想扩展到四元数DMP。(16)中的角加速度被修改如下

$$\begin{array}{l} \ddot{\tau} \dot{\eta} = (1-x) \alpha_z \Big(\beta_z \Big(\widehat{\mathbf{g}}_q - \overline{\mathbf{q}} \Big) + \tau \Big(\widehat{\boldsymbol{\omega}} - \boldsymbol{\omega} \Big) \Big) + \\ \quad \mathbf{f}_q(x), \end{array}$$

其中 $\widehat{\boldsymbol{\omega}}$ 是移动四元数目标 $\widehat{\mathbf{g}}_q$ 的角速度， $(2 \log^q \widehat{\mathbf{g}}_q - \overline{\mathbf{q}})$ 测量当前方向 $\mathbf{q}$ 与 $\widehat{\mathbf{g}}_q$ 之间的误差。预因子 $(1-x)$ 用于避免运动开始时的较大角加速度。四元数DMP的移动目标定义如下

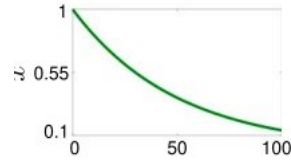
$$\begin{aligned} \widehat{\mathbf{g}}_q &= \exp\left(-\frac{\tau}{n} \ln(x)\right) \\ \widehat{\boldsymbol{\omega}} &= \widehat{\mathbf{g}}_q(0) \cdot \widehat{\mathbf{g}}_q(0) = \exp\left(-\frac{T}{2} \widehat{\boldsymbol{\omega}}\right) \cdot \mathbf{g}_q, \end{aligned}$$

其中 $\mathbf{g}_q$ 是目标四元数， T 是DMP的持续时间，指数映射 $\exp(q \cdot \cdot)$ 定义于(20)。如图9右所示，移动目标 $\widehat{\mathbf{g}}_q$ 在 T 秒后到达目标方向，即 $\widehat{\mathbf{g}}_q(T) = \mathbf{g}_q$ 。这可以通过考虑移动目标 $\widehat{\mathbf{g}}_q(0)$ 的初始值是将目标方向 $\mathbf{g}_q$ 以所需速度 $-\widehat{\mathbf{w}}$ 移动 T 秒计算得到来轻松验证。

所提出的目标穿越方法允许在 T 秒后穿过目标。假设有两个时间长度分别为 T^1 和 T^2 的DMP，可以通过先运行第一个DMP T^1 秒，然后切换到第二个DMP来将它们连接起来。与速度阈值方法类似，通过将 (DMP_2) 的状态初始化为 (DMP_1) 的最终状态，可以防止切换点处可能的不连续性。此过程可重复用于连接 $L \geq 2$ 个连续的DMP。

图10中的结果是在应用速度阈值合并两个单独训练的DMP以拟合最小加加速度轨迹（黑色虚线）时获得的。图10(a)–(e)显示位置，图10(f)–(j)显示运动的方向（单位四元数）部分。合并后的轨迹通过先跟随第一个DMP $T^1 = 5s$ ，然后切换到第二个DMP生成。所需的中间速度在每个方向上设置为 $0.01m/s$ （方向部分为 rad/s ）。生成的轨迹在10秒内到达目标，即演示和执行时间相同。按要求，经过点以所需速度在 $T = 5s$ 处被穿过（图10(c)和(h)）。然而，非零穿越速度引入了轨迹第一部分的变形（图10(e)和(j)）。

3.2.3. 基函数叠加。 Kulvicius等人(2011, 2012)的方法将多个DMP组合成一个复杂的DMP，通过确保(4)中构成 $f(x)$ 的基函数在切换实例处重叠，保证了基元之间的平滑过渡。首先，Kulvicius等人采用了(14)中的S形相位变量，而不是指数衰减的相位变量(3)。如第2.1.1.3节所述，S形相位在运动的大部分时间内为 (≈ 1) ，这使得可以使用较小的强迫项来复现演示。相反，指数相位在 T_s 之前已经接近零（图3），导致更大的强迫项。



经典加速度动力学在(1)中被修改如下：

$$\tau \dot{z} = a_z (\beta_z (\tilde{g} - y) - z) + f(s)$$

与目标穿越类似，Kulvicius等人将运动目标 $\tilde{g}$ 用于加速度动力学中，但将其称为延迟目标函数。(59)中的 $\tilde{g}$ 项通过对下式积分得到：

$$\tau \dot{\tilde{g}} = \begin{cases} \frac{\delta t}{T} (g - y_0), & t \leq T \\ 0, & \text{otherwise} \end{cases}$$

其中 $\tilde{g}(0) = y_0$ 。图9中的延迟目标函数在 T 秒内从 y_0 线性变化至 g ，随后保持恒定，即 $\tilde{g}(t \geq T) = g$ 。

非线性强迫项 $f(s)$ 在(59)中以绿色标出，因其与(4)中的经典形式略有不同。 $f(s)$ 定义如下：

$$f(s) = \frac{\sum_{i=1}^N w_i \Psi_i(t)}{\sum_{i=1}^N \Psi_i(t) s}, \quad \Psi_i(t) = \exp\left(-\frac{\left(\frac{t}{\tau} - c_i\right)^2}{2\sigma_i^2}\right)$$

其中 σ_i 为宽度， c_i 为第 i 个基函数的中心， s 通过对(14)积分得到。在(61)中使用项 (t/τ) 代替相位变量 x 。由于 $(0 \leq t/\tau)$ ，

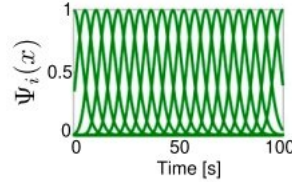
$(t/\tau \leq 1)$ 基函数在0和1之间等间距分布。最后， σ_i 为每个核的宽度，它们为常数且取决于核的数量。

在介绍了与经典方法的主要差异后，可以聚焦于Kulvicius等人(2012)如何解决连接 $(L \geq 2)$ 个DMP的问题。通常， L 个DMP中的每一个都具有不同的时间长度 (T^i) 、期望目标 (g^i) 和初始位置 (y_{0^i}) ，由此可通过积分计算延迟目标函数：

$$\dot{\tau}^i = \begin{cases} \frac{\partial}{\partial t} \{T^i\} \big(g^i - y_{0^i}\big), & \sum_{k=1}^{L-1} T^k \leq t \leq \sum_{k=1}^L T^k \\ 0, & \text{otherwise} \end{cases} \quad (54)$$

注意，由于 $\tilde{g}^i(0) = y_{0^i}$ ，加速度(59)是 s 。因此，(54)中使用的项 $(1 - x)$ 在(59)中不需要。

假设 L 个DMP已训练完毕，且每个DMP有 N 个核，我们可以按如下方式将它们合并为一个DMP：合并后DMP的中心计算如下：



$$c_i = \begin{cases} \frac{T^i(i-1)}{T_{\text{join}}(N-1)}, & i=1 \\ \frac{T^i(i-1)}{T_{\text{join}}(N-1)} + \frac{1}{T_{\text{join}}} \sum_{k=1}^{i-1} T^k, & \text{otherwise} \end{cases} \quad (55)$$

其中 (T^i) 为第 i 个DMP的时长， $T_{\text{join}} = \sum_{i=1}^L T^i$ 为合并运动的时长。合并后DMP的宽度计算如下：

$$\sigma_i = \frac{\sigma_i T^i}{T_{\text{join}}}$$

在(63)和(64)中分别计算的中心和宽度在过渡点处重叠，从而允许连续DMP之间的平滑过渡。合并后DMP的权重通过堆叠 L 个DMP的 N 个权重得到。因此，合并后的DMP拥有 $(N \cdot L)$ 个核和 $(N \cdot L)$ 个权重。相位变量(14)被修改为运行整个联合运动的时长 (T_{join}) 。

Saveriano等人(2019)将基函数覆盖方法扩展至四元数DMP。假设给定一个由 L 个四元数DMP组成的序列。(16)中的角加速度针对每个DMP重新表述为：

$$\dot{\tau}^i = a_z \Big(\beta_z^2 \operatorname{Log}^q \Big(\tilde{\mathbf{g}}_q^i \Big) \overline{\mathbf{m}}_q^i \Big) \quad (56)$$

其中 (i) 表示第 (i) 个四元数DMP， $(\mathbf{f}_q^i(s))$ 如(61)中所定义。项 $(\tilde{\mathbf{g}}_q^i)$ 是四元数延迟目标函数，在 (T^i) 秒内从 $(\mathbf{q}^i(0))$ 变化到 $(\mathbf{g}_q^i)$ （见图9[右]）。为了在保持 $(\mathbf{S}^3)$ 几何结构的同时生成这一运动目标，需要 $(\tilde{\mathbf{g}}_q^i)$ 沿连接 $(\mathbf{q}^i(0))$ 和 $(\mathbf{g}_q^i)$ 的测地线移动。因此， $(\tilde{\mathbf{g}}_q^i)$ 定义如下：

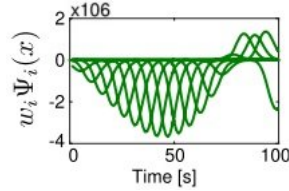
$$\tilde{\mathbf{g}}_q^i(t + \Delta t) = \exp \left(\frac{\tau^i \tilde{\mathbf{w}}^i(t)}{2} \right) \tilde{\mathbf{g}}_q^i(t)$$

其中

$$\tilde{\omega}^i(t) = \begin{bmatrix} 1 \\ \frac{2}{T^i} \operatorname{Log} \Big(\tilde{\mathbf{g}}_q^i \Big) \overline{\mathbf{m}}_q^i \Big) \end{bmatrix} \quad (57)$$

(67)中的角速度针对每个 $\mathbf{q}^l$ 计算。项 $(2\mathrm{Log}^q(\mathbf{g}_{q^l}^* \overline{\mathbf{q}^l(0)}))$ 表示在单位时间内将 $(\mathbf{q}^l(0))$ 旋转至 $(\mathbf{g}_{q^l})$ 的角速度。注意，映射 $(\mathrm{Log}^q(\cdot))$ 和 $(\mathrm{Exp}^q(\cdot))$ 分别定义于(18)和(20)中。延迟目标 $(\tilde{\mathbf{g}}_{q^l})$ 经过所有中间目标 $(\mathbf{g}_{q^l})$ 、 $(l=1, \dots, L-1)$ ，然后到达最终目标 $(\mathbf{g}_{q^L})$ 。

图11中的结果是在应用速度阈值合并两个分别训练以拟合最小加加速度轨迹（黑色虚线）的DMP时获得的。图11(a)–(e)展示运动的位置部分，图11(f)–(j)展示方向（单位四元数）部分。该方法无需切换规则，能自动生成平滑轨迹——如图11(c)和(h)所示具有连续速度——且该轨迹经过中间点附近，有利于整体复现精度（图11(e)和(j)）。然而，与中间点的距离取决于合并后基元的权重，无法单独决定。该方法生成的轨迹往往比演示轨迹持续时间更长，这是由于相位变量在 $(T + \delta_s)$ 秒后衰减为s形（图3）。根据具体应用，时间差异可能导致失败，必须予以考虑。



3.3. 在线自适应

标准的周期性DMP学习方法通过改变高斯核函数的权重来近似输入轨迹 $(\mathbf{y}_d)$ 在(39)中的形状 $(f_d(t))$ （Ijspeert等，2013）。权重的更新方式使得在每个控制步长以及在整个周期性重复过程中，参考轨迹与DMP之间的差异逐渐减小。然而，DMP也可以通过某种外部反馈函数进行重塑，以实现不同应用中的不同功能，例如需要试错法的任务（Kober等，2008）、机器人避障（Ginesi等，2021a；Hoffmann等，2009；Park等，2008；Tan等，2011）、引导（Gams等，2016；Petrić等，2014b）以及辅助外骨骼行为的适应（Peternel等，2016）。另外，现有周期性DMP的频率可以在线调节（Gams等，2009；Petrić等，2011）。

3.3.1. 机器人避障与引导。在Hoffmann等（2009）、Park等（2008）和Tan等（2011）的研究中，检测到的障碍物被拟合了一个势场函数，以改变DMP的形状来避开它。具体而言，Tan等（2011）利用势场计算时变目标并修改了生成的DMP轨迹，而Hoffmann等（2009）、Park等（2008）和Zhai等（2022）则在DMP中添加了一个额外的强迫项。类似地，在Gams等（2016）中，人类手臂被拟合了一个势场函数，用于重塑DMP以执行引导任务。该势场与人类手的位置耦合，以做出指向手势，并指示机器人手臂位置轨迹应改变的方向：

$$\dot{\mathbf{z}} = \Omega(a(\beta(\mathbf{y} - \mathbf{z}) + C_{\mathcal{O}} + f))$$

添加的耦合项 $(C_{\mathcal{O}})$ 是包含势场的避障项，为便于解释，其简化形式如下：

$$[C_{\mathcal{O}}]_s = d_s(\left\| \mathcal{O} - \mathbf{y} \right\|) \exp(-\zeta(\mathcal{O} - \mathbf{y}))$$

其中 $(\mathcal{O})$ 是障碍物（或人类指向手势）， $\mathbf{y}$ 是机器人位置。指数函数和 (ζ) 函数决定了势场，而函数 (d_s) 控制扰动场开始影响DMP的距离。关于 $(C_{\mathcal{O}})$ 及其参数的完整公式，请参见Gams等（2016）。在Rai等（2017）中，该方法被扩展以包含(68)中避障公式的泛化。在最近的一项研究中，Ginesi等（2021a）提出了一种利用超二次势函数理论实现体积避障的新方法。

另一种方法是，碰撞DMP轨迹中有缺陷的部分也可以由人类示范者直接在线调整（Karlsson等，2017b）。另一方面，Kim等（2015）的方法将避障视为优化问题的约束，从而修改DMP轨迹以防止碰撞。

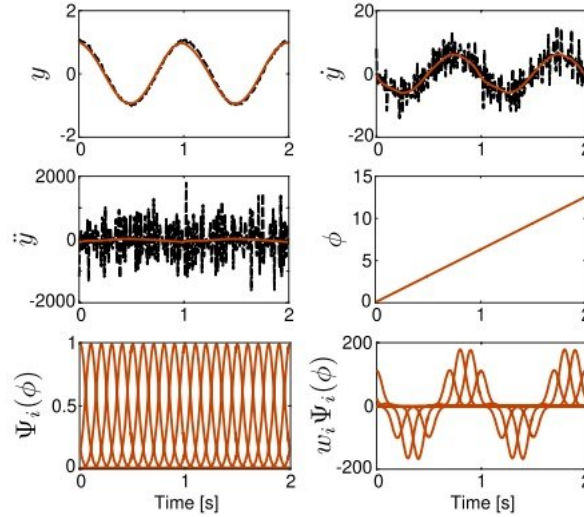
3.3.2. 基于力反馈的机器人适应。与避障类似，任务动力学也可以作为耦合项纳入DMP中。在Gams等（2014）中，任务动力学在DMP的加速度和速度层面进行耦合。所提出的方法被用于交互任务，其中人类根据施加在机械臂上的动力学改变机器人的行为。

$$\dot{\mathbf{z}} = \mathbf{a}_z(\beta_z(\mathbf{g} - \mathbf{y}) - \mathbf{z}) + \dot{\mathbf{C}}_z + \mathbf{f}(\mathbf{x})$$

$$\dot{\tau}\{y\} = z + C_{\{f\}}$$

其中力耦合项 $(C_f = \zeta F)$ 被定义为虚拟力或测量力 (F) ， (ζ) 是缩放因子，它本质上改变了DMP的动态行为，使运动基元能够对耦合的力即时做出反应。随后，Zhou等（2016b）引入了一种基于PD控制器的耦合项公式 $(C_{PD} = \zeta (K^{\mathcal{P}}(F_d - F^e) - D^{\mathcal{V}} \dot{F}^{\mathcal{e}}))$ ，该公式与DMP的速度部分（71）耦合。在该公式中， (F_d) 表示期望力， (F^e) 是测量力， (ζ) 是缩放因子， $(K^{\mathcal{P}})$ 和 $(D^{\mathcal{V}})$ 分别是比例-微分（PD）控制器的比例增益和微分增益。该耦合项公式实现了机器人运动对环境变化的受控适应。

在Kramberger等人（2018）的研究中，该方法得到扩展，将力反馈回路与DMP的速度（2）和目标 g 耦合。该方法的结果类似于导纳控制器（Villani和De Schutter, 2008）的行为，区别在于执行直接发生在轨迹生成层面。



$$\dot{\tau}\{z\} = a_{\{z\}}(\beta_{\{z\}}((g + C_{\{a\}}) - y) - z) + f(x),$$

$$\dot{\tau}\{y\} = z + \dot{C}_{\{a\}}$$

这里， $(\dot{C}_{\{a\}} = \text{varsigma}(F_d - F^e))$ 是导纳耦合项的一阶时间导数，它改变速度，进而改变经积分的耦合项，即DMP的位置输出。所描述的方法可用于笛卡尔空间运动，其中需将力替换为期望力矩和测量力矩。该方法可应用于涉及与环境接触以及与人类接触的机器人任务中。Lu等人（2021）提出的受限DMP公式统一了第3.3.1节和第3.3.2节中描述的公式，同时确保了生成运动的稳定性。

3.3.3. 外骨骼关节力矩自适应。在Peternel等人（2016）的研究中，利用人体努力提供信息，指示辅助外骨骼关节力矩DMP应改变的方向以使其最小化。通过将（41）中的误差计算替换为人体努力反馈项 $(U(E))$ ，将人类纳入机器人控制回路：

$$w_{\{i\}}(t_{\{j+1\}}) = w_{\{i\}}(t_{\{j\}}) + \Psi_{\{i\}} P_{\{i\}}(t_{\{j+1\}}) U(E)$$

其中 $(E(t))$ 是通过肌电图（EMG）信号测量的人体肌肉活动当前努力程度。 $(\{3\})$ 方程（34）–（38）和（42）以原始形式使用。方程（39）–（41）未被使用，因为（74）被用来调制（37）中的权重。

努力反馈项 $(U(E))$ 闭合回路，并作为反馈来调整定义轨迹形状的高斯核权重。正的 $(U(E))$ 增加，而负的 $(U(E))$ 减少编码关节力矩的周期DMP中给定区间的权重值。如果DMP的形状未提供足够的辅助力，人类必须付出努力（即肌肉活动）以产生在给定动力学下完成任务所需的剩余动力。随后，肌肉活动反馈增加DMP的幅度，直到人体努力项 $(U(E))$ 被最小化。请注意，每个关节都有自己的力矩DMP和 $(U(E))$ 项（Peternel等人，2016）。在此之后，除非任务、动力学或条件发生变化，DMP不会改变。若发生变化，人类必须通过额外的肌肉活动来补偿变化，从而使DMP适应新的所需关节力矩。

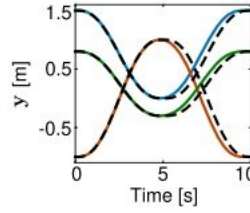
3.3.4. 基于参考速度的轨迹自适应。在许多从演示中学习 (LfD) 的场景中，需要在执行任何阶段同时修改学习到的空间运动和速度。首先在Nemec等人 (2013a) 中提出的速度缩放动态运动基元被用于底层任务表示。原始的DMP公式 (1) 和 (2) 通过在DMP的速度层添加时间缩放因子 ν 进行了扩展。

$$\dot{v}(x)\tau\dot{z}=a_z(\beta_z(g-y)-z)+f(x),$$

$$\dot{v}(x)\tau\dot{y}=z$$

从 (75) 和 (76) 可以明显看出，速度项是相位的函数，因此与 (4) 类似地使用一组RBF进行编码。该方法允许在轨迹执行的任何阶段修改空间运动和速度。作者在一个学习场景中展示了所提出的方法，在每个学习周期 (使用迭代学习控制 (ILC)) 之后，基于力/力矩反馈编码新的速度曲线，从而收敛到特定任务的最优速度。Vuga等人 (2016) 通过引入非均匀加速运动的紧凑表示以及运动参数的简单调制，扩展了该方法。

随后，在Nemec等人 (2018) 的工作中，作者将先前的方法扩展至包含以单位四元数表示的编码方向轨迹的速度缩放。该研究成果提出了一种统一的速度缩放方法，适用于在笛卡尔空间中执行的任务。此外，Gašpar等人 (2018) 提出了一种名为弧长DMPs (AL-DMPs) 的速度方法重构。在这项工作中，他们提出了一种方法，通过基于时间参数化轨迹的弧长，将运动的空间和时间分量分离。弧长基于曲线的微分几何，与运动速度相关，而速度则表示为示教轨迹的时间导数。该方法特别适用于比较多条示教轨迹以提取学习相关信息的情形。继AL-DMPs的思想之后，Simonič等人 (2021) 引入了恒定速度DMPs，以完全解耦任务的空间和时间部分。Pahic等人 (2021) 使用深度神经网络将图像映射为由AL-DMPs表示的空间路径。Weitschat和Aschemann (2018) 添加了一个额外的强迫项，以将速度保持在预设的限值内。该工作的目的是确保机器人任务在与人类交互时的安全执行，并为在机器人位置和速度需随时间变化的环境中实现安全交互提供框架。关于耦合项的完整公式，请参见Weitschat和Aschemann (2018)。



为了在保持轨迹一致性的同时确保每个自由度的速度有界，必须在所有维度上均匀降低速度。这一原理促使考虑时间耦合，即修改DMP的时间常数 (相当于路径遍历的持续时间)，而非空间耦合。通过增大时间常数，所有自由度的时域演化可以同时以相同的因子减缓。基于跟踪误差的时间耦合方法由Ijspeert等人 (2013) 提出，其增强版本由Karlsson等人 (2017a) 提出。然而，该方法需要扭曲路径以实现轨迹减速，因此无法直接用于限制速度或加速度。一年后，作者在Karlsson等人 (2018) 中提供了时间耦合DMPs的稳定性分析。Dahlin和Karayiannidis (2020) 在其工作中提出了一种基于排斥势的时间耦合方法，在保持路径形状不变的前提下，将DMP速度限制在预设的速度界限内。随后，Dahlin和Karayiannidis (2021) 引入了一种DMP时间耦合方法，能够控制生成轨迹的速度和加速度约束。该方法包含一个滤波机制，可在接近加速度约束时主动减速轨迹，从而降低不可行性的潜在风险。

3.4. 柔性关节机器人

DMPs通常用于在具有显著弹性的机械臂中生成轨迹。这一选择通常由相对较低的执行速度所证明，这导致较低的加速度和加加速度水平。然而，当需要高加速度和高加加速度的快速运动时，将DMPs用于轨迹生成与逆动力学用于前馈控制相结合，可能会导致具有柔性模态的机器人出现振荡。在工业环境中，由于节拍时间的要求，执行速度通常快于示教速度，这成为一个重大挑战。为解决此问题，Wahrburg等人 (2021) 提出了一种面向柔性关节机器人的DMP扩展框架，称为FlexDMP，通过引入一个四阶系统来生成轨迹，如下所示

$$\begin{bmatrix} \tau\ddot{z} \\ \tau\ddot{y} \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} (g-y-z) - \dot{z} + f(x),$$

$$\tau \ddot{y} = \ddot{z}, \tau$$

$$\tau \ddot{y} = \dot{z}, \tau$$

$$\tau \dot{y} = z$$

其中 $(y, \dot{y}, \ddot{y})$ 分别为位置、速度、加速度和加加速度。规范系统和强迫项与(3)和(4)中的类似。对于临界阻尼系统， $(\alpha_1 = 4\beta)$ 、 $(\alpha_2 = 3/2\beta)$ 、 $(\alpha_3 = 2/3\beta)$ 和 $(\alpha_4 = 1/4\beta)$ ，其中 $(\beta > 0)$ 。

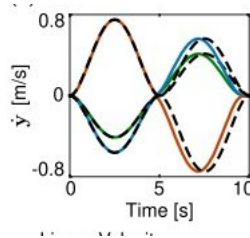
3.5. 其他形式

LfD是一个广泛的研究领域，目前已发展出多种方法用于复现人类示教（Billard等，2016）。如前所述，本教程综述旨在提供对DMP研究的全面概述，因此我们有意跳过了LfD领域丰富的文献。然而，我们发现一些表示方法与DMP公式密切相关。本节将简要回顾这些方法。

Calinon等（2009）以类PD形式计算机器人的加速度指令

$$\ddot{\mathbf{y}} = \mathbf{K}^{\mathcal{P}}(\mathbf{y}_d - \mathbf{y}) + \mathbf{D}^{\mathcal{V}}(\dot{\mathbf{y}} - \dot{\mathbf{y}}_d)$$

其中 $\mathbf{K}^{\mathcal{P}}$ 为刚度， $\mathbf{D}^{\mathcal{V}}$ 为阻尼增益， $\mathbf{y}$ 为机器人的实测状态， $\dot{\mathbf{y}}$ 为其时间导数（速度）， $\mathbf{y}_d$ 和 $\dot{\mathbf{y}}_d$ 分别为通过GMR获取的期望位置和速度。作者随后指出，加速度指令 $\ddot{\mathbf{y}}$ 可视为多个线性动力学的混合，每个动力学收敛至某吸引子。尽管后续工作如Kormushev等（2010）将这种表示称为DMP的“修正版本”，但Calinon等（2012）明确指出了其与标准DMP公式之间的显著差异。



Herzog等（2016）基于线性系统计算机器人的加速度指令

$$\ddot{\mathbf{y}} = \mathbf{u} = \mathbf{K}^{\mathcal{P}}(\mathbf{y}_d - \mathbf{y})$$

其中 $\mathbf{y}$ 为机器人的实测状态， $\mathbf{y}_d$ 为人类示教轨迹， $\mathbf{K}^{\mathcal{P}}$ 为通过线性二次型调节器方法计算的控制增益。随后，利用切比雪夫多项式计算控制输入轨迹 $\mathbf{u}$ 的紧凑表示。该表示无需依赖衰减的相变量来保证收敛，但泛化至不同起始/目标位置时，需重新应用线性二次型调节器方法以寻找新的控制输入序列。

关于周期运动，Ajallooeian等（2013）提出了一种基于动力系统的框架，用于学习任意形状和吸引域的节律性运动。他们利用基于相位的缩放函数来表示已知基础极限环与期望周期轨道之间的映射。基础极限环可取自例如周期性DMP生成的极限环，这使得Ajallooeian等（2013）的方法成为周期基元的一种更通用的公式。

4. DMP在复杂框架中的集成

本节回顾将DMP集成到更大执行框架中的方法。我们将这些方法分为五个主要研究领域：抓取与操作、阻抗学习、强化学习、深度学习以及增量式与终身学习。

4.1. 操作任务

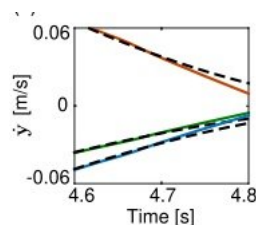
4.1.1. 抓取与工具使用。成功抓取物体是机器人操作的第一步。抓取需要感知环境，特别是视觉感知，以定位物体并根据其几何特征确定合适的抓取点。在此背景下，即使微小不确定性也可能导致物体滑落和抓取失败。为提升视觉驱动抓取的鲁棒性，Krömer等（2010a）通过基于视觉描述子的势场增强DMP，以根据物体的局部几何形状调整手与手指的轨迹。该抓取策略被集成到层次化控制架构中，其中上层确定物体的抓取位置，下层局部调整运动以实现稳健抓取（Krömer等，2010b）。Stein等（2014）提出了一种基于表面凸凹性的点云分割方法，该方法特别适用于识别物体手柄，并采用DMP在实际机器人上执行抓取。

能够抓取和使用工具也是执行日常生活操作所需的能力。为此，Guerin等人（2014）提出了所谓的工具运动基元，该基元将示教转换到工具可供性框架中。其结果是生成一种能够泛化到不同工具姿态以及共享相同可供性工具的运动。Li和Fritz（2015）研究了使用低成本、非灵巧夹爪的工具操作，并提出了一个框架来学习双边操作策略以补偿灵巧性的不足。双边机器人操作是一项具有挑战性的任务，需要手部运动之间的精确协调并遵守空间约束。Thota等人（2016）开发了一种基于DMP的双边操作控制框架，该框架在保证双手时间同步的同时，对空间扰动和目标变化具有鲁棒性。

4.1.2. 运动基元序列化。除了物体抓取外，日常操作还需要精确执行复杂运动。通常，这些复杂运动难以编码为单一运动基元，但可以方便地拆分为更简单的运动（例如，伸手和抓取），这些运动可以被合理排序并执行（图12）。

将DMP作为复杂任务构建模块的可行性在Caccavale等人（2018, 2019）和Ramirez-Amaro等人（2015）的研究中得到了探讨。在这些工作中，人类教师演示了一个相对复杂的任务，该任务包含对多个物体执行的若干动作。随后，示教被自动分割为M个基本运动，用于拟合M个DMP。Ramirez-Amaro等人（2015）利用语义规则（例如，用刀伸向物体意味着切割）来推断高层人类活动，而Caccavale等人则构建了一个层次结构，通过为当前执行上下文选择合适的DMP来安排复杂任务的执行。他们使用了动觉示教和口头指令（开/关夹爪命令）来提供任务演示。Schwaner等人（2021）利用DMP构建了一个外科手术技能库，用于自主执行缝合任务的部分操作。Lemme等人（2014）将分割后的任务示教组织成一个运动基元库，该库由自我生成的轨迹片段学习而来。他们还引入了一种机制来移除未使用的技能并更新该库。Eiband等人（2019）还利用动觉示教和触觉反馈来分割和识别基本运动或技能，并构建一棵描述连续技能之间几何关系（如参考坐标系和目标姿态）的树。在运行时，机器人通过触觉探索定位场景中的物体并更新技能树。随后，技能树中的变换被用于定义DMP的初始和目标姿态，并执行任务。最后，Wu等人（2018）将DMP整合到一个包含语音和本体的对话系统中，以使用自然交互方式学习或重新学习任务。

4.1.3. 数据采集。当无法使用动觉示教或基于标记的运动追踪系统时，采集示教数据便成为一个问题。后者需要昂贵的传感器基础设施，在工厂车间等现实场景中难以搭建。动觉示教则需要力矩控制/协作机器人，这在工业场景中仍不常见。为解决这一问题，Mao等人（2015）利用低成本RGB-D相机，并采用Oikonomidis等人（2011）提出的无标记方法追踪人手。收集到的数据随后被分割为基本运动，用于拟合DMP。此外，Yang等人（2022）的方法无需追踪标记或手动标注。相反，作者利用机器人和人类与物体之间随机非配对交互的视频，进行视觉对应关键点模型的无监督学习。随后，通过贝叶斯优化，仅从单个人类视频示教中，在少量机器人试验内找到节律性DMP的参数。



所述方法假设人类教师始终提供一致且无噪声的任务演示。Ghahramani E. 等人（2015）将带噪声的演示编码为GMM，并利用GMR计算出无噪声轨迹，随后将该轨迹用于拟合DMP，使其能够泛化至不同的起始点、目标点及障碍物配置。Dong等人（2023）提出从正确（正向）与错误（负向）演示中拟合DMP，以增强模型的表征与泛化能力。Niekum等人（2012, 2015）设计了一个框架，通过对任务演示进行分割、识别相似技能并泛化任务执行过程，从而从非结构化演示中学习。有趣的是，Gutzeit等人（2018）对10名志愿者开展的用户研究表明，现有的分割与学习策略已足够稳健，能够在合理时间内实现从人类到机器人的操作技能自动迁移。

4.1.4. 任务学习与执行。部分研究（Deniša 和 Ude, 2013a, 2013b, 2015; Denisa 等, 2021）利用转移图与树结构来嵌入轨迹片段，并借助搜索算法发现部分轨迹序列，从而生成未演示过的运动。依赖分层树状结构表征任务的方法在任务泛化能力上存在局限。Lee 和 Suh（2013）采用概率推理与对象可供性推断出能够应对执行环境中不确定性的适当技能。Beetz等人（2010）通过观

察学习刻板任务解决方案，并利用任务规划与符号推理执行新颖的移动操作任务。Wörgötter 等人（2015）提出了一种生成式学习框架，用于在认知架构的不同层面（包括符号规划以及对对象与动作属性）补充机器人知识库中缺失的信息。Paxton 等人（2016）利用任务与运动规划泛化复杂装配任务的执行，并提出了一种通过演示学习的方法来对符号动作进行具体化。Agostini 等人（2020）结合场景中对象间几何关系的对象中心描述、基于计划中连续三个动作的符号到运动分层分解，以及Caccavale 等人（2019）提出的LfD方法（图12），实现了任务与运动规划。Acin 等人（2013）在三个不同层次上描述操作任务：顶层提供动作、对象及其关系的符号描述；中层使用有限状态机生成由底层具体化的动作基元序列。这些方法的一个共同点是利用DMP在真实机器人上执行任务。

4.2. 变阻抗学习控制

阻抗控制可用于实现柔顺运动，其中控制器类似于环境与机器人末端执行器之间的虚拟弹簧-阻尼系统（Hogan, 1985）。该方法能够实现机器人与环境（可能包括人类）之间平滑、安全且节能的交互。此类交互的标准模型定义如下：

$$\begin{aligned} \frac{d}{dt} \mathbf{M} \dot{\mathbf{y}}_t &= \mathbf{K}_t^{\mathcal{P}} \big(\mathbf{y}_g - \mathbf{y}_t \big) - \\ & \mathbf{D}_t^{\mathcal{V}} \dot{\mathbf{y}}_t + \mathbf{f}_t^{\mathcal{e}}, \\ \dot{\mathbf{I}} \dot{\boldsymbol{\omega}}_t &= \mathbf{K}_t^{\mathcal{O}} \left(\mathbf{R}_g - \mathbf{R}_t \right) \mathbf{I} \left(\mathbf{R}_g - \mathbf{R}_t \right)^{\mathcal{O}} - \\ & \mathbf{D}_t^{\mathcal{W}} \boldsymbol{\omega}_t + \boldsymbol{\tau}_t^{\mathcal{e}} \end{aligned}$$

其中，(81)和(82)分别对应平移与旋转情形， $\mathbf{M}_t^{\mathcal{P}}$ 、 $\mathbf{D}_t^{\mathcal{V}}$ 分别为平移运动的质量矩阵、刚度矩阵和阻尼矩阵， $\mathbf{I}_t^{\mathcal{O}}$ 、 $\mathbf{D}_t^{\mathcal{W}}$ 分别为旋转运动的惯性矩阵、刚度矩阵和阻尼矩阵。 $\widehat{\mathbf{R}}_t$ 、 $\mathbf{R}_t \in \mathcal{SO}(3)$ 为旋转矩阵，分别对应期望旋转目标与末端执行器的实际姿态曲线。 $\mathbf{f}_t^{\mathcal{e}}$ 和 $\boldsymbol{\tau}_t^{\mathcal{e}}$ 分别表示作用于机器人末端执行器的外力与力矩。

事实上，变阻抗控制（VIC）在机器人需要与环境交互以避免高冲击力及对自身或环境造成损伤（即切换至低刚度）时扮演着重要角色（Ajoudani et al., 2012; Abu-Dakka et al., 2018; Peternel et al., 2018a）。另一方面，它在抑制来自环境的意外及不可预测扰动以实现期望的位置跟踪精度（即切换至高刚度）方面同样关键（Yang et al., 2011）。此外，该技术对于协调人机协作运动也具有重要意义（Petemel et al., 2017b）。然而，机器人系统仍需学习如何将此类VIC适应于未见情境，同时避免硬编码。这种学习范式被称为变阻抗学习控制（VILC）。感兴趣的读者可参阅我们近期关于VILC的综述（Abu-Dakka and Saveriano, 2020）或关于远程阻抗的研究（Petemel and Ajoudani, 2022）。

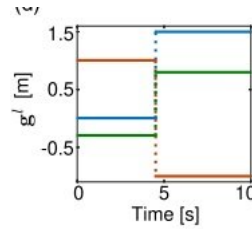
在本综述中，我们将提及部分将DMP与VIC集成于VILC框架的工作。图13展示了一个简单通用示例，其中DMP被集成至VIC控制方案中。

Buchli et al. (2011a) 提出了最早将DMP与基于路径积分的策略改进（PI²）算法（Theodorou et al., 2010）相结合的方法之一，用于在优化阻抗参数的同时学习运动（由DMP表示的位置与速度）。随后，作者利用对角刚度矩阵，并将每个对角元素的变分（时间导数）表示为如下形式

$$\frac{d}{dt} k_{\vartheta_j, t} = a_j \big(\bigwedge_{j=1}^{\text{top}} \big(\vartheta_j + \epsilon_{j, t} \big) - k_{\vartheta_j, t} \big), j=1, \dots, J$$

其中 j 表示第 j 个关节， k_{ϑ_j} 为关节 j 的刚度， $\epsilon_{j, t}$ 是时变探索噪声，每个 $\boldsymbol{\lambda}_j$ 是一个包含 N 个高斯基函数的向量， ϑ_j 是关节 j 的可学习参数。式（83）中的刚度参数化同样关于参数线性，因此可应用 $\mathcal{PI}^2$ 来寻找最优策略。后来，作者利用 $\mathcal{PI}^2$ 在确定性与随机力场中学习VIC（Stulp et al., 2012a）。Nakanishi et al. (2011) 提出了一种方法，用于优化周期性运动及其时变关节刚度。

Basa and Schneider (2015) 对DMP公式进行了扩展，通过添加第二个非线性函数以处理弹性机器人，形式如下



$$\dot{z} = a_z(\beta_z(g-y)-z) + f(x) + f_2$$

其中 f_2 定义为式 (4)，但不含相变量 x 。 f_2 的主要目的是补偿运动时间结束时及其后对移动自由度产生的重力影响。与此不同，Haddadin et al. (2016) 采用最优控制来执行弹性机器人的近似最优运动。

Nemec et al. (2016) 提出了一种协作控制方案，使双臂机器人能够在线调整其刚度，同时执行轨迹以实现精确演化。Umlauf et al. (2017) 使用高斯过程 (GP) 结合 DMP (如 Fanger et al. (2016) 所提出) 来预测轨迹。在执行过程中，其导纳控制器在线同时调整刚度与阻尼。基于能量罐的无源性控制方法已与 DMP 集成，通过在线调整刚度，在未知环境中稳定适应接触并保持无源性 (Shahriari et al., 2017; Kramberger et al., 2018; Kastritsi et al., 2018)。

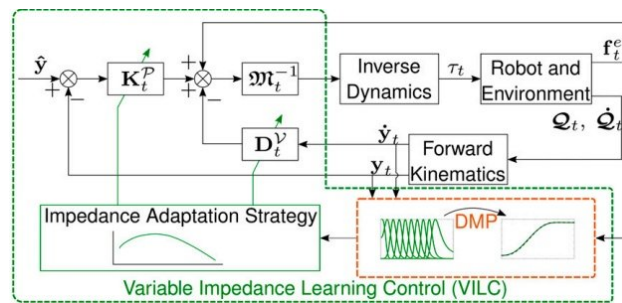


图13. VIC与DMP的通用控制方案。

Bian等人(2019)、Peternel等人(2014, 2018b, 2018a)以及Yang等人(2018, 2019)的方法设计了不同的多模态接口，让人类能够明确地向机器人教授阻抗行为。这些方法大多将基于肌电信号 (EMG) 的可变阻抗技能迁移与基于动态运动基元 (DMP) 的运动序列规划相结合，继承了这两方面在机器人技能获取中的优点。Hu等人(2018)使用协方差矩阵自适应进化策略 (CMA-ES) 更新 DMP 和可变阻抗控制器的参数，以减少机器人在噪声环境中运动时的冲击。Dometios等人(2018)将坐标变换-动态运动基元 (CC-DMP) 与基于视觉的运动规划方法相集成，以调整机器人末端执行器的参考路径，并实现清洗动作的执行。

Travers等人(2016, 2018)首次在运动领域提出了基于形状的柔顺控制器，通过在复杂障碍环境中移动的蛇形机器人上实现振幅柔顺。得益于本体感受步态柔顺技术，他们的方法使蛇形机器人能够盲目适应这类复杂的非结构化地形。

最近，Wang等人(2020)提出了一种自适应导纳控制器，该控制器集成了高斯混合回归 (GMR) 用于提取人体运动特征、DMP用于编码可泛化的机器人运动，以及基于径向基函数神经网络 (RBF-NN) 的控制器用于再现阶段的轨迹跟踪。在其工作中，Spector和Zacksenhouse(2021)在DMP框架内引入了一种残差导纳策略。该策略显式地学习了完整的非对称刚度矩阵，旨在纠正基线策略生成的运动。通过学习策略的有效性，在涉及不同形状和尺寸销钉的销钉插入任务中得到了验证。此外，该策略在处理孔位和销钉方向的不确定性方面表现出鲁棒性，并对新形状具有良好的泛化能力。而且，该学习策略展示了从仿真到真实场景的成功迁移性。

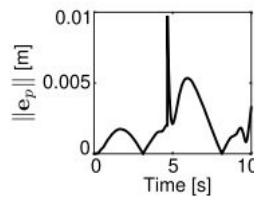
新颖的示教学习 (LfD) 方法明确考虑了训练数据可能由某些带有相关度量的黎曼流形生成。Abu-Dakka和Kyrki(2020)基于黎曼度量重新阐述了DMP，使得所得公式能够处理SPD流形上的对称正定 (SPD) 数据。他们的公式能够适应新的目标SPD点。

最近，仿生控制器已与DMPs相结合 (Zeng等人, 2021)，以学习和适应柔顺技能。

4.3. 强化学习 (RL)

在RL中, 智能体通过探索不同策略(动作)并接收其动作结果的反馈(奖励), 尝试通过试错来改进其行为。动作 a 从策略 $\pi(s, a)$ 中抽取, 该策略表示状态 s 与动作 a 之间的映射。RL的目标是找到最优策略 π^* , 以最大化累积期望奖励, 即可能无限时间区间内期望奖励之和。当智能体是在现实世界中执行任务的机器人时, 状态空间和动作空间本质上是连续的。此外, 机器人智能体还受到不完美(例如噪声)感知和不准确模型(例如接触)的影响。最后, 与现实世界进行大量交互(rollouts)成本高昂且可能存在危险。如Kober等人(2013)所讨论, 机器人特有的挑战需要特定的解决方案才能使RL问题变得可行。

4.3.1. DMP作为控制策略



一种可能性是使用参数化策略, 并利用RL搜索一组最优的有限策略参数。在这方面, DMPs已被广泛用作策略参数化形式。总体思路如图14所示。更详细地, (Peters和Schaal, 2008a; Schaal, 2006)表明, 各种策略梯度和演员-评论家RL方法可以有效应用于改进参数化为DMP的机器人技能。其他研究则专注于开发专门针对参数化策略的策略搜索算法。受随机最优控制启发, Theodorou等人(2010)

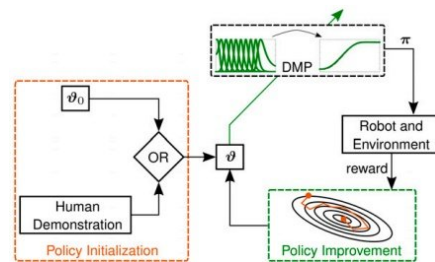


图14. 基于DMP的策略改进通用框图。

提出了 π^* , 该方案将路径积分最优控制应用于DMP。 π^* 与DMP已成功应用于多个领域, 包括VILC (Buchli等, 2011a, 2011b)、接触类任务 (Hazara和Kyrki, 2016)、状态估计不确定性下的抓取 (Stulp等, 2011a)、双臂操作 (Zhao等, 2020)、非抓取操作 (Sun等, 2022) 以及机器人辅助血管介入手术 (Chi等, 2018)。Kober和Peters (2011) 从期望最大化中推导出了所谓的“基于回报加权探索的策略学习”(PoWER)。PoWER与DMP已成功应用于执行高动态任务, 包括杯中小球 (Kober和Peters, 2011) 和煎饼翻转 (Kormushev等, 2010)。

4.3.2. 限制搜索空间。即使采用参数化策略, 寻找最优策略参数所需的 rollout 次数仍可能很大, 尤其是对于拥有多个自由度的机器人。可利用降维技术在缩减后的空间中进行策略搜索 (Colomé和Torras, 2014)。该方法的有效性已在具有挑战性的衣物(即软组织)操作任务中得到验证 (Colomé和Torras, 2018)。模仿学习作为一种有效的策略初始化方法, 通过减少 rollout 次数来加速策略搜索 (Kober和Peters, 2010)。在此方面, Kober等人 (2008, 2010a) 为DMP增加了感知耦合项, 并提议通过人类模仿初始化DMP, 再通过强化学习优化运动技能。模仿学习最终可与降维技术结合使用 (Tan和Kawamura, 2011), 并可先在仿真中进行多次 rollout (Cohen和Berman, 2014), 以进一步加速策略搜索。当提供多个示教时, 可以学习策略参数与查询点(如目标位置)之间的映射, 并利用该映射泛化到新情境(第3.1.3节)。Nemec等人 (2011, 2012, 2013b) 使用这一策略为新情境提供良好的初始策略, 随后通过强化学习进一步优化。由于映射是基于示例查询点估计的, 搜索空间可有效限制在查询点附近, 从而使策略搜索更加高效。Vuga等人 (2015a, 2015b) 将该方法与另一种DMP公式结合, 优化了执行速度。该方法在多种任务上进行了测试, 包括将水倒入杯中, 在运动过程中防止水溅出。Schroecker等人 (2016) 以软经由点(第3.1.2节)的形式提供示教, 将搜索空间缩小至所教经由点的邻域。Reinhart和Steil (2014, 2015) 利用多个示教构建了一个参数化的技能记忆库, 将低维技能参数化与运动基元参数相联系。随后利用这一低维嵌入进行高效的策略搜索。QueiBer等人 (2016) 并非学习从任务到策略参数的映射, 而是利用rollout数据逐步学习一个参数化技能(自助法), 并以此为新任务生成良好的初始策略。

4.3.3. DMP泛化与序列化。部分研究者并非利用泛化来提供更好的初始策略，而是借助强化学习来改进和泛化运动基元。André等人（2015）调整DMP策略以适应斜坡行走。Mülling等人（2010）使用DMP混合模型泛化到新情境。在该方法中，强化学习用于估计形状参数以及每个DMP的最优责任权重。Mülling等人（2013）使用情节强化学习来估计元参数，例如乒乓球任务中球与球拍的时空拦截点。Lundell等人（2017）使用参数化核权重和强化学习搜索最优参数，而Forte等人（2015）则利用基于强化学习的状态空间探索增强给定示教，以自主扩展机器人的任务知识。Hangl等人（2015）利用度量强化学习，在已学习的DMP策略之间平滑切换，并在新情境中执行任务。

强化学习也可用于序列化多个运动基元并执行更复杂的任务；当机器人需要执行例如操作任务时，这是一种成功的策略（第4.1节）。为了序列化多个基元，学习每个运动的目标也至关重要。Tamosiunaite等人（2011）使用连续值函数逼近来优化用于执行倒水任务的DMP的目标参数。Kober等人（2011, 2012）学习了一个元参数函数，该函数将当前状态映射到一组元参数，包括运动的目标和持续时间。Stulp等人（2011b, 2012）没有将形状学习和目标学习分离为不同过程，而是扩展了PI²，使其能够同时学习一系列DMP的形状和目标。Wang等人（2022）将复杂的操作任务和用户执行偏好描述为逻辑和时间约束，并使用强化学习来找到满足这些约束的一组DMP参数。

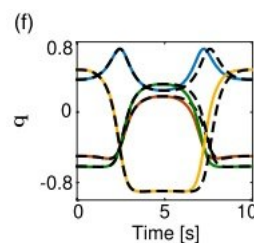
4.3.4. 技能迁移。学到的技能可以潜在地跨不同任务迁移，以加速学习过程并提高机器人的自主性。为此，Fabisch和Metzen（2014）考虑了机器人可以主动选择要学习的任务以在学习中取得最佳进展的情况。主动选择任务的过程被视为一个非平稳赌博机问题，对此存在合适的算法解决方案，同时利用内在动机启发法在任务选择后对智能体进行奖励。Cho等人（2019）基于多个演示的时间和空间熵来定义运动技能的复杂度，并使用测量到的复杂度生成学习和迁移运动技能的顺序。他们的实验结果提供了技能学习和迁移的有用指南。简而言之，人类应在可能的情况下演示最复杂的任务，然后机器人能够迁移运动技能。反之，如果未提供演示，则先学习简单技能，再将简单技能迁移到更复杂的任务中会更有效。

4.3.5. 分层技能学习。强化学习在处理高维连续状态和动作空间时往往缺乏可扩展性。为解决此问题，分层强化学习采用分而治之的方法，将强化学习问题分解为层次化的子任务，以缩小搜索空间。层次结构中的不同层级代表不同时间和/或空间尺度上的信息。

Stulp和Schaal（2011）提出将不同的选项表示为待序列化的DMP。 PI^2 被扩展以优化每个DMP在时间抽象不同层级上的形状和（子）目标。具体而言，形状根据序列中下一个基元之前的代价进行调整，而子目标则考虑两个DMP整个序列的代价。End等人（2017）中的分层直接策略搜索不依赖于预定义的一组子策略和/或子目标，而是利用信息论原理来揭示一组多样化的子策略和子目标。

减少发现最优策略所需的rollout次数在分层强化学习（HRL）中同样重要。如前所述，模仿学习（IL）是寻找良好初始策略的有效方法。然而，在某些应用中（例如使用多指机器人手进行操作），很难甚至无法提供专家示范。为了提高策略搜索效率，Ojer De Andres等人（2018）采用了HRL方法，其中上层考虑离散的动作和状态空间，以搜索最优的手指步态及手指间的同步。该信息被传递给下层，由节律性DMP和 PI^2 生成连续的手指指令。另一种提高数据效率的方法是采用基于模型的强化学习方法。Colome等人（2015）利用摩擦模型改进了DMP策略，并实现了对软组织（围巾）的操作。Kupcsik等人（2017）提出了一种基于模型的HRL方法，用于数据高效地学习上层策略，这些策略能够很好地泛化到不同的执行情境中。Li等人（2018）提出了一种混合分层框架，其中上层在笛卡尔空间中计算最优规划，并通过高效求解器将其转换为期望的关节目标。下层则负责在不确定性条件下，利用强化学习和DMP学习关节空间轨迹。最近，Davchev等人（2022）提出了残差LfD框架，该框架结合了DMP和强化学习，用于学习装配任务中的残差修正策略。在该论文中，他们展示了直接在任务空间应用残差学习，并对机器人的完整位姿进行操作，能够显著提升DMP的整体性能。

4.4. 深度学习



一种流行的机器学习方法是神经网络（NN）。由于其非参数特性，它们能够有效表示非线性映射。神经网络过去的一个主要缺点是学习过程的计算复杂度。近年来，人们对神经网络重新产生了兴趣。新的深度学习方法已成功（LeCun 等人，2015）应用于机器视觉和语言处理。

近年来，深度学习也已应用于机器人领域，用于学习任务动力学（Yang 等人，2016）和运动维度降维（Chen 等人，2015）。作者（Chen 等人，2015, 2016）提出了一种名为AutoEncoded DMP（AEDMP）的框架，该框架使用深度自编码器来寻找以潜在特征空间表示的运动。在该空间中，DMP能够最优地泛化到新任务，同时该架构使得DMP可以作为一个整体进行训练。Pervez 等人（2017b）在其工作中，将用于物体分类的视觉感知数据与由DMP表示的任务特定运动定义相结合。这些数据通过卷积神经网络（CNN）进行建模，其中图像和相关运动直接由深度神经网络处理，从而保留了相关的DMP属性，并消除了运动复现过程中提取任务参数的需要。此后，Kim 等人（2018b）将深度强化学习与DMP相结合，从示范中学习和泛化机器人技能。该框架建立在强化学习方法的基础上，基于示范来学习和优化新的DMP技能。强化学习方法由一个分层搜索策略支持，减少了机器人的搜索空间，从而能够更高效地学习复杂任务。此外，Pan 和 Manocha（2018）提出了一种深度学习方法，用于复杂环境中高维可变形机器人的运动规划。运动技能由DMP编码，并训练一个神经网络用于避障和导航。这些数据进一步通过深度Q学习进行优化，表明所学习的规划器能够实时高效地为高维机器人规划并执行导航任务。

Pahic 等人（2018）提出了一种用于感知-动作耦合的深度学习方法，展示了基于视觉的图像与相关运动轨迹之间的耦合。随后，他们扩展了该方法以纳入CNN，并为其给出了一个区分性的性质表述（Pahic 等人，2020），该方法利用一个损失函数来测量物理距离

运动轨迹，而非测量无物理意义的DMP参数之间的距离，从而提升了算法的性能。最近，他们扩展了GPR的用途，以创建训练自编码器神经网络进行降维所需的数据库（Lončarević等，2021）。Mavsar等人（2022）在他们的工作中提出了一种循环神经架构，能够将变长的输入运动视频转换为描述机器人轨迹的一组参数，这些参数随后通过DMP进行编码，从而在仅接收少数几帧后即可进行预测。此外，利用仿真环境扩大训练数据库并提升网络的泛化能力，该网络用于机器人交接任务。此外，Jaques等人（2021）在他们的研究中引入了牛顿变分自编码器（Newtonian VAE），这是一种用于学习潜在动力学的框架。受牛顿第二定律启发，他们在隐藏空间中定义了一个线性动态系统。该系统基于具有J个自由度的刚体模型。Newtonian VAE框架能够形成PD可控的状态空间，从而在学习的潜在空间内利用DMP基于视觉演示实现鲁棒的路径跟踪。

4.5. 终身/增量学习

终身（增量）学习是一种框架，能够对顺序到达的任务进行持续学习（Chen和Liu，2018；Fei等，2016；Thrun，1996）。该框架的核心组成部分是一个数据库，用于维护从先前学习任务中获得的知识 $\{TSK_1\}$ 、 $\{TSK_2\}$ 、...、 $\{TSK_{N-1}\}$ 。增量学习从任务管理器将新任务 $\{TSK_N\}$ 分配给学习代理开始。在这种情况下，代理利用数据库中的知识作为先验数据，以增强其模型在新任务上的泛化性能。当新任务 $\{TSK_N\}$ 学习完成后，数据库会使用从学习 $\{TSK_N\}$ 中获得的知识进行更新。实际上，增量学习框架为代理提供了三种能力：（i）持续学习，（ii）知识积累，以及（iii）重用先前的知识以增强未来的学习。图15展示了DMP与终身学习框架集成的总体结构。

Churchill和Fernando（2014）提出了一种认知架构，能够以允许重新组合和重用任务特定能力的方式，在多个任务中积累适应性和技能。Lemme等人（2014）基于几何相似性对示教进行分割，随后创建了一个运动基元库。该库通过移除未使用的技能并添加新技能来更新。Reinhart和Steil（2014，2015）利用多次示教构建了一个参数化技能记忆，将低维技能参数化与运动基元参数联系起来。然后利用这种低维嵌入进行高效的策略搜索。分段线性相位被用于提升增量学习的性能（Samant等，2016）。

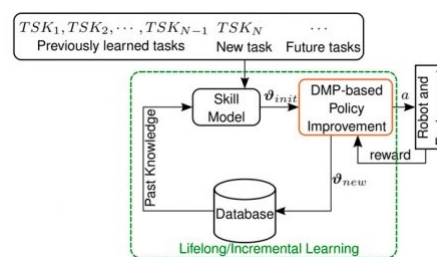


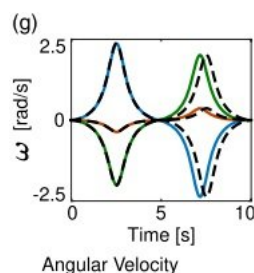
图15. 终身/增量学习方法的一般框架。

Duminy等人（2017）设计了一个框架，用于学习哪种数据收集策略对于获取运动技能以实现多种结果最为有效，并对其经验进行泛化，以实现累积学习的新结果。

提出了一种生成式学习框架，用于在认知架构的不同层次（包括符号规划以及对象和动作属性）上，用缺失信息补充机器人的知识库（Wörgötter等，2015）。上述方法均使用DMP来表示学习到的技能并在真实机器人上执行。

Wang等人（2016）提出了一种改进的DMP公式，称为DMP+，它能够通过提高现有基元的可用性并减少用户在模仿学习过程中的疲劳，有效地修改已学习的轨迹。随后，DMP+被集成到一个带有语音和本体的对话系统中，以利用自然交互方式学习或重新学习任务（Wu等，2018）。

学习力轨迹的另一种方法是学习机器人的阻抗，即通过学习期望的刚度轨迹。改变机械臂阻抗的能力对于简化在不可预测和非结构化环境中的物理交互至关重要（Burdet et al., 2001; Hogan, 1984）。在Peternel等人（2018a）的研究中，通过按钮界面的遥控操作来控制机器人阻抗，该阻抗由DMP学习，使机器人能够执行各种协作装配任务。例如，学习到的位置和刚度DMP被用于将销钉插入槽中以连接两个部件，或拧紧螺栓（Peternel et al., 2018a）。Yang等人（2018）采用了类似的方法来学习用于蔬菜切割任务的DMP。



尽管基于遥控操作的方法在教授机器人用于交互任务的DMP方面非常有效，但通常涉及复杂且昂贵的系统。Abu-Dakka等人（2018）的方法使机器人能够通过测量与环境的交互力来学习刚度轮廓，从而执行阀门转动任务。Peternel

等人（2017a）的方法利用人类演示和EMG，从人类肌肉活动测量中学习刚度DMP，以执行锯切和擦拭（图17）任务。

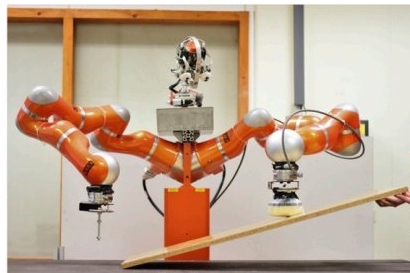


图17. 使用DMP适应变化表面（如擦拭任务）（Kramberger等人，2018）。

然而，单一轨迹的适应不太可能为更一般的情况生成合适的解决方案，在这些情况下，任务执行需要显著变化。在通过动作引导学习初始DMP运动轨迹后，机器人可以在执行任务时根据测量的交互力来调整这些轨迹。Pastor等人（2011）提出了一种根据测量传感数据实时调整演示DMP轨迹的方法。他们开发了一种基于估计和实际力数据的轨迹自适应调节器。Prakash等人（2020）扩展了实时自适应方法，结合了模糊分数阶滑模控制器，以高效稳定地将演示的DMP轨迹适应于快速运动，如乒乓球挥拍。最近，Cui等人（2022）提出了一种耦合多个DMP的方法，用于对可变形物体的运输任务进行机器人任务建模。

Sutanto等人（2018）提出了一个数据驱动的框架，从演示中学习反馈模型。他们使用RBF-NN来表示运动原语的反馈模型。与此研究类似，Gams等人（2010）提出了一种根据期望力（机器人应以此力作用于环境）来调整演示运动的方法。因此，他们确保了所学运动对不同表面的适应。该方法后来被扩展（Pastor et al., 2011），以提供统计上最可能的力-力矩轮廓（Pastor et al., 2012），此外，力-力矩数据被用于训练分类器（Straizys et al., 2020），以调节演示轨迹，用于组织或水果切割等精细任务。

从策略学习进一步推进，Do等人（2014）提出了一个自适应框架，其中不仅可以学习期望的自适应力或轨迹，还可以学习整个技能。他们通过在不同环境条件下的擦拭任务演示了该方法。

5.1.2. 装配任务

装配是实现自动化的更具挑战性的任务之一，不仅需要考虑位置轨迹，还需要考虑任务动力学。为应对这一挑战，提出了多种方法。Abu-Dakka等人(2015a)提出

复杂且易变化的装配任务无法即时演示并执行，因此需要自适应方法来确保成功执行。Nemec 等人(2020)采用异常策略(以DMP建模)来处理复杂装配案例。Sloth 等人(2020)提出了一种异常策略框架，结合离散与周期性DMP，并耦合力控制，以在紧公差条件下学习装配任务。Angelov 等人(2020)通过考虑任务动力学与顺序，整合了多种不同的控制策略。该方法使用DMP生成自由运动，并利用卷积神经网络进行装配。



图 18. DMP 在装配任务(例如轴孔装配)中的应用示例 (Kramberger 等人, 2016b)。

在某些情况下，主动探索与自主数据库扩展可用于自动学习装配策略。在 Petric 等人(2015)的研究中，所提出的算法能够以自主方式从数据库中构建并组合 CMP 运动知识。

与装配任务相辅相成的是，仅凭演示轨迹进行拆卸同样具有挑战性。如 Ijspeert 等人(2013)所述，DMP 在指定的运动目标参数处具有唯一点吸引子，本质上排斥了可逆性。因此，Nemec 等人(2018)提出了一种框架，通过从单个演示运动中分别学习两个独立的 DMP(一个正向、一个反向)来应对拆卸挑战。Iturrate 等人(2019)进一步发展了这一思路，利用逻辑斯蒂微分方程重新表述了 DMP 的相位系统，从而获得了两个稳定的点吸引子。这种可逆动态运动基元(RevDMP)方法提供了动力系统的可逆性表述，并证明了该算法在轴孔装配任务中的有效性。

5.1.3. 接触自适应的学习方法。期望的力-力矩曲线可通过 ILC 进行跟踪(Gams 等人, 2014, 2015b)。在重复性机器人任务中，迭代学习因其有效性和鲁棒性而日益受到关注(Bristow 等人, 2006)。然而，为获得有效结果，需要在学习参数进行仔细调整。Norrlöf(1991)和 Tayebi(2004)提出了一种自适应学习方法，用于自动调整学习参数。

另一种方法是使用强化学习来调整 DMP。例如，在 Buchli 等人(2011b, 2011a)的研究中，通过强化学习在任务执行过程中调整了刚度参数。

除基于反馈的 DMP 自适应和强化学习外, 另一种替代方案是扩展性与泛化方法。Matsubara 等人 (2011) 提出了一种从现有知识生成新控制策略的算法, 从而实现了 DMP 的扩展性增强; 与此同时, Mülling 等人 (2010) 利用运动基元的混合来生成乒乓球挥拍动作。另一方面, Krug 和 Dimitrov (2015) 将 DMP 的泛化与模型预测控制相结合, 而 Gams 等人 (2015a) 则将泛化应用于 DMP 耦合项, 通过学习并随后将其添加到演示轨迹中, 以生成新的关节空间轨迹。

Stulp 等人 (2013) 提出通过在相位和任务参数的完整空间中进行一次回归来学习函数逼近器, 从而避免了两次连续回归的需求。Forte 等人 (2012) 对 LWR 和 GPR 在轨迹泛化方面进行了比较研究。该研究表明, 使用 LWR 轨迹逼近可以获得更高的精度。Koropouli 等人 (2015) 提出了一种力控制策略的泛化方法。通过同时学习策略及策略差异数据 (使用 LWR), 他们能够通过训练数据的叠加来估计新输入下的策略。

Deniša 等人 (2016a) 在 CMPs 框架中, 使用基于 GPR 的泛化方法对联合位置轨迹和力矩指令进行处理。为展示该方法的通用性, Petric 等人 (2018) 将其应用于基于机器人的装配任务。最后, Kramberger 等人 (2017) 对该方法进行了扩展, 以应对期望任务的变化, 例如相似物体的装配。这使得机器人能够利用 LWR, 从演示的成功任务执行数据库中自动生成运动轨迹, 这些轨迹包括用 DMPs 编码的运动学和动力学演示轨迹。新获得的数据用于适应工作空间的变化。然而, 统计学习中的一个主要问题是如何有效处理方向轨迹的无奇异性表示。为解决这一问题, Kramberger 等人 (2016a) 提出了笛卡尔空间 DMPs 的公式, 其中方向用单位四元数表示。

5.2. 人机协同操作

虽然机器人与被动态环境的交互控制可以解决大多数任务, 但在某些情况下, 机器人需要与主动体 (例如人类或另一台机器人) 进行交互。人机协作正在成为机器人领域的关键方向之一 (Ajoudani 等人, 2018)。为了实现成功的物理人机协作, 机器人必须能够与人类伙伴协调地控制复杂运动。在这一方向上, 调节阻抗的能力对于协调人机共操作工具过程中的物理交互至关重要 (Peternel 等人, 2017b)。DMPs 提供了一种优雅的方法来编码这种协调的动态运动。

在 Peternel 等人 (2014) 的研究中, 协作机器人通过遥操作在线学习如何与人类同事进行协作锯切。阻抗通过使用 EMG 测量的肌肉活动来控制机器人。DMPs 被用于编码由人类遥操作员演示的协调的相位相关运动和阻抗。通过遥操作进行教学是将物理交互技能传授给协作机器人的有效方式, 但该设置成本高昂且未广泛普及。

一种直观的替代遥操作的方法是让机器人在与人类伙伴协作的过程中, 直接通过物理交互学习技能。众多研究集中于学习协作伙伴之间的同步运动 (Gams 等人, 2014; Kulvicius 等人, 2013; Lu 等人, 2022; Peternel 等人, 2018b; Prada 等人, 2013; Sidiropoulos 等人, 2019, 2021; Ugur 和 Girgin, 2020; Umlauf 等人, 2014; Wu 等人, 2022; Zhou 等人, 2016a)。例如, 在 Kulvicius 等人 (2013) 的研究中, 交互运动被编码为 DMPs, 并根据来自协作过程中代理之间分歧的测量力进行调整。

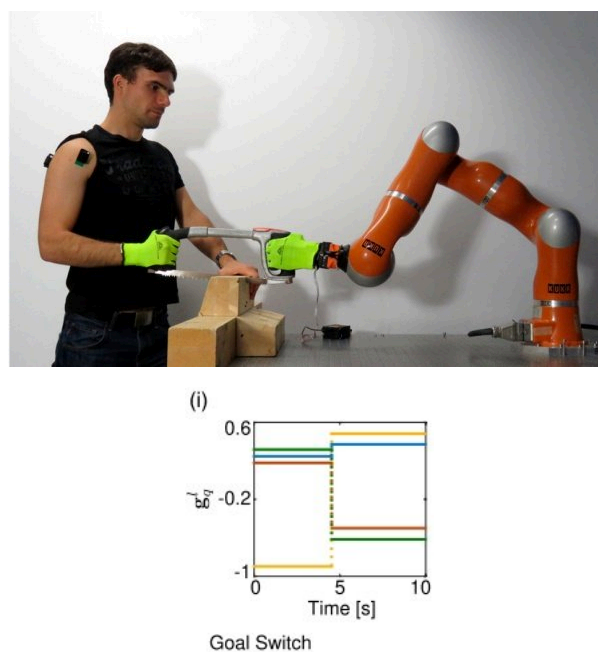


图19. 使用DMPs进行人机协作锯切的示例 (来自Peternel等人, 2018b).

类似地,在Gams等人(2014)的研究中,协作运动被编码为DMPs,并利用力反馈和ILC进行适应。Zhou等人(2016a)的方法结合了两个DMPs来编码每个合作者手臂的运动,并以领导者-跟随者的方式耦合。

除了适应协作运动外,在Peternel等人(2018b)的研究中,机器人还使用DMPs直接从与人类的共操作中在线学习阻抗(图19)。机器人从一组基本技能开始,使其能够以纯粹的跟随者角色与人类协作。通过协作任务的执行,机器人随后在线学习运动和阻抗轨迹,并用DMPs进行编码。当人类感到疲劳时,机器人利用学习到的高级技能接管大部分任务执行。

Ben Amor 等人 (2014) 的方法提出了一种标准DMP的升级版,称为交互基元,能够解释协作运动中的概率特性。该DMP不再使用单一的权值,而是包含权重分布。这一分布使机器人能够学习合作行为的内在相关性,并在协作过程中推断人类伙伴的行为。Cui 等人 (2016, 2019) 利用视觉信息提取上下文相关参数,以增强交互基元,提高任务执行过程中的鲁棒性。

还存在其他类型的协同操作场景,例如手内双操作或人机物体交接。例如,在 Amadio 等人 (2022)、Gao 等人 (2019) 和 Koene 等人 (2014) 的研究中,DMP被用于执行双操作;而在 Abdelrahman 等人 (2020)、Iori 等人 (2023)、Lafleche 等人 (2019)、Prada 等人 (2014) 以及 Solak 和 Jamone (2019) 的研究中,DMP则被用于人机物体交接。

当人类工作者的工作环境存在危险,或者机器人数量相对于人类工作者过多时,显而易见的解决方案是让机器人之间进行协作。Peternel 和 Ajoudani (2017) 的方法利用DMP使新手机器人通过协同操作向专家机器人学习。初始阶段,新手机器人保持柔顺,让专家机器人引导任务执行。在第一阶段,新手机器人通过DMP学习参考运动。第二阶段,它变得僵硬以执行新学到的运动,而专家机器人则启动协作任务执行中预期的僵硬/柔顺阶段。最后,新手机器人学习在任务的哪些阶段增加或降低阻抗,并将这种阻抗行为通过DMP进行编码。

5.3. 人类辅助、增强与康复

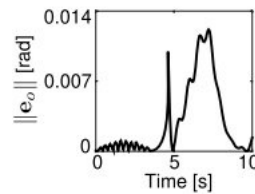
最常见的协同操作类型是经典的人机协作,即人类和机器人代理共同执行工业或日常任务。另一种协同操作发生在人类使用可穿戴机器人(如外骨骼)时。外骨骼可用于多种不同功能。其中一种功能是增强,即放大当前人类运动,以增强现有(健康)人类的能力,例如在涉及重负载的任务中。当人类能力受损时,外骨骼需发挥辅助功能。若人类能力受损程度更大,则可利用外骨骼的康复功能进行物理治疗。在增强功能中,可在机器人上采用DMP来减轻健康人类工作者的繁重和/或重复性运动;而在辅助和康复功能中,DMP可用于帮助受损人类完成日常任务,或对患者进行重复性物理治疗以促进康复。

除外骨骼功能类型外,另一个重要方面是物理人机交互过程中外骨骼与人类之间的负载分担。例如,在康复早期的严重受损患者案例中,外骨骼生成的DMP可能承担几乎所有的运动负载,扮演领导者角色,执行被动理疗。随着康复进展,当更偏好主动训练时,大部分负载可转移至引领运动的人类,而DMP系统则作为追随者发挥支持作用。当完全康复不可能时,DMP可在日常任务中提供辅助,实现负载部分分担。在增强现有(健康)人类能力的情况下,DMP系统可适应人类运动,并在人类完成特定重负载任务所需努力的基础上增加额外力量。在某些情况下,外骨骼可从人类运动中学习DMP,从而完全接管重复性动作。

Lauretti等人(2017, 2018)的方法通过示教学习离线获得了DMPs,随后由上肢外骨骼用于支持人体运动。在Peternel等人(2016)的研究中,控制方法利用DMPs交互式地调整执行上肢外骨骼辅助所需的关节力矩,并补偿周期性运动中的所有底层动力学(图21-左)。相位依赖的力矩轨迹在线实时更新,以最小化由EMG测量的肌肉活动反馈。Lanotte等人(2021)提出的基于DMPs的方法实现了针对离散外骨骼运动的类似自适应能力。在Petrič等人(2016)的研究中,机器人利用DMPs编码辅助运动,然后通过费茨定律考虑人体运动控制的方面进行自适应。Li等人(2021a)引入了一种分层控制策略,通过整合DMPs与GMM实现人机协作操作的学习。所提出的方法利用定义良好的基元,为基于DMPs的控制方法提供了全面的数学公式以及稳定性和收敛性的严格分析。

基于外骨骼的步态辅助与康复是DMPs非常常见的应用领域,存在大量实例(Abu-Dakka等人, 2015; Amatya等人, 2020; Escarabajal等人, 2023; Huang等人, 2016a; Hwang等人, 2019, 2021; Schaal, 2006; Yuan等人, 2020; Zou等人, 2021)。在Abu-Dakka等人(2015, 2020)的研究中,采用并联机器人进行踝关节康复,其运动由DMPs生成(图20)。Escarabajal等人(2023)将类似的并联机器人应用于膝关节康复,其中使用RevDMPs使患者能够逆转运动以维持自身期望的节奏。在Huang等人(2016a)的研究中,DMPs用于学习下肢外骨骼的步态运动轨迹。该方法随后通过强化学习方法进行了扩展,以自适应一个力耦合项(类似于第3.3.2节中介绍的早期方法),从而实现运动轨迹的在线自适应(Huang等人, 2016b)。在Luo等人(2022)的研究中,DMPs被调整为适应步态摆动相中脚部不同的起始和终止位置。Xu等人(2023)的方法在镜像疗法概念中利用DMPs实现腿部外骨骼运动,将健康肢体的运动转移至受损肢体。Hong等人(2023)使用DMPs规划佩戴假肢行走时的避障腿部运动。

除正常步态外，DMPs还被应用于下肢外骨骼的爬楼梯（Xu等人，2020）和坐站转移（Kamali等人，2016）辅助运动。在Joshi等人（2019）的研究中，使用机械臂辅助人体穿衣，运动由DMPs生成。Ding等人（2022）开发了一个面向老年人辅助的框架，结合了DMPs的导纳控制以提供移动辅助和操作支持。该框架在配备机械臂的移动平台上实现，用于示教学习。



除辅助身体运动和康复外，DMPs还被应用于放松目的。例如，在Li等人（2020）的研究中，机械臂通过DMPs提供按摩运动。

5.4. 遥操作

遥操作是机器人学的主要领域之一，使人类能够直接实时控制（远程）机器人。通常，控制通过接口完成，这些接口可以捕获发送给机器人的人类指令，并能提供来自机器人的触觉反馈。遥操作侧重于赋予人类操作员对机器人的完全或共享控制，而DMPs用于编码自主机器人行为。因此，我们在此主要考察遥操作被用于教授机器人由DMPs编码的新自主行为的情况。

在Kormushev等人(2011)的研究中，结合了动作教学与远程操作，以形成基于DMP的熨烫机器人技能。在通过动作引导学习运动轨迹后，使用触觉设备和远程操作系统记录了相应的力。在Peternel等人(2014)的研究中，

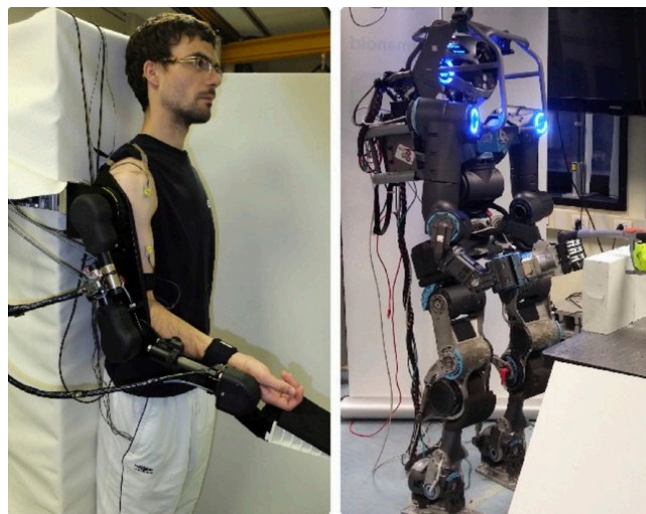


图21. 左侧照片展示了Peternel等人(2016)中的手臂外骨骼应用。右侧照片展示了高自由度人形机器人Walk-man (Tsagarakis等人, 2017)在Peternel和Ajoudani(2017)中执行锯切操作。

远程操作用于教导机器人如何与另一名人类进行物理协作。由于没有触觉反馈，该远程操作设置是单向的，但人类除了能够教导运动外，还能教导机器人的阻抗。前者通过EMG测量肌肉活动来指令，后者则通过光学运动捕捉系统测量的人类操作员手臂运动来指令。类似基于远程操作的DMP学习方法也用于Lentini等人(2020)和Yang等人(2018)的研究中。

在Peternel等人(2018a)中，人类操作员通过远程操作教导机器人如何执行自主装配动作（图16-右）。DMP用于编码所指令的阻抗和运动；然而，与EMG相比，此处采用了更实用的按钮式阻抗指令接口。更重要的是，该远程操作设置是双向的，触觉接口为人类操作员提供了机器人所感受到力的反馈。通过远程操作学习的DMP还可泛化以提升适应性（Yang等人, 2018）。在Luo等人(2023)中，DMP被用于编码熟练操作员的动作，随后用于培训新手操作员。Zhang等人(2023)的方法利用DMP部分自动化了阑尾切除手术过程。

获取新技能并不总是需要真实的机器人。在Beik-Mohammadi等人(2020)中，机器人和环境均为仿真，人类操作员使用虚拟现实系统。结合DMP与强化学习（RL）形成了自适应技能。Abu-Dakka等人(2015a)提出的场景本质上基于远程操作；然而，人类演示者并非假装化身于机器人中，而是将机器人任务环境克隆到人类一侧（图16-右）。这消除了对力反馈和触觉设备的需求，因

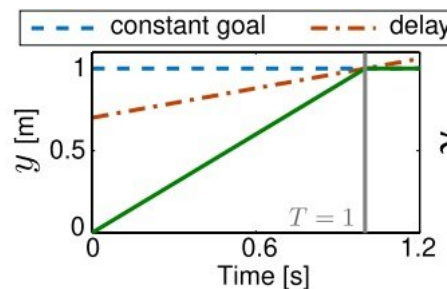
为人类在其一侧感受到真实环境，同时运动由非接触式传感系统（即磁力追踪器）捕捉，并镜像至机器人。

通过远程操作进行多次演示可能不一致，尤其是在多智能体共享控制场景中。Pervez等人(2019)提出的方法能够通过共享控制远程操作同步不一致的演示，并用DMP对其进行编码。Maeda (2022)研究了利用DMP隐式融合人类与机器人策略的可能性，而无需设计任务特定的仲裁函数或提供多次（可能不一致的）演示。

5.5. 高自由度机器人

DMP通过在所有自由度间共享一个规范系统（3）并仅维护一组独立的变换系统，为处理高维空间系统提供了一种优雅且快速的方法。这里所指的高维空间是指具有10个或更多自由度的系统（即图21右侧的Walkman人形机器人）。在本节中，我们将简要提及一些涉及高自由度数量的潜在工作。

Ijspeert等人(2002b, 2002a)在模仿学习框架中使用DMP，通过30自由度的类人机器人学习网球正手、网球反手和有节奏的击鼓。Pastor等人(2009)使用DMP编码了一个10自由度的外骨骼



机器人臂。Luo等(2015)将DMP与随机策略梯度RL和GPR相结合，设计了一种在线自适应推恢复控制策略。该方法已应用于具有20个自由度的PKU-HR5仿人机器人。André等(2015, 2016)基于关联技能记忆与周期性运动及DMP，实现了一种传感器轨迹预测模型，用于仿人机器人的早期故障检测。他们在仿真中将该算法应用于具有20个自由度的DARwIn-OP机器人。Pfeiffer和Angulo(2015)通过将DMP应用于具有23个自由度的REEM机器人平台来表征手势。Nah等(2020)提出了一种优化DMP参数的方法，以应对像鞭子这样的高自由度系统的复杂性。他们在仿真中针对10、15、20和25自由度系统进行了测试。为了减少在新任务条件下适应所需的滚转次数，Queißer和Steil(2018)使用CMA-ES优化DMP参数。此外，他们还引入了一种混合优化方法，该方法将策略参数流形上的快速粗优化与动作无约束空间中的细粒度参数搜索相结合。该方法在使用10自由度机器人臂的仿真中得到了成功验证。Liu等(2020)提出了基于DMP的轨迹生成方法，使具有10个自由度（针对两条腿）的全身仿人机器人能够实现自适应行走。Liang等(2021)开发了一种高效方法，使具有26个自由度的服务机器人具备执行手语动作的技能。

Travers等(2016, 2018)提出了一个框架，将DMP与高斯形状的空间激活窗口相结合，通过连接底层控制器与高层规划器，为高自由度机器人系统（例如蛇形机器人）在复杂环境（存在障碍物）中规划运动。DMP也可应用于软体连续体机器人，其中自由度的划分更为模糊（Salem等，2023）。

5.6. 运动分析与识别

DMP倾向于用相似的形状参数 $\{w_i\}$ 拟合拓扑相似的运动轨迹（Ijspeert等，2013）。由于DMP的时间和空间不变性，这一特性使得形状参数成为识别相似运动的有用描述符。事实上，Strachan等(2004)已经证明，从5次重复的4类离散手势（使用3自由度加速度计测量）计算出的形状参数是线性可分的，即易于分类。Lantz和Murray-Smith(2004)对10类周期性手势得出了类似结论。Xu等(2005)利用两个DMP参数向量之间的相关性来衡量原始运动之间的相似性，并识别步态模式。类似地，Ijspeert等(2013)利用参数向量之间的相关性识别Graffiti字母表中的26个字母。

形状参数 $\{w_i\}$ 也适用于拟合更复杂的分类器，例如支持向量机。该策略被成功用于对单目相机（Liu等，2014）或双目相机（Wang和Payandeh，2015）观察到的手势进行分类。Zhang等(2017)没有采用固定数量的基函数（形状参数数量），而是使用快速动态时间规整（Salvador和Chan，2007）来对齐不同长度的参数向量，然后使用K近邻对不同运动进行分类。

运动识别还可用于通过比较传感数据与运动模板来确定机器人是否正确执行任务。在此方面，André等(2016)使用了一种关联技能记忆（类似于Pastor等(2011)中的记忆结构）作为传感器轨迹的预测模型，从而实现早期故障检测。在该工作中，DMP被用于紧凑地编码关联技能记忆并加速故障检测。所描述的方法表明，DMP是手势识别（尤其是对于计算能力有限的系统）的一个有

价值的选择。Liang等(2021)提出了针对生成双臂手语运动的运动重定向问题的解决方案。他们的方法涉及一种离线约束优化技术，该技术最小化与DMP生成轨迹（该轨迹编码了人类演示）的偏差。需要指出的是，他们的方法仅在离线环境下应用和测试。

人类在执行相同任务时往往采用略有差异的方式。有时，执行风格上的差异包含了将运动适应于不同执行情境的有用信息。例如，在无障碍物的到达运动中便是如此。为捕捉执行风格，Matsubara 等人 (2010) 在 DMP 的强迫项中增加了一个从多次示教中学习到的风格参数。在运行时，可使用不同的风格参数在示教行为之间进行平滑插值。Zhao 等人 (2014) 不仅采用了不同风格的运动，还学习了风格参数与目标之间的平滑映射，以提升泛化能力。

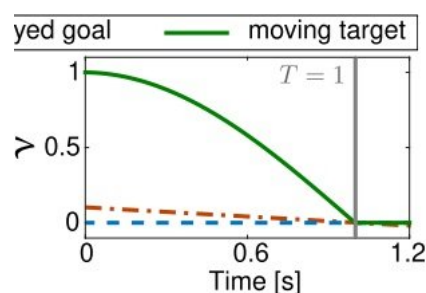
当人类提供无缝示教时，DMP 可用于在线分割与识别。为此，Meier 等人 (2011) 假设给定一个 DMP 库，并利用该库在任务示教过程中识别运动片段。与为每类基元使用示例模板不同，Chang 和 Kulić (2013) 通过运动到非运动的转换对视频流进行分割，对分割后的数据拟合 DMP，并以无监督方式对相似运动片段进行聚类。Song 等人 (2020) 利用关键点（即不同任务示教中的共享特征）的概念进行无监督轨迹分割。Mandery 等人 (2016) 通过检测与环境接触来分割全身运动，并利用这些接触构建一个概率语言模型，其中单词表示姿态，句子表示姿态序列。学习到的语言模型用于规划通过拼接多个 DMP 执行的全身运动轨迹（见第 3.2 节）。Kordia 和 Melo (2021) 提出了一种方法，用于从预学习运动库中识别观测到的轨迹并预测其目标位置。他们利用观测轨迹的显著特征辅助识别，并使用 DMP 表示运动。

DMP 已被发展为神经生物学运动基元的计算模型（Schaal 等人，2007）。与青蛙脊髓力场相关的神经生理学实验发现，启发了 Hoffmann 等人 (2009) 对 DMP 公式的修改。如第 3.1.1 节所述，这种多维表示克服了经典 DMP 的局限性，例如轨迹过冲以及轨迹对描述运动的参考系的依赖。Hoffmann 等人 (2009) 还受人类在手臂运动中避免碰撞的方式启发，推导了一种 DMP 的避碰策略。DeWolf 等人 (2016) 研究了人类在运动控制过程中应对手臂动力学和运动学结构变化的能力。他们提出了一个运动控制系统的脉冲神经元模型，该模型使用 DMP 实现前运动皮层的准备与规划功能。Kuppuswamy 和 Alessandro (2011) 研究了机器人动力学参数变化对 DMP 轨迹跟踪性能的影响。他们的研究表明，在 DMP 学习过程中应显式考虑身体参数的变化。Hotson 等人 (2016) 将捕捉神经信号的脑机接口与一个 DMP 模型相结合，该模型表示非人灵长类动物执行末端轨迹。该系统用于解码操纵四种不同物体的灵长类动物的真实轨迹。

5.7. 自动驾驶与现场机器人

DMPs 可应用于机器人领域的多种自主非平稳场景。Perk 与 Slotine (2006) 利用 DMPs 为无人机 (UAV) 定义飞行路径并实现障碍物规避，其轨迹基于控制无人机电机油门的摇杆运动生成。随后，Fang 等人 (2014) 将该方法扩展到编码用户演示的无人机数据，提取并编码飞行轨迹中的节律性和线性片段，将其组合为飞行控制技能。此外，Tomić 等人 (2014) 将无人机运动建模为最优控制问题，最优控制求解器的输出通过 DMPs 进行编码，使其能够实时泛化并应用于飞行中无人机轨迹的修改。类似地，Lee 等人 (2018) 与 Kim 等人 (2018a) 提出了一种基于自适应控制器的无人机协作面操作框架，该控制器根据负载的质量和惯性特性自适应调整无人机运动。同时，DMPs 被融入控制方案以实时修改飞行轨迹并规避障碍物。该框架后续扩展至包含路径优化，其中 DMPs 在实时避障中发挥关键作用 (Lee et al., 2020)。

如前所述，DMPs 是一种通用的运动表示方法，可应用于多种任务与场景。该领域近期应用之一涉及自主水下航行器 (AUVs)。Carrera 等人 (2015) 将 DMPs 集成至 AUV 的示教学习场景中，演示数据包含机械臂与航行器的传感输出，这些数据被有效用于演示水下阀门旋转任务。



6. 讨论

DMPs 同样出现在自动驾驶领域。在 Wang 等人 (2018, 2019) 的近期工作中，作者提出一种框架，将复杂驾驶数据分解为以运动基元表示的更基础驾驶技能组合。在该框架中，DMPs 被用于以可接受的精度表示驾驶员轨迹，并可泛化至不同场景。

本节提供关于如何从本文讨论的多种方法中为特定应用选择最合适方案的指南。一个有用的判断标准是代码的可用性，它能极大简化实现过程。我们搜索了开源 DMP 实现，并将其收录于一个 Git 仓库（见 6.2 节）。为继续回馈社区，我们还发布了表 4 所列的实现。本节最后讨论了 DMP 公式本身的局限性、开放问题以及可能的研究方向，这些内容总结于表 5。

6.1. 不同应用场景的指南

前文介绍了不同的 DMP 公式及其扩展，以及可能的应用场景。通常，不存在一种适用于所有范围与目标的统一公式，采用何种方法取决于要实现的目标与应用条件。因此，我们提供一些指南，以帮助用户选择合适的公式。

6.1.1. 离散与周期。对于具有明确起始点和终止点的任务，离散 DMPs 是编码其间运动轨迹的逻辑选择。此类任务包括抓取与拾放 (Caccavale et al., 2019; Deniša et al., 2016a; Forte et al., 2012; Stulp et al., 2009)、装配中的特定动作 (Krüger et al., 2014; Abu-Dakka et al., 2014; Nemec et al., 2020; Angelov et al., 2020) 以及切割 (Straizys et al., 2020; Yang et al., 2018)。

当起始点与终止点重合时，周期 DMPs 是逻辑选择，因为编码的运动可以反复重复。其典型应用包括重复性任务，如运动 (Nakanishi et al., 2004; Rückert and d'Avella, 2013; M. Wensing and Slotine, 2017)、人体增强/

表 5. DMP 功能与限制总结: 已解决 (✓) 或部分解决 (□) 的特性。

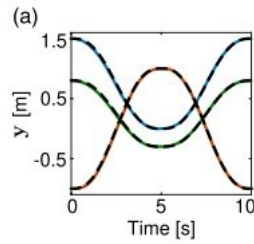
局限性	相关工作	状态
途经点	Ning et al. (2011, 2012), Weitschat and Aschemann (2018), Saveriano et al. (2019), and Zhou et al. (2019)	✓
起始点	Hoffmann et al. (2009), Ijspeert et al. (2013), Weitschat et al. (2013), and Dragan et al. (2015)	✓
目标点	Ijspeert et al. (2013), Weitschat et al. (2013), Abu-Dakka and Kyrki (2020), Dragan et al. (2015), and Weitschat and Aschemann (2018)	✓
避障	Park et al. (2008), Hoffmann et al. (2009), Tan et al. (2011), Kim et al. (2015), and Rai et al. (2017)	✓
几何约束数据	Pastor et al. (2009), Abu-Dakka et al. (2015a), Ude et al. (2014), Saveriano et al. (2019), and Abu-Dakka and Kyrki (2020)	\ (I [∧] {5})
概率性	Ben Amor et al. (2014)	I
外推	Pervez and Lee (2018) and Zhou et al. (2019)	I
高维输入	Pervez et al. (2017a) and Pahić et al. (2020)	I
闭环	Peternel et al. (2016) and Kramberger et al. (2018)	I
多吸引子	Nemec et al. (2018) and Iturrate et al. (2019)	I

康复 (Petenerl et al., 2016)、擦拭表面 (Gams et al., 2016; Kramberger et al., 2018; Petenerl et al., 2017a) 以及锯切 (Petenerl et al., 2018b)。尽管如此，即使是那些仅偶尔执行一次的典型非重复性任务，只要其起始点与终点重合，仍然可以用周期性DMP进行编码 (Petenerl et al., 2018a)。

在某些情况下，无法清晰地区分运动是周期性的还是离散的。例如，Ernesti et al. (2012) 已表明，仿人机器人步态中的第一步是向周期性运动过渡的暂态过程。他们所提出的表示方法是编码收敛至极限环轨迹的暂态过程的理想候选方案。最后，在诸如复杂装配等情况下，任务需要结合离散型与周期型DMP (Sloth et al., 2020)。

6.1.2. 空间表示。DMP的原始公式在过去和现在都成功应用于多维独立数据，其中每个自由度 $\in \mathbb{R}$ (见2.1.1节和2.2.1节)。这些数据可以是关节或笛卡尔位置、力、力矩等，其中数据的每个自由度可以独立于其余部分进行演化。然而，若不进行数据的预处理和/或后处理，此类公式不足以成功编码具有特定几何约束的数据。此类数据的示例包括：i) 方向，其中数据受到附加约束（例如，旋转矩阵表示中的正交性约束或四元数表示中的单位范数约束）的紧密束缚；ii) 完整刚度/阻尼矩阵和可操作性矩阵，它们被封装在SPD矩阵中。

在早期许多工作中，方向轨迹的学习和自适应忽略了其几何约束 (Pastor et al., 2009)，导致方向不当，因而需要额外的重新归一化。另一个例子中，Umlauf等人 (2017) 采用特征分解进行阻抗自适应。



为满足此类几何约束，研究者提出了DMP的新公式，确保在方向自适应过程中生成正确的单位四元数或旋转矩阵（Abu-Dakka等人，2015a；Koutras和Doulgeri，2020a；Saveriano等人，2019；Ude等人，2014），以及在SPD剖面（如刚度或可操作性椭圆）的自适应过程中生成正确的SPD矩阵（Abu-Dakka和Kyrki，2020）。我们认为，在编码具有底层几何约束的数据时，采用这些几何感知型DMP更为可取。

6.1.3. 权重学习方法。DMP将运动轨迹表示为稳定的动力系统，其中包含可学习的权重，用于定义运动的形状。在LFD范式中，DMP权重通常利用人类示范以监督学习方式获得。将人类示范转化为DMP强迫项训练数据的过程见2.1.1.1节。权重的数量（即用于逼近强迫项的RBF数量）是一个超参数，通常由用户提供。作为实用的调参指导，需考虑RBF数量随轨迹长度及其复杂度的增加而增加，复杂度取决于凹凸性变化以及峰值的频率和幅度。给定训练数据和RBF数量后，可采用不同技术来拟合权重。

当强迫项如公式(4)所示由RBF组合构成时，LWR被广泛使用。然而，在文献中，也可采用Si等人（2021）中的RBF-NN，或在提供多条示范时，利用Li等人（2021b）和Pervez等人（2017a）中的GMM/GMR，或Fanger等人（2016）中的GPR来表示强迫项，并使用期望最大化算法来拟合（超）参数。深度神经网络通常通过反向传播训练，似乎是实现输入图像到强迫项映射的诱人方案（Pervez等人，2017b），从而模拟人类的感知-动作循环。尽管诱人，但利用深度学习技术作为运动基元的可能性仍需进一步研究。

在实际应用中，DMP轨迹与机器人运动之间可能存在错位。典型示例包括装配或其他需要与环境进行物理交互的任务（见5.1节）。在此情况下，可通过增量调整DMP运动来改善机器人性能。迭代学习控制（ILC）作为一种有趣的方法，可迭代更新DMP权重，确保快速收敛到期望性能（Abu-Dakka等人，2015a；Gams等人，2014；Kramberger等人，2018）。然而，ILC假设给定待复现的目标行为。当目标行为难以明确指定且机器人性能不理想时，需采用强化学习（RL）解决方案。如4.3节所述，DMP是有效的控制策略，结合策略搜索算法（如 π^2 或 PoWER），能够解决复杂且高度动态的任务。

6.1.4. 在线自适应。在现实世界中执行机器人任务需要具备自适应能力。当需要基于某种反馈对DMP进行自适应时，应采用其中一种扩展方法。例如，为了根据检测到的障碍物改变现有运动，可以使用Gams等人(2016)、Hoffmann等人(2009)、Park等人(2008)以及Tan等人(2011)提出的方法（参见第3.3.1节）。如果需要基于实时力反馈自适应地学习运动动力学，则可以采用Peternel等人(2016)的方法（参见第3.3节）。

此外，对于装配或抛光等工业任务，可以采用将力控制与示教轨迹相结合的自适应策略（Abu-Dakka等人，2015a；Gams等人，2010；Kramberger等人，2016），确保系统遵循预定义轨迹并适应环境不确定性。对于在线自适应，DMP可用作轨迹生成器，其输出作为力控制算法的输入；另一方面，力反馈可以直接作为耦合项纳入DMP公式中（参见第3.3.2节），从而无需额外的力控制器。类似的方法也可用于基于速度的运动自适应（参见第3.3.4节）。

6.1.5. 阻抗与力。在物理交互任务中，DMP可用于学习力或阻抗（Petenerel等人，2017a）。如果任务需要位置控制，则应结合参考位置使用DMP学习阻抗。如果任务需要控制特定力，例如在擦拭和钻孔过程中对表面施加推力，则力控制或阻抗控制均可行。然而，若安全性是最关键的方面，则应使用DMP学习阻抗控制，以便使机器人变得柔软。

此外，为克服任何非期望运动，控制策略可以结合基于储能器的无源性方法进行增强（Shahriari等人，2017）。该方法监控建模子系统（例如DMP轨迹生成、阻抗控制与环境）之间的能量流。在发生能量违规时，系统首先尝试被动补偿违规行为；随后，若违规无法补偿（例如能量储存器耗尽），则停止系统。在任务特性不完全已知的情况下，可以在无源性方法之上添加学习策略（Kramberger等人，2018），以学习任务的整体能量需求。

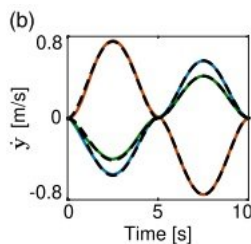
6.2. 资源与代码

代码和数据集的可用性有助于加速新应用的搭建，而无需从头重新实现一种有前景的方法。我们搜索了现有的DMP实现，发现多位研究人员已在各种开源仓库中发布了他们的DMP代码。我们决定将可用的实现列于本文附带的Git仓库中（<https://gitlab.com/dmp-codes-collection/third-party-dmp>）。对于每个实现，我们注明DMP类型、作者、代码下载URL以及所使用的编程语言。我们还提供关键特性的简要描述。

除了列出已有方法外，本文附带的Git仓库还包含我们决定向社区发布的实现。所提供实现的列表如表4所示。

6.3. 局限性与开放问题

与任何运动基元表征方法一样，DMP既有优势也有固有的局限性。前文已广泛讨论了DMP的优点。在此，我们介绍DMP的主要局限性，并讨论需要进一步研究的开放问题。这些局限性的总结见表5。



6.3.1. 隐式时间依赖性。用于抑制非线性强迫项并确保收敛到给定目标的相位变量，在DMP公式中引入了一个隐式时间依赖性。将时间依赖性隐式表示为动力学系统的原因在于，此类相位变量可以方便地进行操控。例如，在第2.1.1.2节中，我们展示了如何操控相位变量来减慢（甚至停止）执行过程。时间依赖性的一个缺点在于，DMP运动的形状会受到相位变量时间演化的显著影响。若相位过早消失，轨迹的最后部分将以收敛到目标的线性动力学方式执行；若相位持续时间过长，轨迹可能发生 overshoot，无法在期望时间内达到目标。在这两种情况下，DMP运动可能显著偏离示范动作。设计合理的相位停止机制可以解决这个问题，但所采用的合适相位停止策略取决于具体应用。

为了克服这一局限性，多位学者致力于从示范中学习稳定且无时间依赖（即自治）的动力学系统。一个全局稳定的自治系统会生成一个从任意初始状态收敛到给定目标的向量场。无需相位变量，所生成的运动仅取决于系统的当前状态。值得关注的稳定自治系统学习方法利用了李雅普诺夫理论（Khansari-Zadeh 和 Billard, 2011, 2014）、收缩理论（Blocher 等, 2017; Ravichandar 和 Dani, 2015）、微分同胚变换（Perrin 和 Schlehuber-Caissier, 2016; Neumann 和 Steil, 2015）以及无源性考量（Kronander 和 Billard, 2015）。这些方法已被有效用于从示范中学习复杂运动。

总体而言，自治系统有潜力表示比DMP更为复杂的运动。例如，自治系统可以在状态空间的不同区域编码不同的运动。在这一点上，DMP只能生成连接起点与终点的刻板轨迹，无论初始状态在状态空间中位于何处。然而，这种刻板运动生成也是DMP的一个优势，因为它使得在训练数据覆盖不足的状态空间区域中预测所生成的运动更为容易。相反，对于自治系统在仅有少量或没有训练数据时的泛化行为，则难以预测。DMP在高维空间中具有良好的扩展性，因为所学习的强迫项始终依赖于一个共享的标量相位变量。而自治系统直接在高维状态空间上进行学习，这会带来数值挑战并需要更多的训练数据。综上所述，每种表示方法各有优缺点，选择时间依赖还是自治运动基元取决于具体应用。

6.3.2. 随机信息。将示范运动表示为概率分布具有若干优势。例如，在概率框架中，通过以新目标（或路径点）为条件进行条件化，可实现对新目标（或路径点）的泛化，而由概率分布计算出的协方差可表示不同自由度之间的耦合关系（Paraschos 等, 2013）。事实上，经典DMP是确定性的，缺乏关于所建模运动的随机信息。

Ben Amor 等人（2014）提出了一种方法，用于估计将DMP权重 $\mathbf{w}$ 与在 $\mathbf{T}$ 个时间点观测到的部分轨迹 $\mathbf{y}_{1:\mathbf{T}}$ 联系起来的预测分布 $\mathcal{P}(\mathbf{w}|\mathbf{y}_{1:\mathbf{T}})$ 。利用 $\mathcal{P}(\mathbf{w}|\mathbf{y}_{1:\mathbf{T}})$ 来估计给定部分运动下最可能的权重，并重建轨迹的缺失部分。然而，DMP的完整概率特征刻画仍然缺失。

ProMP框架（Paraschos等人，2013）提出了一种替代性的运动基元表示方法，该方法以协方差矩阵的形式包含不同演示之间的变异性以及不同自由度（DoF）的信息。这使得能够显式编码不同方向之间的耦合关系，并通过在期望目标、中间点或中间速度上进行条件化来增强泛化能力。ProMP计算的协方差表示了演示中的变异性与相关性。而在其他表示方法（如GPR）中，协方差则是由于训练数据不足而导致的模型不确定性的度量。Li等人（2023）尝试将ProMP与DMP的表述进行统一。核化运动基元（KMP）（Huang等人，2019；Silvério等人，2019）提供了在同一框架内对变异性、相关性和不确定性进行建模的可能性。然而，与DMP相比，KMP在较长轨迹上的计算成本可能较高，这源于核矩阵逆的计算。

6.3.3. 闭环实现及其问题。绝大多数方法仅将DMP用作闭环控制器的参考轨迹生成器，而由控制器实际执行轨迹。然而，DMP也可以作为闭环控制器本身的一部分，此时传感器测量值（例如力和力矩）被用作DMP中的耦合项以改变其行为。换句话说，在开环情况下，DMP作为计划，在执行过程中不会在线改变（可能每次执行后迭代更新），而在闭环情况下，DMP作为动作生成器，在执行过程中会在线变化。迄今为止，只有少数方法探索了闭环概念。例如，在Peternel等人（2016）的研究中，DMP直接作为控制回路中外骨骼执行器的力矩生成器，该回路通过人类用户的肌肉活动反馈进行闭环。尽管如此，在此类场景中，闭环稳定性和无源性成为关键考量，必须在广泛应用之前加以解决（Kramberger等人，2018）。

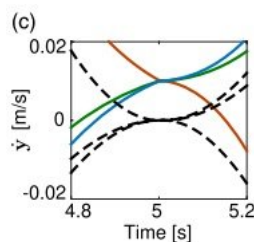
6.3.4. 应对高维输入。DMP的主要局限之一在于它显式地以时间（即1-D输入）对人和机器人的轨迹进行编码，这可能导致同步问题，因为新评估中的人体运动可能与演示的运动存在显著差异（例如速度更快/更慢）。为避免同步问题，Ben Amor等人（2014）设计了一种时间对齐策略，而Pervez等人（2017a）则在训练过程中使用期望最大化方法（Bishop, 2006）估计相位信号。

由于DMP使用基函数对轨迹进行建模，该方法在学习时间驱动的轨迹（即1-D输入）时非常有效。然而，当演示包含高维输入时，指定基函数的中心向量和宽度变得相当繁琐。具体而言，正如Bishop（2006）所讨论的，当输入维度增加时，基函数的数量往往呈指数级增长。为缓解这一局限，一些研究探索了现代深度学习技术。Pahič等人（2020）使用神经网络从输入图像中合成DMP权重，然后利用经典DMP公式生成运动轨迹。Pervez等人（2017b）使用CNN（LeCun等人，2015）从输入图像预测2-D任务参数（例如目标位置），并使用全连接神经网络从2-D参数和相位变量中恢复强迫项。CNN与全连接神经网络分两个阶段进行训练。该方法具有前景，但两个网络的独立训练增加了预处理工作并使学习过程复杂化。

文献中的其他替代方法，如GMM/GMR（Calinon, 2016）、任务参数化GMM（TP-GMM）（Calinon, 2016）和KMP（Huang等人，2019, 2021），可直接用于学习包含高维输入的演示。

6.3.5. 多吸引子系统。DMP广为人知的二阶动力学特性使其趋向于单一吸引子系统（Ijspeert等，2002a）。这些特性，例如运动的收敛性与调制，已得到充分研究，其实现方法可见于众多研究论文中。由于二阶动力学的存在，若在运动执行过程中出现方向反转等情况，系统将变得不稳定。过去几年中，针对可逆性问题主要引入了两种解决方法。第一种方法（Nemec等，2018）将可逆性视为学习两个独立基元，每个方向对应一个。该方法虽前景可观，但并未体现真正的可逆性，因其每个基元仅使用一个吸引点。

另一方面，Iturrate等人（2019）提出了一种包含两个稳定吸引子系统的替代公式。第一个吸引子定义在轨迹起点 y_0 处，第二个定义在目标点 g 处，二者之间的动力学系统保证了根据所选吸引子实现稳定收敛。该方法在保持DMP所有特性的同时，展现了真正的可逆性。然而，所有问题尚未完全解决，该方法仅在任务空间和关节空间位置轨迹上进行了评估。针对方向（例如任务空间中的四元数）的适当公式化表述仍有待完善。



7. 结论

自21世纪初被引入以来，DMP已成为机器人运动指令生成系统中应用最广泛、最受欢迎的方法之一。多位研究者对经典公式进行了扩展与利用，以克服其局限性并满足不同需求。这些研究在过去二十年中产生了大量发表论文。

本文的目标之一是对DMP领域的大量文献进行分类与综述。我们采用了系统性综述方法，在一个常用数据库中自动检索与DMP相关的论文。通过清晰且无偏的标准对检索结果进行人工筛选，最终确定了本教程综述所涵盖的论文。

本工作的另一个目标是提供一份DMP教程，以严谨的数学术语呈现经典公式及其关键扩展。我们努力统一不同方法中的符号表示，使其更易于理解。此外，我们提供了实用指南，帮助读者针对特定应用选择合适的方法。遵循教程宗旨，我们还搜索了所描述方法的开源实现，并向社区发布了多种基于DMP的实现的代码。

本文讨论了DMP的优势、局限性以及未解决的问题。我们在表5中进行了总结，同时指出了已解决的问题和仍需进一步研究的问题。鉴于DMP研究仍然十分活跃，我们提供了全面的讨论，以帮助读者了解该领域已取得的进展以及未来可聚焦的研究方向。

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利益冲突声明

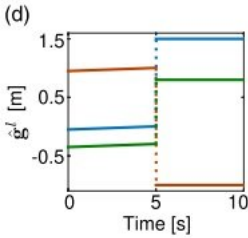
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ORCID iD

Matteo Saveriano  <https://orcid.org/0000-0002-9784-3973>



Fares J Abu-Dakka  <https://orcid.org/0000-0001-9062-9416>

Aljaž Kramberger  <https://orcid.org/0000-0002-4830-4885>

Luka Peternel  <https://orcid.org/0000-0002-8696-3689>

注释

正如Pastor等人(2009)所讨论的，使用(48)的变换系统在每次 $\{(g_{\text{new}} - y_0)\}$ 和 $\{(g - y_0)\}$ 的符号不同时，会生成一条镜像轨迹，同时达到一个新目标 $\{g_{\text{new}}\}$ 。

Saveriano等人采用了Hoffmann等人(2009)提出的多维DMP公式，用于位置DMP和四元数DMP。在本综述论文中，我们重新阐述了Saveriano等人(2019)中的合并方法，使其符合第2.1.1节和第2.1.2.1节中的公式。

注意，可以使用其他测量人体力度的反馈替代EMG，例如关节力矩或肢体力。

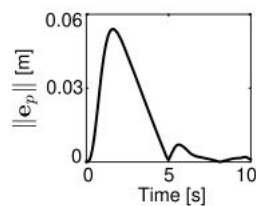
阻抗与导纳之间存在对偶性，即力产生运动，运动也产生力；因此，如果其中一个输入，另一个只能是控制系统的输出 (Peternel等人2017a)。

该工作将经典DMP扩展到了不同的空间，如 $\mathcal{SO}(3)$ 或 $\mathcal{S}^{m}_{++}$ 。尽管形式上相似，但扩展到其他黎曼流形（如格拉斯曼流形或双曲流形）并不简单，且尚未得到充分解决。

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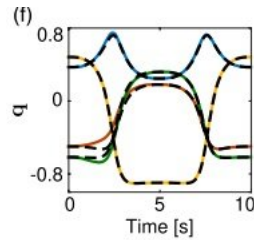
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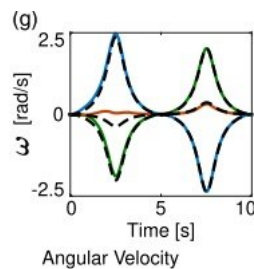
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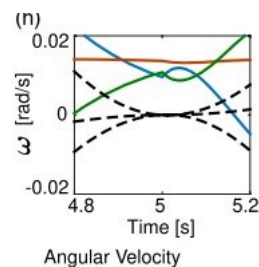
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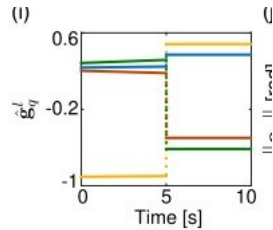
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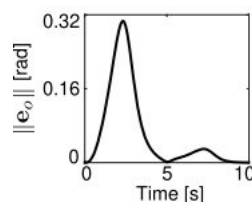
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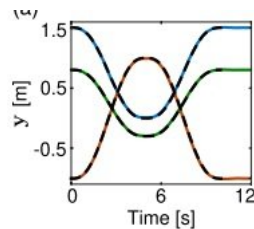
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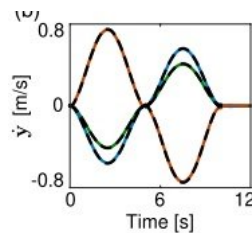
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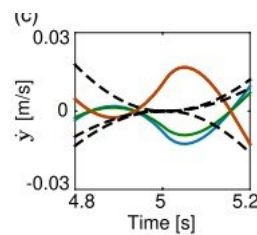
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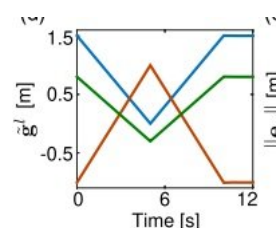
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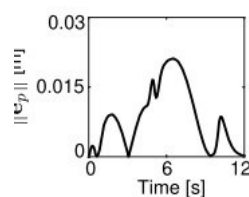
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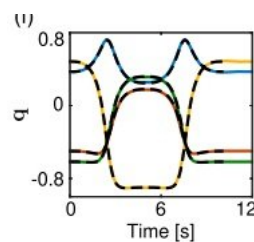
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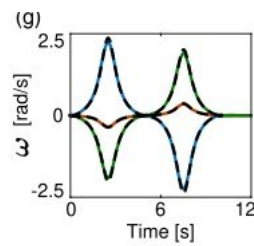
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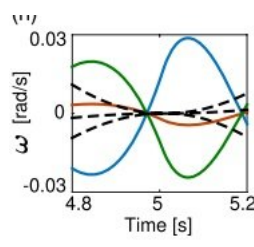
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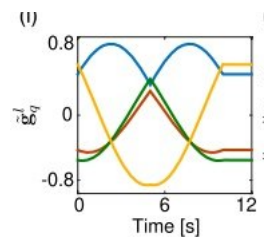
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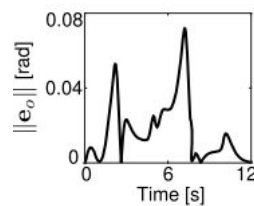
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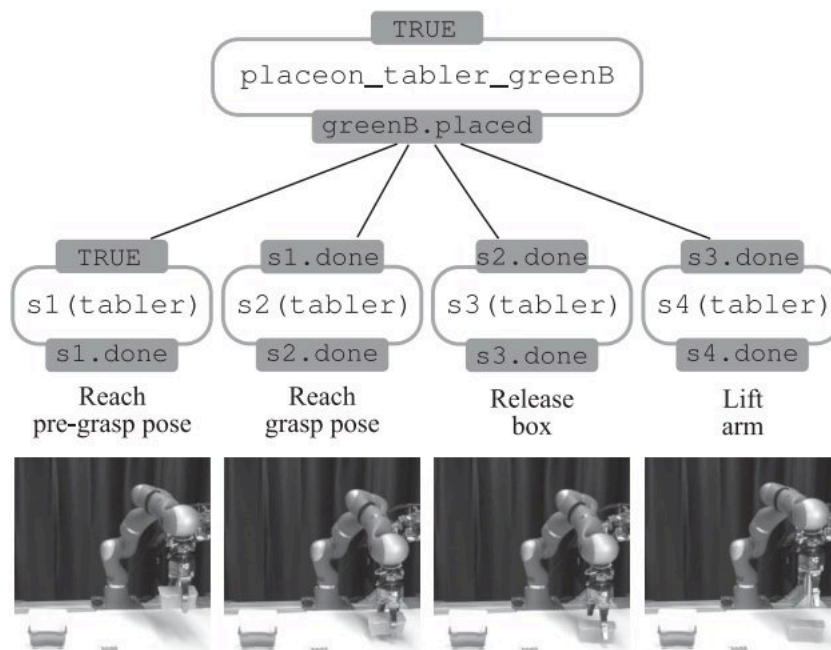
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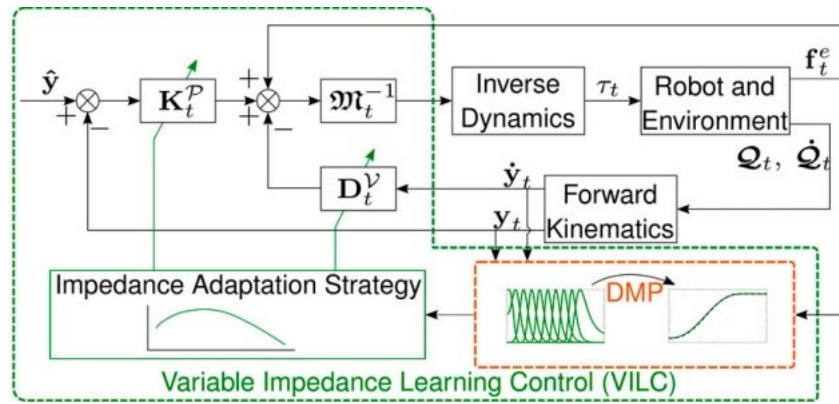
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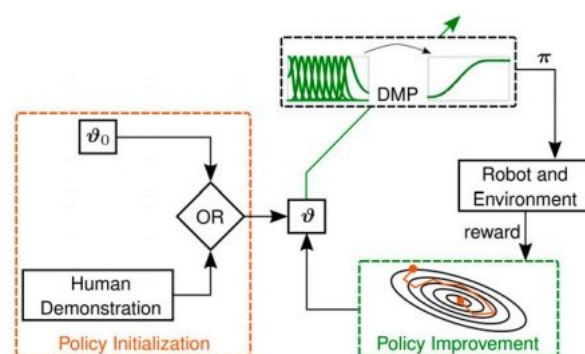
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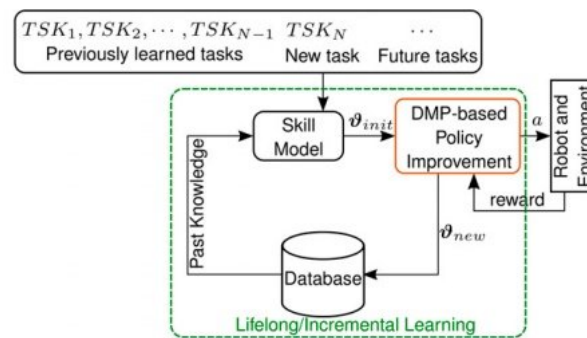
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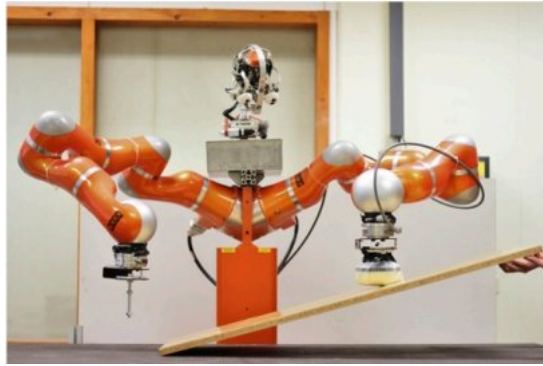
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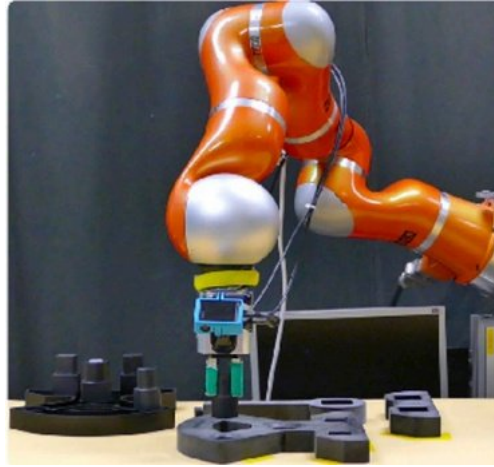
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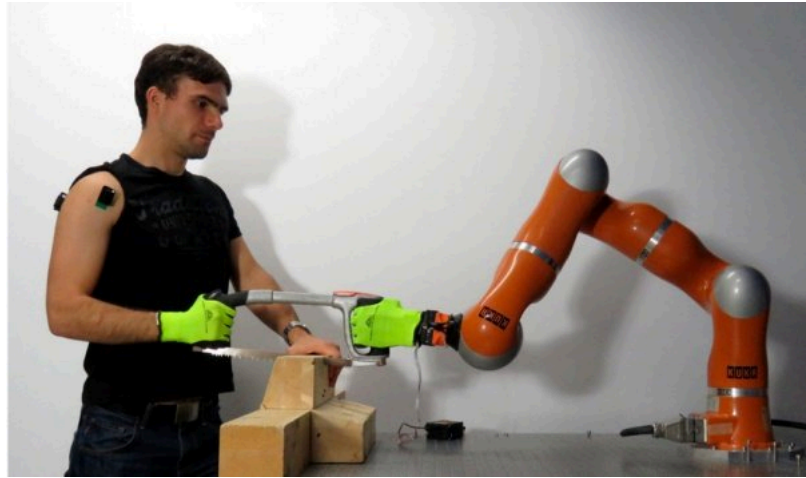
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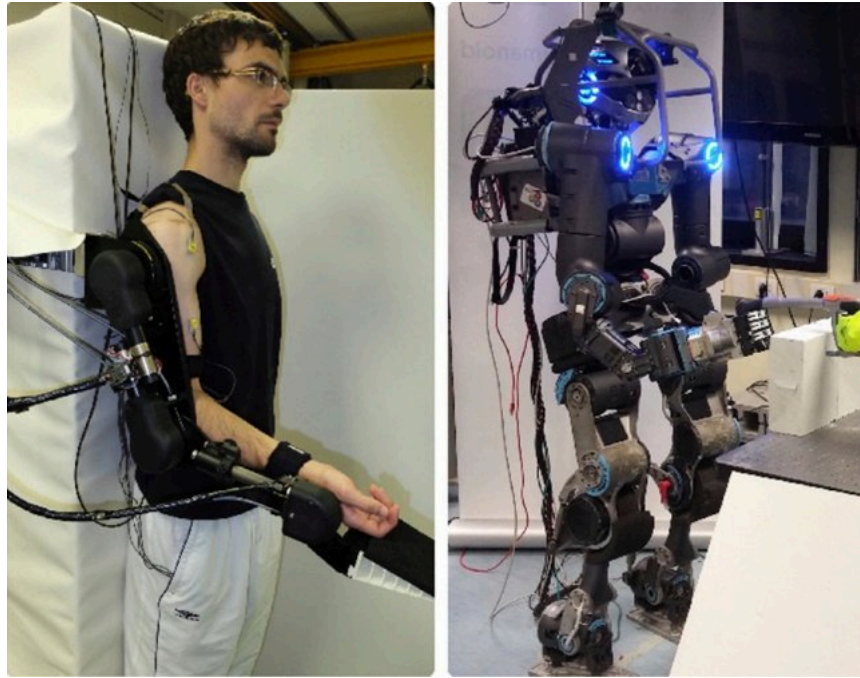
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